Differential Equations

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Preface

These are lecture notes for a 2nd year undergraduate course given to students in the Physics Department, Imperial College London. This course comprises 18 one-hour in-person lectures.

These notes have been significantly adapted from those written by Prof. Alan Heavens and Dr. Robert Kingham, the previous course lecturers.

Differential equations are widespread in physics. Indeed, key laws of physics, such as Newton's laws in mechanics, Maxwell's equations in electromagnetism, and the equations governing fluid and gas dynamics are all differential equations. The latter two examples are intimately linked to conservation laws, also expressed as partial differential equations. The study of quantum mechanics involves Schrödinger's wave equation, a partial differential equation. Most of these are, or combine to yield, second-order differential equations. Furthermore, they are often linear differential equations. If not, approximations that make them linear still provide powerful insights into the physics being described. Therefore, understanding linear second-order differential equations is very important in physics.

The broad aims of this course are twofold: to provide the tools to find solutions to such equations and to provide an understanding of the general nature of the solutions. The general insights unveiled are quite profound. We will see how quantisation can arise from mathematics. We will appreciate the origins of Fourier mathematics and see that it is but one example of orthogonal basis functions that can be used in the study of physics.

Special thanks to Spencer Kelly for proofreading this document and providing the solutions to the problem sheets.

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Introduction

Aims and Learning Outcomes

Aims – To introduce a range of techniques for solving linear ordinary and partial differential equations and eigenvalue problems. To provide practice in applying these techniques. To provide an understanding of the general properties of linear differential equations and their solutions, and how these underpin the behaviour of key equations in physics.

Learning Outcomes – On completing the Differential Equations course, students will:

- be able to classify differential equations (DEs) as ordinary or partial, as linear or non-linear, as homogeneous or inhomogeneous, and by their order and degree
- be able to solve common types of 1st order ordinary differential equations (ODEs); separable equations, linear equations by means of an integrating factor
- be able to apply the method of variation of parameters to solve ODEs
- be able to solve linear 2nd-order ODEs with constant coefficients by finding the complementary function and a particular integral
- be able to assess whether two functions are linearly independent by evaluating the Wronskian and understand how to extend this to more functions
- be able to solve particular examples of 2nd-order ODEs with variable coefficients by means of a power series or generalised power series (Frobenius method)
- be able to determine the radius of convergence of a power series
- have encountered Legendre's equation and Bessel's equation and appreciate how their solutions (Legendre polynomials and Bessel functions) can be used to deal with the Laplacian (∇^2) in spherical and cylindrical coordinates
- be able to identify separable partial differential equations (PDEs), including the diffusion, wave, Poisson, and Laplace equations, and to apply the technique of separation of variables
- understand how boundary conditions (BCs) may restrict the possible values of the separation constant to a discrete set of eigenvalues
- know the meaning of a self-adjoint problem and a Sturm-Liouville problem
- know and be able to show that the eigenvalues of an ODE with BCs that together form a Sturm-Liouville problem are real, and eigenfunctions are orthogonal
- understand how this enables an arbitrary function to be represented as a series of these orthogonal eigenfunctions, and be able to use this in simple cases
- understand and be able to apply the Green's function method for DEs
- understand and be able to apply Fourier methods for DEs

How to Study for this Course

This is *not* a flipped course. The material is presented for the first time in the lectures, and the lecture is intended as your first exposure to it. Understanding comes afterwards, through your own work. For this reason the lecture notes are deliberately posted *after* each lecture.

During term

1. **Attend the lecture.** Follow the argument as it is developed. Do not expect to have mastered everything by the end of the hour; that is not the purpose of the lecture.
2. **Review the notes.** Soon after the lecture, work through the corresponding section of the notes with pen and paper, filling in any steps that are not immediately obvious to you.
3. **Attempt the problem sheet.** The problem sheets are where understanding is consolidated, and some concepts are only fully developed there. Keeping up with every question during term is demanding, but you should at least attempt the first few questions on each sheet in the week it is released.

Lecture recordings. Avoid watching the Panopto recordings obsessively. Re-watching a lecture feels productive but is a slow and largely passive way of learning. Move on quickly from the lecture to practice and consolidation using the notes and the problem sheets.

AI chat tools. These can be very useful for clarifying a concept you are stuck on, or for extending your understanding beyond what is covered in the lectures. Current models are very good on core undergraduate material, but they still make mistakes: check what they tell you against the notes. Use them to support your own work, not to replace it, and avoid being drawn down rabbit holes.

During revision

1. **Review the notes and make your own condensed version.** Distilling the course into a few pages of your own is one of the most effective ways to identify what you do and do not yet understand.
2. **Re-do the problem sheets, thoroughly.** Work each problem from scratch, without consulting the solutions until you have a complete answer.
3. **Avoid Panopto (and YouTube).** The lectures are not intended as reference material, and watching them again is very time consuming. The same applies to online video lectures.
4. **Only then, move on to past exam papers.** Attempt them *blind*, under exam conditions. Resist the temptation to look at the answers at the first hurdle. Struggling with a problem is better than being shown a solution and you must learn to know when an answer is correct. Recognising a solution is not the same as being able to produce one.

Stop viewing and start doing – you can't doom scroll your way to understanding physics!

Course Arrangements

Differential Equations is part of the 'Differential Equations and Electromagnetism' module (with code PHYS50003), or 'DEEM' for short!

Lecturer – Porf. Carlo Contaldi

- Room: 505 Huxley (Theoretical Physics Group)
- Email: c.contaldi@imperial.ac.uk

Lectures

18 in total in Term 1 – Two per week during weeks 2–11 (but one per week in weeks 5 & 11)

Lectures	Chapter	Title
Part 1 – Ordinary Differential Equations (ODEs)		
4	1, 2 3	Classification of DEs, First-Order ODEs, Second-Order ODES: General Features and Simple Solutions.
4	4	Series Solutions of Second-Order ODEs.
2	5	Boundary Value and Eigenvalue Problems.
Part 2 – Partial Differential Equations (PDEs)		
4	6, 7 8	Introduction to PDEs, Separation of Variables, Solving PDEs with Fourier methods.
2	9	Laplacian in Spherical and Cylindrical Polar Coordinates,
2	10	Inhomogeneous Equations: Green's Functions.

Revision lecture: TBC – Likely to be toward the end of Term 2.

Canvas

Lecture notes, presentation slides and handouts will be placed in DEEM Canvas site - PHYS50003 - Differential Equations and Electromagnetism 2026-2027.

Problem sheets

9 in total – Released weekly (except week 5) before the 1st lecture of the week.

Working through the problems is an essential part of taking this course. In some cases, concepts will be introduced in the lectures/notes and fully developed in the problem sheets. Problem Sheets are released on the Lambda Feedback

Continual Assessment

In DEEM, continual assessment accounts for 20% of the course. See DEEM Canvas site > **General Information > Assessments** for more details.

- **Assessed problem sheets** – Released weeks 4 & 8 –
  The two APS in Term 1 will combine all three modules (DEEM, QP, TPSoM). Deadlines and links to the problem sheets will be available through DEEM Canvas site **> General Information > Assessments** .
- **Seminars** – Weeks 6 & 10 cover DE.

Keeping in Touch with the Lecturer

Office hours

Office hours for DE will run twice a week through Term 1 and are advertised in the DEEM Canvas site. They will take place in my office, 505 Huxley.

If you cannot make these please email me to arrange other times.

Discussion Board

'Ed Discussion' will be used for moderated questions & answers. It can be accessed through the link on Canvas and directly at Ed Discussion.

Expect responses from the Lecturer to posted questions around the middle and end of the week.

Books

The course is self-contained. However, the following three books are recommended to consolidate your knowledge of differential equations. (These books are also on the DEEM Leganto Reading List.) Between them, these 3 books cover the material in this course (and more!):

- **Mathematical methods in the physical sciences** (3rd ed.); *Boas, M.* –
  Chapters 8, 12, 13 (e-book available) – This is at the right level.
- **Mathematical methods for physics and engineering** (3rd ed.); *Riley, Hobson & Bence* – Chapters 14–18, 21 (e-book available)
  This covers the course material but in a quite thorough fashion. The presentation of concepts can become more advanced than required. (Provides a comprehensive list of methods for tackling nonlinear first- and second-order ODEs in chapters 14 & 15, respectively, which is beyond the scope of this course. But now you know where to look if you need this.)
- **Elementary differential equations and boundary value problems** (11th ed., Global ed.); *Boyce, DiPrima & Mead* – (e-book available (12th ed.))
  Don't be misled by 'elementary' in the title! Being a book that focuses on differential equations, it has space to lead the reader from the basics right through to quite advanced proofs and treatments (yielding deeper insight afforded by this course). It has plenty of fully worked-through examples and lots of exercises. There is something here for everyone.

  (*Tip: The web version of the Kortext reader rendered unreadably HUGE equations for me, but using the Kortext app fixed this for me.*)

Also worth consulting:

- **Advanced engineering mathematics** (10th ed.); *Kreyszig, E.* –
  Chapters 1–3, 5, 12 (e-book available)
  This is at the right level but also includes more depth about the key theorems. (It also covers matrix methods for systems of coupled, linear ODEs in chapter 4, which is (slightly) beyond the scope of this course.)
- **Mathematical methods for physicists a comprehensive guide** (7th ed.); *Arfken, Weber & Harris* – Chapters 9–12. (e-book available)
  The presentation is terse in some areas (e.g. basics) but is more immediate to physics (rather than engineering). It goes to higher dimensions (uses vector calculus) quite quickly. (It also covers Integral Equations in chapter 16, which is related to differential equations. This is beyond the scope of this course.)

The following books are useful resources once you understand the material:

- **The Cambridge Handbook of Physics Formulas** (Cambridge Uni. Press, 2000); *Woan, G.* – (e-book available) – A great resource for undergraduate physicists (and even postgraduate and research-level physicists)! Chapter 2 lists key mathematical formulae and definitions, including differential equations. Other chapters provide a resource for mechanics, quantum physics, thermodynamics, electromagnetism, optics, solid state physics!
- **Schaum's outline of theory and problems of differential equations** (2nd ed.); *Bronson & Ayres* – You wouldn't want to learn the subject from this book alone, but it provides compact reviews of the subject and plenty of worked examples (solved problems).

Examinable material

Everything in the lecture notes, lectures and problem sheets, unless explicitly stated.

The focus of this course is not about memorising lots of standard cases, but knowing the techniques and how to use them, plus understanding the general nature of differential equations and their solutions. Bear in mind that the formula sheet in the exams will be a resource to help you do the integrals needed to obtain closed form expressions.

Non-Examinable Appendices

Here you can find more detail on some topics. Some of this content was previously in the course, but removed to trim it down.

Part I

Part One – ODEs

Chapter 1

Classification of Differential Equations

1.1 Ordinary versus Partial Differential Equations

A differential equation (DE) is an ***ordinary differential equation*** ODE if the solution $y(x)$ has only one ***independent variable***, which is x in this case. y is the ***dependent variable***. The exponential decay equation should be a familiar example;

$$\frac{dy}{dx} = -ky \ . \tag{1.1}$$

If the equation has derivatives with respect to more than one independent variable (so the solution is, e.g., $y(x,t)$), it is classed as a ***partial differential equation*** (PDE), e.g.,

$$\frac{\partial y}{\partial t} + v\frac{\partial y}{\partial x} = 0 \ , \tag{1.2}$$

the advection equation, describing the pulse $y(x,t)$ moving at speed v in the positive (+ve) x-direction.

For PDEs, derivatives are usually written ∂, whereas for ODEs, they are written d. Be disciplined about this, as they mean different things. Note that here

$$\frac{\partial y}{\partial x} = \left.\frac{\partial y}{\partial x}\right|_t \ , \tag{1.3}$$

which means the derivative of $y(x,t)$ with respect to x, **keeping t constant**. Often, the bare form on the LHS is used, in which case it is implicit that all other independent variables are held constant[1].

In this first part of the course, $y(x)$ will be our canonical choice for dependent and independent variables. However, in physics, the variables are often different. Familiar equations which illustrate this are Newton's 2nd law

$$m\frac{d^2x}{dt^2} = -b\frac{dx}{dt} + F(x) \ , \tag{1.4}$$

an ODE for position $x(t)$ (in this case, with a damping term and another (arbitrary) position dependent force $F(x)$) and the 1D time-dependent Schrödinger equation

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi = \imath\hbar\frac{\partial\psi}{\partial t} \ . \tag{1.5}$$

[1]The bare form is convenient if you know what the other independent variables are. However, it can lead to confusion and misuse if not!

for wave function $\psi(x,t)$.

The rest of this first half of the course will just consider ordinary differential equations.

1.2 Order

The order of a differential equation is the maximum number of times the dependent variable has been differentiated. In the examples above, (1.1) and (1.2) are first order, whilst (1.4) and (1.5) are second order DEs.

1.3 A Very Simple ODE

Let us consider the simplest concrete example of interest; the trivial 2nd-order ODE

$$\frac{d^2y}{dx^2} = 0 \ . \tag{1.6}$$

By definition, this ODE tells us that the solutions $y(x)$ have no curvature (second derivative) so, from intuition, we know that they should be straight lines. We can verify this easily for this ODE because it can be integrated twice straight away. (It is unusual to be able to do this. We will not be so lucky in the rest of the course.) The ***general solution*** is

$$y(x) = c_1 x + c_0 \ , \tag{1.7}$$

as expected. It has two arbitrary integration constants, c_0 and c_1. It is a straight line of arbitrary gradient and intercept.

To find a ***particular solution***, we provide two ***initial conditions*** to determine values for the two constants. For instance, insisting that $y = -1$ and $y' = 1$ at $x = 0$ requires that $c_0 = -1$ and $c_1 = 1$, yielding

$$y_1(x) = x - 1 \ .$$

A different particular solution, that instead satisfies $y = 2$ and $y' = \frac{1}{2}$ at $x = 0$, is

$$y_2(x) = \frac{x}{2} + 2 \ .$$

It is a different straight line that crosses $y_1(x)$.

It is easy to verify that $y = y_1(x) + y_2(x) = \frac{3}{2}x + 1$, a linear combination of the two solutions is also a solution;

$$\frac{d^2}{dx^2}(y_1 + y_2) \quad = \quad \frac{d^2}{dx^2}\left(\frac{3}{2}x + 1\right) = \frac{d}{dx}\left(\frac{3}{2}\right) = 0 \ .$$

This occurs because $\frac{d^2y}{dx^2} = 0$ is a linear differential equation.

1.4 Most General ODE

The most general form of an ODE is an equation involving the independent variable (x), the dependent variable (y) and its derivatives;

$$F\left(x, y, y', y'', ..., y^{(n)}\right) = 0 \ . \tag{1.8}$$

Here, the prime notation denotes differentiation

$$y' \equiv \frac{dy}{dx} \quad , \qquad y'' \equiv \frac{d^2y}{dx^2} \quad , \qquad y^{(n)} \equiv \frac{d^ny}{dx^n} \ . \tag{1.9}$$

Thus (1.8) is the most general form of an nth-order ODE, and F could be anything! Usually, the highest derivative can be isolated such that

$$y^{(n)} = G\left(x, y, y', y'', ..., y^{(n-1)}\right) \ , \tag{1.10}$$

where G is a different function than F. (This is the *explicit form* of a DE. $F = 0$ is the *implicit form*.)

1.5 Most General Solution

The most general solution of (1.10) has the form

$$y = g(x, c_1, c_2, \ldots, c_n) \ , \tag{1.11}$$

where c_1, c_2, …, c_n are ***parameters*** (often referred to as ***integration constants***) and g is a (perhaps complicated!) function of x and the parameters. The key point here is that the **number of constants is the same as the order**. At this point, values for the parameters are not specified and could be any real number $c_k \in \mathbb{R}$ or complex number $c_k \in \mathbb{C}$. For the implicit ODE (1.8), the most general possible ODE, the general solution[2] has the form

$$f(x, y, c_1, c_2, \ldots, c_n) = 0 \ . \tag{1.12}$$

In physics, we usually want to find solutions given that we know the DE that governs our system. Mathematically, it is insightful to turn the problem on its head: Given a specific function of the form (1.11), what is the DE that yields that function? Formally, this can be done by differentiating equation (1.11) n times w.r.t. x and eliminating all the n parameters between the resulting set of $n+1$ equations (i.e. including eqn (1.11) itself). Further consideration is beyond the scope of this course.

1.6 Linear/Nonlinear

An ***linear ODE*** is one where F in equation (1.8) can be written as a linear function of y, y', …, $y^{(n)}$. The most general form of a linear n^{th}-order ODE is

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \ldots + a_1(x)\frac{dy}{dx} + a_0(x)y = f(x) \tag{1.13}$$

[2] In mathematics, this is known as the 'primitive' of the ODE.

where the 'coefficients' $a_0(x)$, $a_1(x)$, $\ldots a_n(x)$ in front of the derivatives are only functions of the independent variable.[3] This includes the possibility of some or all just being constants. A key point here is that the right-hand side (RHS) $f(x)$ is a function of x only (and not y).[4]

If an ODE cannot be written in the form above, then it is a ***nonlinear ODE***. Simple examples are

$$\frac{dy}{dx} = (1+y^2) \quad , \tag{1.14}$$

$$\frac{d^2y}{dx^2} + \sin(y) = 0 \ . \tag{1.15}$$

This is due to the appearance of y^2 and $\sin(y)$ (rather than just y) in the first and second examples, respectively.

Most of the key differential equations in physics (certainly those appearing in the core UG programme) involve linear differential equations.

1.7 Homogeneous/Inhomogeneous

Collect all the terms involving the dependent variable (y) on the left-hand side (LHS) and the other terms on the RHS. Equation (1.13) is in this form, with the RHS given by $f(x)$. If $f(x) = 0$ *everywhere* the equation is ***homogeneous***. If $f(x) \neq 0$ *somewhere*, it is ***inhomogeneous***. In the examples immediately above, equation (1.14) is an inhomogeneous ODE and (1.15) is homogeneous.

1.8 Differential Operators & Linearity

The LHS of (1.13) can be viewed as a ***differential operator***, denoted by $\hat{O}$, acting on $y(x)$, i.e., $\hat{O}(y)$. The RHS of (1.13), i.e., $f(x)$ cannot be included in $\hat{O}$. In fact the LHS of (1.13) is a ***linear differential operator***, also frequently referred to simply as a 'linear operator'. We will usually write linear operators as $\hat{L}$. The general form of a linear operator is therefore

$$\hat{L} = a_n(x)\frac{d^n}{dx^n} + a_{n-1}(x)\frac{d^{n-1}}{dx^{n-1}} + \ldots + a_0(x) \ . \tag{1.16}$$

A ***linear combination*** of y_1 and y_2 is $ay_1 + by_2$, where a, b are arbitrary constants. Suppose y_1 and y_2 are both solutions to $\hat{L}(y) = 0$, which is a DE. A linear operator is *defined* in the following way;

$$\hat{L}(ay_1 + by_2) = a\hat{L}(y_1) + b\hat{L}(y_2) \ . \tag{1.17}$$

This property is known as ***linearity***. Thus $\hat{L}(ay_1 + by_2) = 0$ and therefore $y = ay_1 + by_2$ is also a solution of $\hat{L}(y) = 0$.[5]

Conversely, rearranging equation (1.14) yields $y' - y^2 = 1$. Here, $\hat{O}(y) = y' - y^2$ is a nonlinear operator. You can readily verify that if y_1 and y_2 are solutions of $\hat{O}(y) = 0$, then $\hat{O}(ay_1 + by_2) \neq a\hat{O}(y_1) + b\hat{O}(y_2)$ in this case, verifying that this particular differential operator is indeed not linear.

The above property of linear operators and the homogeneous DE they form has profound consequences for physics. It allows the use of the ***superposition principle*** whereby a linear combination of two

[3] A similar definition applies to PDEs, as will be seen later.
[4] The f here is different from that in eqn (1.12).
[5] These concepts extend to operators involving partial derivatives, too.

solutions is also a solution. This is crucial for allowing electromagnetic waves to pass unimpeded through each other in vacuum (or another linear medium) or for a superposition of travelling waves to result in a standing wave. In conjunction with suitable boundary conditions, linear homogeneous equations give rise to the mathematics you learned in the Fourier course preceding this one. We will later see how it also gives rise to analogous but different sets of orthogonal basis function for other prominent differential equations in physics.

1.9 Degree

The ***degree*** of a DE is the power to which the *highest-order* derivative is raised when the DE is rationalised so that only integer powers of derivatives of y occur. [6] All the previous ODE and PDE examples so far are **first-degree** differential equations. Two examples of 2nd-degree equations are

$$y' = \frac{x \pm \sqrt{x^2 - 4y}}{2} \quad \rightarrow \quad (y')^2 - xy' + y = 0 \tag{1.18}$$

and

$$\frac{d^3y}{dx^3} + x\left(\frac{dy}{dx}\right)^{\frac{3}{2}} + x^2y = 0 \quad \rightarrow \quad \left(\frac{d^3y}{dx^3}\right)^2 + 2x^2y\frac{d^3y}{dx^3} - x^2\left(\frac{dy}{dx}\right)^3 + x^4y^2 = 0 \ . \tag{1.19}$$

The first example is a 1st-order, 2nd-degree ODE. The second example is a 3rd-order, 2nd-degree, non-linear, homogeneous ODE! This is a very pathological case, just for illustrative purposes. Don't worry; we will not consider such tricky cases any further in the course as they do not arise in the physics courses you will be taking.

[6] In terms of classifying the degree, it does not matter how $y^{(0)}$, i.e., the *undifferentiated* dependent variable, appears in the DE. In principle, one could have a function $\phi(x, y)$, e.g., e^y or $x\sin(y)$, etc.

Chapter 2

First-order ODEs

The general form for a first-order ODE of first-degree is

$$\frac{dy}{dx} = f(x, y) \ . \tag{2.1}$$

It can be linear or nonlinear. We will limit ourselves to first-degree ODEs from here on, except for a brief look at a 2nd-degree ODE at the end of this chapter.

The general solution to first-order ODEs are essentially obtained by integrating the equation up once, using an ***indefinite integral***. This introduces the *single* arbitrary constant that the general solution to a first-order ODE requires. A word on notation: Indefinite integrals, e.g., $\int f(x)dx$ will often be written using the following notation

$$\int^x f(z)dz \ , \tag{2.2}$$

with a dummy integration variable (e.g. z here) and the lower limit omitted. Remember that an arbitrary integration constant arises when evaluating an indefinite integral, e.g., for $f(x) = x$

$$\int^x z dz = \frac{x^2}{2} + c \ . \tag{2.3}$$

In general

$$\int^x f(z)dz = F(x) + c \quad \text{where} \quad \frac{dF}{dx} = f(x) \ , \tag{2.4}$$

and $F(x)$ is known as the ***antiderivative*** of the function $f(x)$.

2.1 Methods for Solution

Four methods for solving first-order ODEs are covered below. An additional, ***non-examinable*** method for solving a class of 1st-order ODEs known as 'exact' can be found in appendix A.

2.1.1 Separable equations

This technique works for equations of the form

$$\boxed{\frac{dy}{dx} = f(y)g(x)} \tag{2.5}$$

where the RHS is a *separable* function of x and y. Such equations can be nonlinear or linear. The above equation can be solved by direct integration;

$$\boxed{\int \frac{dy}{f(y)} = \int g(x)dx \ .} \tag{2.6}$$

Of course, performing the actual indefinite integrals to obtain a closed-form expression may be tricky. If you can, each will produce an arbitrary integration constant. These can be combined into the *single* integration constant required for any first-order ODE.

This technique was covered in the Complex Analysis course that you took last year.

2.1.2 Integrating factors for linear first-order ODEs

The general form of a linear first-order ODE is

$$a_1(x)\frac{dy}{dx} + a_0(x)y = h(x) \ . \tag{2.7}$$

It can be written in ***standard form*** by dividing through by $a_1(x)$ so that the coefficient of the highest-order derivative is unity;

$$\boxed{\frac{dy}{dx} + f(x)y = g(x) \ .} \tag{2.8}$$

All such equations can be solved with an ***integrating factor*** (IF). Again, this technique was covered in the Complex Analysis course that you took last year.

The IF is a simple trick that allows us to use reduce the exercise to that of a separable ODE. To see this, let's introduce a to-be-determined function $\mu(x)$ that allows us to write (2.8) in a simpler form with a single derivative term on the LHS

$$\frac{d}{dx}[\mu(x)y(x)] = \mu(x)g(x) \ .$$

This can be arranged if

$$\frac{d}{dx}[\mu(x)y(x)] = \mu\frac{dy}{dx} + \frac{d\mu}{dx}y = \mu(x)\frac{dy}{dx} + \mu(x)f(x)y \ . \tag{2.9}$$

Whic is true if μ satisfies

$$\frac{d\mu}{dx} = \mu(x)f(x) \ . \tag{2.10}$$

We can then write (2.8) as (2.1.2)

$$\frac{d}{dx}\left[\mu(x)y(x)\right] = \mu(x)g(x) \ . \tag{2.11}$$

What we need to do now is actually determine $\mu(x)$. This can be done by solving (2.10) using separation of variables:

$$\int \frac{d\mu}{\mu} = \int f(x)dx \ .$$

The LHS indefinite integral evaluates to $\ln|\mu|+c_\mu$ where c_μ is the integration constant. The RHS integral is integrated up to x, so **taking care to relabel the dummy (integration) variable**;

$$\ln|\mu(x)| = \int^x f(u)du + c$$

(where $c = -c_\mu$) so that

$$\boxed{\mu(x) = \exp\left[\int^x f(u)du\right] \ .} \tag{2.12}$$

Technical but important points:

1. We can ignore the integration constant since it just multiplies μ, and any multiple of μ is a good IF. So we choose to set $c = 0$ so that exponentiation results in the multiplicative factor $\exp(c) = 1$.
2. We dropped the $|\ldots|$ on μ in (2.12), for the following reason. If we keep it, we have two possibilities: $\mu = \pm\exp(\int^x f(u)du)$. In other words, μ can be positive or negative with magnitude $\exp(\int^x f(u)du) > 0$. Multiplying an IF by -1 does not change its utility. This is thanks to the linearity of the DE that the IF works on, i.e. (2.8) has the form $\hat{L}(y) = g(x)$. So, we are safe to choose the positive version.

Now we integrate both sides of equation (2.11) :

$$\mu(x)y(x) + c' = \int^x \mu(u)g(u)du$$

which yields the final solution

$$\boxed{y(x) = \frac{1}{\mu(x)}\int^x \mu(u)g(u)du + \frac{c}{\mu(x)}} \tag{2.13}$$

where the constant of integration here ($c = -c'$) is certainly important[1]. This expression involving an integral is considered to be a solution. Ideally, of course one would evaluate the integral and $\mu(x)$ analytically, but if not, a computer can do it numerically to high precision.

[1] Note that there is no need to introduce a further integration constant when evaluating the indefinite integral (for some specific integrand) in the IF method equation (2.13). The one constant that is required when integrating a 1st-order DE to get the solution has already been introduced via c. If one does introduce a further arbitrary integration constant, say d, it can just be added to c and changes nothing. One still has a single arbitrary constant c'' (which equals $c+d$). [If the constant c were *not* present in (2.13), then of course one *would* need the integration constant when evaluating the indefinite integral.]

2.1.3 Variation of parameters

This is a powerful method for inhomogeneous equations (IEs) that are linear. Here, we will look at its application to first-order IEs. Sections 3.3 and 3.6 will demonstrate its use with 2nd-order linear ODEs.

Say that we can solve the HE to get, say, $y(x) = c\,u(x)$. Here c is a constant. Whatever its value, $y(x) = c\,u(x)$ will be a solution of the HE. However, it is not a solution for the IE. To get a solution to the IE, we 'elevate' (or vary) the parameter c to a function of the independent variable x

$$\boxed{y(x) = v(x)u(x)\ .} \tag{2.14}$$

We can eliminate some terms using the HE, and the remaining equation for $v(x)$ may be soluble. This method is also useful for higher-order ODEs (see *non-examinable* appendix B).

For now, we will demonstrate that we can recover the IF method this way. Consider

$$\frac{dy}{dx} + f(x)y = g(x)\ .$$

We first solve the HE, writing the solution as $u(x)$, so $du/dx = -f(x)u$, and

$$\int \frac{du}{u} = -\int^x f(s)ds$$

so that

$$u(x) = \exp\left[-\int^x f(s)ds\right] \tag{2.15}$$

to within a multiplicative constant e^{-c}, where c is the integration constant arising from evaluating the LHS of the preceding equation. Note the minus sign. Also note that u is closely related to the IF; $u(x) = 1/\mu(x)$. Now substitute $y = vu$ into the IE:

$$\frac{dy}{dx} + fy = \frac{d}{dx}(vu) + fuv = v\cancel{\frac{du}{dx}} + u\frac{dv}{dx} + \cancel{fuv} = g(x)\ ,$$

where the HE has been used to cancel the indicated terms: Those terms are $v(du/dx + fu) = v\cdot(0)$. This leaves

$$u\frac{dv}{dx} = g(x)\ .$$

Now we have an equation whose variables we can separate:

$$v(x) = \int^x \frac{g(s)}{u(s)}ds + c \tag{2.16}$$

and hence the general solution is

$$y(x) = \frac{1}{\mu(x)}\int^x g(s)\mu(s)ds + \frac{c}{\mu(x)}$$

which is the integrating factor solution we found before, with IF $\mu(x) = 1/u(x) = \exp\left[\int^x f(s)ds\right]$.

2.1.4 Substitutions

The general idea is to substitute for one of the variables to transform an intractable ODE into one that can either be tackled using one of the previous methods or is well-known. Then the solution, say $u(x)$, for the new variable can be transformed into the desired form $y(x)$. This applies not only to 1st-order equations but also to 2nd-order and higher ones. The difficulty is to know what substitution works for a particular ODE. (Or you might be lucky and quickly guess the right one!)

Sometimes, substituting to change the dependent variable helps. A prime example is when the $f(x,y)$ in the general ODE $y' = f(x,y)$ is such that x and y appear together as x/y everywhere. Such functions are termed 'homogeneous'.[2] Then the substitution $u = x/y$ results in a separable equation $du/dx = F(u)G(x)$. This can be solved for $u(x)$, and then the desired solution is simply $y(x) = x/u(x)$. Other cases are well known. (Problem sheet 1 has a few problems of this kind.)

Sometimes, changing the independent variable helps. A key example is ***Euler's differential equation***

$$x^2y'' + \alpha xy' + \beta y = 0 \tag{2.17}$$

(where α and β are constants), a 2nd-order ODE, which can be transformed into an easier 2nd-order ODE with constant coefficients via the substitution $t = \ln x$.

2.2 General Characteristics of Solutions

The general solution to a first-order, first-degree ODE represents a family of ***integral curves***. A particular value of the integration constant c picks out a particular curve. In a **domain of the x-y plane** where the Existence and Uniqueness theorems (to be covered imminently in section 2.3) are satisfied, **different integral curves do not intersect**. The complete family of integral curves covers the whole of the domain. For the simple case of the decay equation $dy/dx = -ky$ and a simple separable equation $dy/dx = -x/y$, the general solutions are

$$y(x) = C\exp(-kx) \qquad \text{and} \qquad x^2 + y^2 = R^2 \ ,$$

respectively. Figure 2.1 shows a selection of integral curves for these ODEs and demonstrates that the solutions for different c do not intersect. For the decay equation, the domain is the upper half of the x-y plane (assuming we allow $x < 0$). For $dy/dx = -x/y$ the circles fill the entire plane.

Direction fields are useful tools to get a qualitative understanding of the solutions of first-order, first-degree ODEs. Essentially, a line segment with gradient y' is plotted on a grid of positions in the x-y plane. It would be straightforward to do this in Python (perhaps taking care of where y' will be infinite). Even if you cannot find the solution in closed form, the integral curves can be approximately traced from the direction field. It can provide a qualitative picture of the behaviour of the whole family of solutions. Figure 2.2 shows the direction field for the logistic equation

$$\frac{dy}{dx} = y(1 - y/a) \ , \tag{2.18}$$

a nonlinear equation that you saw in your Complex Analysis course in Year 1. Here, an inspection of the RHS of (2.18) easily shows that the steady-state populations, where $dy/dx = 0$, occur at $y = 0$ and $y = a$. The direction field visually shows that $y = 0$ is unstable ($y(x)$ is 'repelled' away from the *vicinity* of $y = 0$ as x increases) and $y = a$ is stable ($y(x)$ tends towards the value a with increasing x).

[2] Homogeneous in this context has no connection to the previous meaning of homogeneous we encountered!

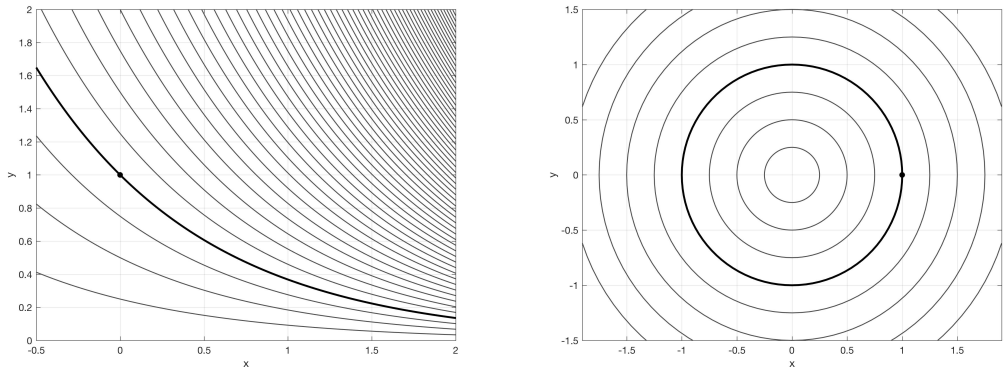


Figure 2.1: Integral curves for (left) $dy/dx = -kx$ (with $k = 1$) and (right) $dy/dx = -x/y$. The particular solution for initial condition $y(x = 0) = 1$ is highlighted for the former. The particular solution for $y(x = 1) = 0$ is highlighted in the latter.

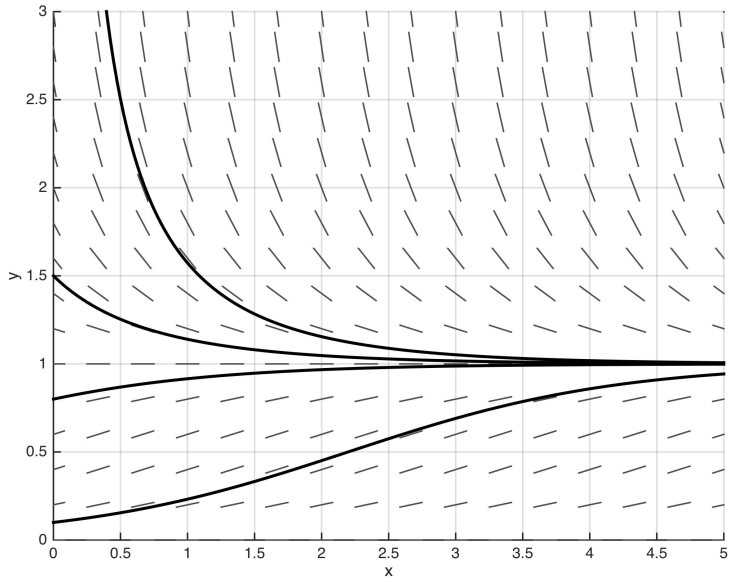


Figure 2.2: Direction field for the logistic equation $dy/dx = y(1 - y/a)$, with $a = 1$. Solutions for four different initial conditions are overlaid.

2.3 Existence and Uniqueness of Solutions (without proof)

We concentrate on the general, linear first-order ODE and ask under what conditions the solution can go through the point x_0, y_0 where $y_0 = y(x_0)$, and how many solutions there will be.

Theorem. (Existence and Uniqueness) *The solution to*

$$\frac{dy}{dx} + f(x)y = g(x) \ , \quad y(x_0) = y_0 \tag{2.19}$$

exists in the range $\alpha < x < \beta$, and is unique if $f(x)$ and $g(x)$ are continuous in the same range, providing it contains x_0.

There is also an existence and uniqueness theorem for the general, nonlinear first-order ODE given by equation (2.1). It essentially requires that both f and $\partial f/\partial y$ are continuous and is more restrictive in the range of x where a unique solution can exist. The technicalities will not be covered in this course.[3]

These theorems only apply to first-degree ODEs as we will see below.

2.4 (*) Higher Degree Equations and Singular Solutions: a brief look

When the degree of a first-order equation is higher than one, two new things can happen.

1. The family of solutions can cross: A breakdown of uniqueness.
2. A solution arises that has no arbitrary integration constant: The ***singular solution***.

(*) Non-examinable – *The rest of this section in non-examinable. You will not be required to solve higher degree ODEs in the exam.*

The example of a 2nd-degree ODE seen in section 1.9

$$(y')^2 - xy' + y = 0$$

exhibits both behaviours. You can verify by substitution into the ODE that a solution is

$$y_{GS} = c(x - c) \ ,$$

a straight line, where c is the expected constant for a first-order ODE. This is, therefore, the general solution. However, trying

$$y_{SS} = \frac{x^2}{4}$$

shows that it is also a solution. Observe that it has no integration constant![4] It is known as a ***singular solution***. The plot of these solutions in figure 2.3 illustrates that the straight-line solutions sweep in a way that at a given location x, y, it is possible for two integral curves to go through the point. You can also see that each particular solution is tangent to the singular solution. In other words, the singular solution lies on the envelope formed by the complete set of general solutions. This behaviour is quite typical.

[3] See Boyce, DiPrima & Meade §2.8 (pg 83) for more details, including a proof of the theorem, if you are interested.

[4] Verify that $y = \frac{x^2}{4} + d$ (where d is an arbitrary constant) is *not* a solution (unless $d = 0$).

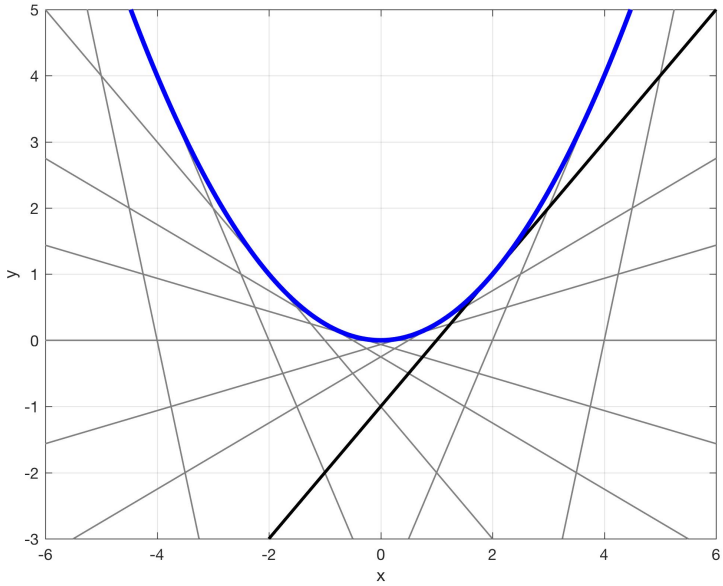


Figure 2.3: Solution of the second-degree ODE $(y')^2 - xy' + y = 0$. The thick blue line is the singular solution. The thick(ish) black line is the particular solution for $c = 1$.

Chapter 3

Second-Order ODEs: General Features and Simple Solutions

Second order differential equations are very important as they are very common in physics. Most second order DEs arising in physics are also linear, or at least a meaningful understanding of the physical system can be obtained when the nonlinear terms are simplified to yield a linear DE. Although the equations of mechanics governing the motion of point particles are 2nd-order ODEs, many other equations are 2nd-order PDEs (e.g. Schrödinger's wave equation). Nevertheless, key methods for solving such PDEs ultimately end up with having to solve one or several 2nd-order ODEs. Therefore this chapter and the next (which focusses on solving 'tricky' 2nd-order ODEs) will lay the foundations necessary to later allow for the solution of the key PDEs in physics.

Some of the material below was already covered in your Year 1, Complex Analysis course. But there will be plenty of new concepts introduced around them. This chapter will cover general properties that apply to all ***linear*** 2nd-order ODEs. However when we come to finding specific solutions in order to illustrate these principles, we will specialise to constant coefficients in this chapter.

3.1 A Very Simple ODE – continued

As a warm up exercise, let us continue the simple example of the 2nd-order equation $y'' = 0$ started back in section 1.3. Adding a simple RHS to it, so that now we have an inhomogeneous ODE (which we shorten to inhomogeneous equation (IE) for convenience)

$$\frac{d^2y}{dx^2} = 1$$

and integrating twice yields a different general solution to before;

$$y(x) = \frac{x^2}{2} + c_1 x + c_0 \ .$$

There are two parts to the solution; the new part $x^2/2$, known as the ***particular integral*** (PI), and the part $c_1 x + c_0$ which we obtained previously when the RHS of the ODE was zero. That part is known as the ***complementary function*** (CF). Substituting $y_{\text{PI}} = x^2/2$ into $d^2y/dx^2 = 1$ yields $1 = 1$, verifying that it is indeed a solution of the IE. Adding the CF to the PI makes no difference to the ability of $y = y_{\text{PI}} + y_{\text{CF}}$ to solve the IE, since the LHS differentiates the CF away to zero. The ***general solution*** of this is thus $y_{\text{GS}} = y_{\text{CF}} + y_{\text{PI}}$.

3.2 General Solutions of 2nd-order ODEs

3.2.1 Existence and Uniqueness (without proof)

In ***standard form***, a linear 2nd-order ODE is

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = g(x) \tag{3.1}$$

or $\hat{L}(y) = g(x)$. Notice that the coefficients of y' and y can be functions of x.

Theorem. (Existence and Uniqueness) *If $p(x)$, $q(x)$, $g(x)$ are continuous over the interval $\alpha < x < \beta$ containing x_0 (IVP), there exists a unique solution that satisfies the initial conditions $y(x_0) = y_0$ and $y'(x_0) = y_0'$.*

This theorem applies to inhomogeneous ($g(x) \neq 0$) and the homogeneous ($g = 0$) cases.

Note that this theorem strictly applies to ***initial value problem***s (IVP) where the two conditions are specified at a *single* point x_0. These so called ***initial conditions*** (ICs) are the values of y and y'. When the two conditions are specified at *different* points, it is a *boundary value problem* (BVP), e.g., $y(x_a) = y_a$ and $y(x_b) = y_b$ referred to as *boundary conditions* (BCs). As we will see in chapter 5, a BVP may have a unique solution, no solution or an infinity of solutions that meet the BCs. For a given ODE, the solvability of the 'problem' can vary depending on the choice of BCs. So the existence and uniqueness theorem above does not apply to BVPs.

3.2.2 Form of the General Solution

The solution to *any* linear 2nd-order ODE is the sum of the ***complementary function*** (CF) and a ***particular integral*** (PI).

The CF is the general solution of the HE (where $g(x)$ is set to zero). Call it y_{CF}. The general solution of a 2nd-order HE must contain two integration constants, therefore the CF will have the form

$$\boxed{y_{CF}(x) = c_1 y_1(x) + c_2 y_2(x)} \tag{3.2}$$

where y_1 and y_2 are two different (independent) solutions that arise and c_1 and c_2 are the arbitrary constants. The two solutions are called ***basis functions***. The solution to the HE satisfying *any* set of ICs can be found via a superposition of the basis functions. A given set of ICs can be satisfied by setting c_1 and c_2 to appropriate values, which yields a particular solution that satisfies them. (The concept of independent solutions will be explored in section 3.5.)

A PI is *any* solution to the inhomogeneous equation. Call it y_{PI} . The general solution is

$$y_{GS}(x) \quad = \quad y_{CF}(x) + y_{PI}(x) \ . \tag{3.3}$$

The PI does not need to satisfy the ICs. The CF (with appropriate c_1 and c_2) ensures that the general solution satisfies the ICs.

3.3 CF for 2nd-order ODEs with Constant Coefficients

(Mostly review of material from Complex Analysis) The HE is now

$$\boxed{a_2\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_0 y = 0 \ ,} \tag{3.4}$$

where a_0, a_1 and a_2 are all constants: They do not depend on x. Solve using the ***trial function***

$$y = e^{mx} \ . \tag{3.5}$$

By repeatedly differentiating the trial function, substituting into the DE and taking out the factor y that appears in every term (on the LHS), we find that m satisfies

$$a_2 m^2 + a_1 m + a_0 = 0 \ .$$

This is the ***characteristic equation***, which is a quadratic equation in m, with the usual solutions:

$$\boxed{m = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0a_2}}{2a_2} \ .} \tag{3.6}$$

Case 1: $a_1^2 > 4a_0a_2$ – There are two, distinct, real roots, m_+, m_- (where $m_+ \neq m_-$). Thus

$$\boxed{y = c_1 e^{m_+ x} + c_2 e^{m_- x} \ .} \tag{3.7}$$

Here we identify the two basis functions as $y_1 = \exp(m_+ x)$ and $y_2 = \exp(m_- x)$ (or vice versa).

Case 2: $a_1^2 < 4a_0a_2$ – There are two complex roots, m_+, m_-, which are complex conjugates.

$$m_\pm = p \pm iq$$

where $p, q \in \mathbb{R}$ and

$$\boxed{y = c_1 e^{(p+iq)x} + c_2 e^{(p-iq)x} = e^{px}[c_1 e^{iqx} + c_2 e^{-iqx}] \ .} \tag{3.8}$$

Note that $c_{1,2}$ are typically complex. We can also write

$$y = e^{px}[c_3 \cos(qx) + c_4 \sin(qx)] \ . \tag{3.9}$$

The 2 complex exponentials, or the 2 products of exponentials and trigonometric functions, constitute different sets of basis functions for the solution. You can use either.

Case 3: $a_1^2 = 4a_0a_2$ – There are equal roots.

$$m = -\frac{a_1}{2a_2} \ . \tag{3.10}$$

This is only one solution, but we need 2! At this point we can only write $y = c_1 \exp(mx)$.

If you only have one arbitrary constant in your answer, you are missing part of the solution and have to

work harder to find it!

For the second solution, let us try our trick of variation of parameters:

$$y = u(x)e^{mx} \tag{3.11}$$

which we substitute into the DE:

$$\begin{aligned} a_2\frac{d^2}{dx^2}\left[ue^{mx}\right] + a_1\frac{d}{dx}\left[ue^{mx}\right] + a_0 ue^{mx} &= 0 \\ a_2\left[u''e^{mx} + 2u'me^{mx} + um^2e^{mx}\right] + a_1\left[u'e^{mx} + ume^{mx}\right] + a_1\left[ue^{mx}\right] &= 0 \\ a_2u'' + u'[2a_2m + a_1] + u[a_2m^2 + a_1m + a_0] &= 0\ . \end{aligned}$$

Now, the terms in both square brackets are zero; the first from the solution for m [equation (3.10)] and second because it is the characteristic equation (which evaluates to zero when m is a root, as it is here). So

$$u'' = 0$$

with solution $u = ax + b$, so the full solution is

$$\boxed{y(x) = (a + bx)e^{-a_1x/(2a_2)}} \tag{3.12}$$

where a and b are constants and the set of basis functions is $\{e^{mx}, xe^{mx}\}$.

Using variation of parameters to obtain the second solution from a known first solution is known as ***reduction of order***. It is applicable to all linear 2nd-order ODEs, even ones with variable coefficients as will be encountered in chapter 4. It is also applicable to higher order linear ODES.

3.4 Particular Integral

In this section we seek to find PIs for the IE

$$a_2y'' + a_1y' + a_0y = g(x)\ , \tag{3.13}$$

and to understand the general properties of PIs for *any* linear 2nd-order IE,

$$\hat{L}(y) = g(x)\ . \tag{3.14}$$

There is a general expression for a PI, in terms of something called the Wronskian. This will be covered a little later in section 3.5. It works for any linear 2nd-order IE. But for 2nd-order equations with constant coefficients, using trial functions is usually quicker. This is known as the ***method of undetermined coefficients***. You used this in your Year 1, Complex Analysis course. Sometimes it is even possible to just deduce by inspection.

3.4.1 Exponentials

If $g(x) = Ae^{\alpha x}$, then $y(x) = ce^{\alpha x}$ is a solution. The so-called undetermined coefficient c is determined by differentiating this trial solution twice (w.r.t. x) and substituting it into eqn (3.13). This readily yields

$$c = A/\left(a_2\alpha^2 + a_1\alpha + a_0\right)\ . \tag{3.15}$$

So we have found a PI

$$\boxed{y_{PI}(x) = c\,e^{\alpha x}\ .} \tag{3.16}$$

Important: If $\alpha = m_+$ or m_-, then what we obtain is part of the CF (solution to the H.E.), so is in fact not a PI! To obtain a PI, for this case, we must instead use the trial solution

$$\begin{aligned} y_{PI}(x) &= c\,x\,e^{\alpha x} && \text{when} && m_+ \neq m_- \\ y_{PI}(x) &= c\,x^2\,e^{\alpha x} && \text{when} && m_+ = m_-\ . \end{aligned}$$

These modified solutions (i.e. with the x or x^2 factor) are automatically found by the formal methods for finding a PI to be covered a little later in section 3.5. However, it doesn't matter how the PI is found. Any method will do. In this case of an IE with constant coefficients, the procedure of adding an x or x^2 factor is well-known and is sufficient.

3.4.2 Cos & Sin functions

Consider $g(x) = a\cos\beta x + b\sin\beta x$ (where a and b are constants). We generally need a PI that includes *both* cos and sin terms, even if $g(x)$ only contains the cos term or sin term alone. This is because the derivative changes sin into cos and vice versa. The PI is

$$\boxed{y_{\text{PI}}(x) = c\cos\beta x + d\sin\beta x\ .} \tag{3.17}$$

Differentiating twice and substituting into the ODE gives two simultaneous equations for c and d, from the coefficients of cos and sin. They can be solved for c and d. Again, if β coincides with a root of the characteristic equation, then special care will need to be taken, similar to in section 3.4.1 above.

3.4.3 x^n

If $g(x) = ax^n$ and n is a *non-negative integer* (i.e. $n = 0, 1, 2, \ldots$) then the PI is

$$\boxed{y_{\text{PI}}(x) = d_n x^n + d_{n-1}x^{n-1} + d_{n-2}x^{n-2} + \ldots + d_1 x + d_0\ .} \tag{3.18}$$

Differentiating twice and substituting into the ODE yields (a perhaps involved) expression with terms involving x^0, x, x^2, $\ldots x^n$. Collect terms in each power of x. The coefficients (i.e. expressions involving the d_k coefficients and a_0, a_1, a_2) of every power of x must each vanish. This gives $n+1$ simultaneous linear equations for the $n+1$ undetermined d_k coefficients. They can, therefore, be determined! This is straightforward when $n = 1$ or $n = 2$. For $n > 2$, it is tedious but possible.

3.4.4 General properties

The properties below hold for *any* linear inhomogeneous ODE, not just ones with constant coefficients.

Additive – If g is a sum of terms $g = g_1 + g_2$, then the PI is the sum of the PIs for g_1 and g_2 individually. In other words, find a PI that satisfies the IE, $\hat{L}(y) = g_1(x)$; call this $p_1(x)$. Find a PI that satisfies the IE, $\hat{L}(y) = g_2(x)$; call this $p_2(x)$. Then a suitable PI for the IE under consideration, $\hat{L}(y) = g$, is $p(x) = p_1(x) + p_2(x)$.

To show that this works;

$$\begin{aligned} \hat{L}(p_1 + p_2) &= \hat{L}(p_1) + \hat{L}(p_2) = g_1 + g_2 \\ \therefore \quad \hat{L}(p_1 + p_2) &= g \ . \end{aligned}$$

Thanks to the linearity of the DE, we can build up the PI to the full IE in an additive fashion.

Not unique(!) – There are infinitely many PIs. If you have found one PI, say $p_a(x)$, then another viable PI, say $p_b(x)$, can be obtained by adding arbitrary bits of the CF to it, e.g., $p_b = p_a + (ay_1 + by_2)$ where a and b are constants (and unrelated to the c_1 and c_2 in the CF itself), and y_1 and y_2 are the basis functions (see equation 3.2). The proof of this starts by noting that $\hat{L}(p_a) = g(x)$ and $\hat{L}(p_b) = g(x)$ by definition, since they are PIs. Therefore,

$$\hat{L}(p_b - p_a) = \hat{L}(p_b) - \hat{L}(p_a) = g - g = 0$$

since the equation is linear. This shows that $p_b - p_a$ is a solution of the HE, hence $p_b - p_a = ay_1 + by_2$ as claimed.

3.5 Linear Independence and the Wronskian

We now consider the general second-order linear homogeneous ODE,

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = 0 \ , \tag{3.19}$$

with initial values $y(x_0) = y_0$, $y'(x_0) = y_0'$. Remember that if $p(x)$ and $q(x)$ are continuous on an interval $\alpha < x < \beta$ that contains x_0, a unique solution exists to an IVP.

Suppose we choose a method, apply it to solve the equation and find a solution made of two parts:

$$y(x) = c_1 y_1(x) + c_2 y_2(x) \tag{3.20}$$

where $c_{1,2}$ are constants. Is it really the *general* solution, or has the method somehow broken down and provided us with an incomplete set of solutions? That is, is it possible to find specific values c_1 and c_2 so that (3.20) can satisfy **any** set of initial conditions (ICs)? In other words, will those $y_1(x)$ and $y_2(x)$ actually give us the unique solution to the IVP that we know should exist, whatever y_0 and y_0' are specified?

To find out, we note that satisfying the ICs implies

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= y_0 \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) &= y_0' \ . \end{aligned}$$

Let us write it in matrix form,

$$\begin{pmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_0 \\ y_0' \end{pmatrix} \ . \tag{3.21}$$

To find c_1 and c_2, we invert the matrix, *if it is invertible*. It is, provided the determinant, known as the ***Wronskian***

$$\boxed{W_{y_1,y_2}(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix}} \tag{3.22}$$

is non-zero, i.e., $W_{y_1,y_2}(x_0) \neq 0$.[1] Thus, the unique solution to the IVP can be found if $W(x_0) \neq 0$. This is because y_1 and y_2 are linearly independent when $W \neq 0$.

3.5.1 Linear dependence and independence of functions

To determine whether any two functions $f_1(x)$ and $f_2(x)$ are ***linearly independent*** on an interval $\alpha < x < \beta$, we consider the equation

$$\boxed{k_1 f_1(x) + k_2 f_2(x) = 0} \tag{3.23}$$

where k_1, k_2 are constants. If (3.23) can be satisfied at *every* x in the interval by a *single choice* of non-zero constants (i.e. $k_1 \neq 0$ and $k_2 \neq 0$) then f_1 and f_2 are ***linearly dependent*** on that interval. If not, they are ***linearly independent***. With just two functions, linear dependence implies that f_1 and f_2 are proportional.

The concept of linear independence of functions is a general concept. It applies to any functions, not just solutions to differential equations.

3.5.2 Wronskian test for linear independence

The Wronskian can be used to check for linear independence, if the functions being tested are *differentiable*. We resume labelling our functions as y_1 and y_2. Rewriting (3.23) and differentiating w.r.t. x yields

$$\begin{aligned} k_1 y_1 + k_2 y_2 &= 0 \\ k_1 y_1' + k_2 y_2' &= 0 \ . \end{aligned}$$

The two equations can be written in matrix form as

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \ .$$

Notice the same matrix appears as in equation (3.21), except now with x instead of x_0. If the Wronskian

$$\boxed{W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}} \tag{3.24}$$

is non-zero (i.e. $W(x) \neq 0$) anywhere in the interval then

$$\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

and there are only trivial solutions to (3.23) in that interval. Hence y_1 and y_2 are linearly independent.

Conversely, if $W(x) = 0$ *everywhere* in the interval, then the functions are linearly dependent in that interval.

We will see later that in connection with testing whether two solutions to a 2nd-order homogeneous

[1] We will now just refer to the Wronskian by W and remember that it depends on the functions y_1 and y_2.

ODE are independent, determining the Wronskian at one location, say x_0, is sufficient.[2] If you know $W(x_0) = 0$ then $W(x) = 0$ over the entire interval (where the linear ODE satisfies the existence and uniqueness theorem). Similarly, if you know $W(x_0) \neq 0$ somewhere then $W(x) \neq 0$ over the whole interval.

Summary

For a pair of *differentiable* functions y_1, y_2 on an interval $\alpha < x < \beta$:

- If $W = 0$ for *all* x in the interval, the functions are linearly dependent on this interval.
- If $W \neq 0$ somewhere in the interval, the functions are linearly independent on this interval.
- If $y_{1,2}$ are the two solutions to a 2nd-order homogeneous ODE and they are linearly independent, then they are basis functions.
- Then we can construct the solution of the linear ODE for any initial conditions via a superposition of them: We can write the general solution as $y = c_1 y_1 + c_2 y_2$.

If one or more of the functions are not sufficiently differentiable over the interval then the Wronskian cannot be used to check for linear independence and equation 3.23 – the fundamental definition of linear independence – has to be used instead.

Example: non-differentiable function

Consider $y_1(x) = x^2|x|$ and $y_2(x) = x^3$ on the interval $-\alpha < x < \alpha$. The function $y_1(x)$ is not differentiable at $x = 0$, because $|x|$ is not differentiable there.[3] They are clearly independent over this interval (sketch them). However restricting the interval to $x \geq 0$ or $x \leq 0$, it is clear that $y_{1,2}$ are linearly dependent on both. $W = 0$ on both restricted intervals and the Wronskian test works on both. However using the Wronskian test over the whole interval $-\alpha < x < \alpha$ does not work since W is undefined at $x = 0$. $y = x^2|x|$ is an example of a seemingly smooth function – one without jumps or corners – that is not differentiable. It is also not a solution of any linear ODE that satisfies the existence and uniqueness theorem.

Generalising to *n* functions

Note that the concept of linear independence extends to n functions:

$$k_1 y_1(x) + k_2 y_2(x) + \ldots + k_n y_n(x) = 0 \ . \tag{3.25}$$

If $k_j = 0$ for all $j = 1, 2, \ldots, n$ is required to satisfy this, then the n functions $y_j(x)$ are linearly independent. If there is a non-trivial solution (i.e. *some* k_j are not zero) then they are linearly dependent. If these n functions are sufficiently differentiable (i.e. $n-1$ times differentiable), then a Wronskian test can be used to check for linear independence. The Wronskian for n functions is the $n \times n$ determinant,

$$W(x) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix} \tag{3.26}$$

[2] The two solutions to the HE should be linearly independent. If not, there is either a mistake in your working or more work needs to be done to find the second independent solution.

[3] This can be understood from the limit definition of the derivative; a unique value for $\lim_{\Delta x \to 0}[f(x+\Delta x) - f(x)]/\Delta x$ must exist independent of the direction that Δx is taken to zero. With $f(x) = |x|$, one gets 1 if $\Delta x > 0$ and decreased to zero but -1 if $\Delta x < 0$ and shrunk to zero 'from the left'.

where all the entries are functions of x, i.e., $y_1(x)$, $y_2^{(n-1)}(x)$, but (x) has been dropped for brevity. The significance of $W = 0$ and $W \neq 0$ is the same as for $n = 2$. While it is easy to tell if two functions are dependent or independent, it is not so obvious for more than two. Then the Wronskian method becomes very powerful indeed.

Non-examinable – More details about the Wronskian and its uses can be found in the appendix B.

3.6 Obtaining the Second Solution

If your method for solving a second-order homogeneous ODE, $y'' + p(x)y' + q(x)y = 0$, yields only one independent solution, all is not lost – there is a procedure for obtaining the 2nd solution from the existing solution. Moreover, it guarantees the result is indeed independent of the 1st solution:

$$\boxed{y_2(x) = y_1(x) \int^x \frac{\exp\left(-\int^z p(s)ds\right)}{[y_1(z)]^2} dz \ .} \tag{3.27}$$

Note that this works for *any* linear 2nd-order ODE, including ones with variable coefficients (which we will see how to solve in section 4.)

Proof: We use ***variation of parameters***[4] so assume

$$y_2(x) = v(x)y_1(x) \ .$$

Differentiating twice; $y_2' = v'y_1 + vy_1'$ and $y_2'' = v''y_1 + 2v'y_1' + vy_1''$. Then substituting into the ODE yields

$$v''y_1 + v'(2y_1' + p(x)y_1) + v\underbrace{(y_1'' + py_1' + qy_1)}_{\to 0} = 0$$

where we used the fact that y_1 is a solution (so $\hat{L}(y_1) = 0$) to show that the final term on the LHS vanishes. This yields a 1st-order ODE[5] for v'

$$-\frac{1}{v'}\frac{dv'}{dx} = \frac{2}{y_1}\frac{dy_1}{dx} + p(x) \ . \tag{3.28}$$

If this can be solved for v' (and thus v), then $y_2 = vy_1$ will indeed be a solution of $y'' + py' + qy = 0$. Equation (3.28) can easily be solved by integrating:

$$\begin{aligned} -\ln|v'| &= \ln y_1^2 + \int^x p(s)ds = \ln\left[y_1^2 \exp\left(\int^x p(s)ds\right)\right] \\ \therefore \quad v'(x) &= \frac{\exp\left(-\int^x p(s)ds\right)}{y_1^2(x)} \ . \end{aligned}$$

Integrating this up once more with respect to x, one can find $v(x)$ and then multiplying through by y_1 yields equation (3.27).

[QED]

Now to verify that y_2 obtained via equation (3.27) and y_1 are indeed linearly independent. Evaluating

[4] Variation of parameters is also known as 'Lagrange's Method'. The method can also be extended to apply to any nth-order *linear* ODE.

[5] Since variation of parameters reduced a 2nd-order ODE to a 1st-order one in this case, it is sometimes known as *reduction of order* in this context.

the Wronskian determinant $W(x)$ yields

$$\begin{aligned} W(x) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} y_1 & vy_1 \\ y_1' & v'y_1 + vy_1' \end{vmatrix} = v'y_1^2 \\ \therefore \quad W(x) &= \exp\left(-\int^x p(z)dz\right) . \end{aligned} \tag{3.29}$$

Since $\exp(a) > 0$ for any $a \in \mathbb{R}$, consequently $W \neq 0$ and y_1 and y_2 are linearly independent. Furthermore, one sees that $W \neq 0$ over the *whole* interval $\alpha < x < \beta$ where the conditions for existence and uniqueness of solutions apply. This justifies the previous assertion that when checking whether solutions to an ODE are linearly independent or not, if you determine that $W \neq 0$ at one point then this applies across the whole interval where the solutions exist.

3.7 Higher-Order Equations

These are not common in physics. Nevertheless, the concepts seen in this chapter extend to n^{th}-order linear ODEs

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \ldots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

in a natural way:

$$y_{\text{CF}}(x) = \sum_{j=1}^{n} c_j y_j(x) \quad , \qquad y_{\text{GS}}(x) = y_{\text{CF}}(x) + y_{\text{PI}}(x) \ , \tag{3.30}$$

so that there are n basis functions, which of course are linearly independent (i.e. $W \neq 0$).

For ODEs with constant coefficients, the trial solution method $y = e^{mx}$ still works. For example, for third-order:

$$\begin{aligned} a_3 y''' + a_2 y'' + a_1 y' + a_0 y &= 0 \\ a_3 m^3 + a_2 m^2 + a_1 m + a_0 &= 0 \end{aligned} \tag{3.31}$$

with (normally) 3 distinct solutions, which then yields the CF

$$y_{\text{CF}} = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} \tag{3.32}$$

with 3 arbitrary constants. This extends to higher-order. Moving to n^{th}-order ODES, the difficulty becomes solving the nth-order characteristic polynomial for its n roots m_j. Repeated roots can be dealt with using the procedures seen for 2nd-order equations.

There is an intimate link between an nth-order ODEs of a single dependent variable, and a system of coupled first-order ODEs in n dependent variables. If you look back at your Complex Analysis notes, you will see this. You also routinely use this in mechanics when expressing an equation of motion, e.g.,

$$m\frac{d^2x}{dt^2} = -b\frac{dx}{dt} + F(x)$$

as

$$\begin{aligned} \frac{dx}{dt} &= v \\ m\frac{dv}{dt} &= -bv + F(x) \ . \end{aligned}$$

Systems of coupled first-order ODEs in n dependent variables are particularly amenable to numerical solution and revealing the characteristic behaviour of the system. These will be covered in the Computational Physics and Advanced Classical Physics courses in year 3, respectively.

Chapter 4

Series Solutions of Second-Order ODEs

Chapter 3 covered general properties of linear 2nd-order ODEs and their solutions. It also reviewed the complementary functions (i.e. solutions to the homogeneous equation) that arise in the case of constant coefficients.

This chapter extends the determination of complementary functions to the more general and interesting case of linear 2nd-order ODEs with variable coefficients that are functions of x (the independent variable). They can be found assuming that $y_{\text{CF}}(x)$ assumes the form of a ***power series***. A motivation for this is that the transcendental functions[1] $\exp(mx)$, $\cos(qx)$ and $\sin(qx)$ forming the basis functions for the constant coefficient case (seen in section 3.3) *are* power series expansions

$$\begin{aligned} \exp(mx) &= 1 + mx + \frac{(mx)^2}{2!} + \frac{(mx)^3}{3!} + \ldots \\ \cos(px) &= 1 - \frac{(px)^2}{2!} + \frac{(px)^4}{4!} - \frac{(px)^6}{6!} \ldots \quad . \end{aligned}$$

The method to be covered here will recover these power series expansions for constant coefficient ODEs. It will also yield analogous power series expansions for many, but not all, variable coefficient ODEs. These expansions define other types of transcendental functions which are often known as ***special functions***. Here we will encounter Legendre polynomials (which are not transcendental) and Bessel functions (which are), though there are many others. The ODEs that can be solved by the power series method turn out to be essential for solving PDEs, especially the most important ones in physics, as will become apparent later, in chapter 7. Thus this chapter underpins the solution of PDEs. As in the previous chapter, only first-degree ODEs will be considered.

Our focus will be the general 2nd-order, linear, homogeneous ODE:

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \quad . \tag{4.1}$$

Under certain conditions (see sections 4.3.1 and 4.6.1 for the requirements), we can make a ***power series expansion*** about a *point* x_0:

$$y(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + \ldots = \sum_{n=0}^{\infty} c_n(x - x_0)^n \quad . \tag{4.2}$$

[1] Transcendental functions are ones that cannot be expressed in terms of a *finite* sequence of algebraic operations (addition/subtraction, multiplication/division, raising to a power). For instance, $\exp(x)$ and $\cos(x)$ are expressed as an infinite series, so are transcendental.

Under a wider range of conditions, a more generalised power series

$$Y(x) = (x - x_0)^r y(x) \tag{4.3}$$

will be needed, where the power r needs to be determined. However, such series solutions are only useful if the series converge. So we first need to be able to determine if a series converges.

4.1 Radius of Convergence

This is the range of x for which the power series solution exists. (The name radius arises since a similar definition applies to problems where the independent variable is complex, and there is often a circle in the Argand diagram within which a solution exists).

Consider a ***partial sum***

$$s_N \equiv \sum_{n=0}^{N} t_n \tag{4.4}$$

where t_n is a term in the series expansion. If $\lim_{N\to\infty} s_N = s$ exists, then the series is said to *converge*. The series ***converges absolutely*** if

$$s'_N \equiv \sum_{n=0}^{N} |t_n| \tag{4.5}$$

converges. If so, the original series converges as well.

4.1.1 Ratio test

To test whether a series converges, we use the ***ratio test***. Define

$$\boxed{r_n \equiv \left|\frac{t_{n+1}}{t_n}\right| \qquad \text{and let} \qquad r = \lim_{n\to\infty} r_n \ .} \tag{4.6}$$

If

- $r < 1$, the series converges absolutely, so the series converges.
- $r = 1$, the situation is unclear and needs further investigation.[2]
- $r > 1$, the series diverges absolutely.

In a power series expansion, terms depend on $x - x_0$, so the series may converge for small $x - x_0$ and diverge otherwise. The ***Radius of Convergence*** R is the limiting value of $|x - x_0|$ for which the series converges.

Example — Consider the sum (which is a power series about $x_0 = 0$)

$$\sum_{n=1}^{\infty} (-1)^{n+1} n x^n.$$

[2] Gauss' test for convergence can usually successfully determine whether or not a series converges when $r = 1$. See *non-examinable* appendix C if interested.

$$r_n \equiv \left|\frac{t_{n+1}}{t_n}\right| = \left|\frac{(-1)^{n+2}(n+1)x^{n+1}}{(-1)^{n+1}nx^n}\right| = \left|\frac{n+1}{n}x\right|$$

so $r = \lim_{n\to\infty} r_n = |x|$, and $r < 1$ for $|x| < 1$, so $R = 1$. The series converges when x lies within this radius of convergence.

4.2 Working with Power Series

This is a reference for use with the following sections.

4.2.1 Differentiating

Take

$$\begin{aligned} y(x) &= \sum_{n=0}^{\infty} c_n(x-x_0)^n \\ &= c_0 + c_1(x-x_0) + c_2(x-x_0)^2 + \dots \ . \end{aligned} \tag{4.7}$$

Then dy/dx is found by differentiating each term, which lowers the power by one:

$$\begin{aligned} \frac{dy}{dx} &= c_1 + c_2(2)(x-x_0) + c_3(3)(x-x_0)^2 + \dots \\ &= \sum_{\mathbf{n=1}}^{\infty} c_n(n)(x-x_0)^{n-1} \ . \end{aligned} \tag{4.8}$$

A key point here is that the constant term for $n = 0$ (i.e. c_0) vanishes and the lower limit of the summation $\sum \dots$ can be changed to $n = 1$ to reflect this. Similarly,

$$\begin{aligned} \frac{d^2y}{dx^2} &= c_2(2)(1) + c_3(3)(2)(x-x_0) + c_4(4)(3)(x-x_0)^2 \dots \\ &= \sum_{\mathbf{n=2}}^{\infty} c_n(n)(n-1)(x-x_0)^{n-2} \ . \end{aligned} \tag{4.9}$$

4.2.2 Shifting and relabelling the summation index

This will be used later to collect terms from different summations into a single summation. Consider the series

$$s = \sum_{n=0}^{\infty} t_n = t_0 + t_1 + t_2 + t_3 + \dots \ .$$

The name of the index is arbitrary. If we use l instead then $s = \sum_{l=0}^{\infty} t_l = t_0 + t_1 + t_2 \dots$, i.e., nothing changes. What does this look like if we change to a shifted version of the index, say, $k = n + 1$? There will be a corresponding value of k for each value of n:

n	0	1	2	3	…	l	…
k	1	2	3	4	…	$l+1$	…

Noting that $n = k - 1$ in this example, the above series can be expressed in terms of k as

$$s = \sum_{k=1}^{\infty} t_{k-1} = t_0 + t_1 + t_2 + t_3 + \dots \quad .$$

You can see from the expanded forms that changing index from n to k *and* suitably adjusting the lower limit, does not change the series. When the upper limit is infinite, no adjustment is needed when shifting the index.

Shifting the index variable will be useful for combining power series, by aligning powers of x. Consider adding together the series expansions of $y(x)$ and dy/dx given by equation (4.7) and (4.8), respectively. Using the expanded forms and collecting in powers of x:

$$y' + y = [c_0 + c_1]x^0 + [c_1 + 2c_2]x^1 + [c_2 + 3c_3]x^2 + \dots \tag{4.10}$$

In terms of the summations:

$$y' + y = \sum_{n=0}^{\infty} c_n(x - x_0)^n + \sum_{l=1}^{\infty} c_l(l)(x - x_0)^{l-1} \quad ,$$

where the second sum is written with index l to emphasise that it is independent to the index in the first sum. But we can still add the series, term by term and merge into a single summation over a general term with a *single* power of x. Let $l - 1 = m$ (and therefore $l = m + 1$) to align the powers of x in the second summation with the first. When $l = 1$, $m = 0$. Then

$$\begin{aligned} y' + y &= \sum_{n=0}^{\infty} c_n(x - x_0)^n + \sum_{m=0}^{\infty} c_{m+1}(m+1)(x - x_0)^m \\ &= \sum_{n=0}^{\infty} c_n(x - x_0)^n + \sum_{n=0}^{\infty} c_{n+1}(n+1)(x - x_0)^n \\ &= \sum_{n=0}^{\infty} [c_n + (n+1)c_{n+1}](x - x_0)^n \quad . \end{aligned} \tag{4.11}$$

Note that we have relabelled $m \longrightarrow n$ on the second line above. You can readily see that the general expression for terms in (4.11) will yield (4.10) as n takes the values $0, 1, 2, \dots$ in turn.

4.3 Power Series Solution about an Ordinary Point

4.3.1 Conditions for Existence

Writing the equation in standard form

$$y'' + p(x)y' + q(x)y = 0 \tag{4.12}$$

when does the power series solution exist? It depends on whether p and q are ***analytic*** or not.

Analytic functions and ordinary points – A function $f(x)$ that has a Taylor expansion about $x = x_0$, with a radius of convergence $R > 0$ (i.e. not zero), is said to be *analytic* at $x = x_0$. An important point is that we need *all* the derivatives of f to exist at x_0. So, e.g., $f(x) = 1/x$ is not analytic at $x_0 = 0$.

If p and q are *analytic* functions at x_0, then x_0 is classified as an ***ordinary point*** and the power series solution exists. We will find that the solution splits into two:

$$y(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n = a_1 y_1(x) + a_2 y_2(x) \tag{4.13}$$

for 2 arbitrary constants a_1 and a_2.[3] It turns out that the radius of convergence of y_1 and y_2 are *at least as large* as the smaller of those of p and q.

4.3.2 Method 1 – Tables & Patterns

To motivate the power series solution technique, we will apply it to an ODE that we know the solution to; the simple harmonic oscillator:

$$y'' + y = 0 \qquad \text{with GS} \qquad y(x) = a\cos(x) + b\sin(x) \ ,$$

where a and b are arbitrary constants. Since $p(x) = 0$ and $q(x) = 1$ and are both analytic everywhere, this technique will work. We choose to expand about $x_0 = 0$.

The first method here involves directly working out the coefficients, arranging them by powers of x in a table and then deducing the terms in the series from this.

The power series and its second derivative are

$$\begin{aligned} y(x) &= c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots \\ y''(x) &= c_2(2)(1) + c_3(3)(2)x + c_4(4)(3)x^2 + c_5(5)(4)x^3 \dots \ . \end{aligned}$$

Building a table

	x^0	x^1	x^2	x^3	…
y	c_0	c_1	c_2	c_3	…
y''	$2\ c_2$	$2\times 3c_3$	$3\times 4c_4$	$4\times 5c_5$	…

We require the overall coefficient of each and every power of x to vanish. These are formed by adding entries in each row and setting to zero.

[3] We switch notation to a_1, a_2 to avoid confusion with the coefficients c_n of the power series.

$$\begin{aligned} x^0: &\qquad c_2 = -c_0\frac{1}{2} &&= -c_0\frac{1}{2!} \\ x^1: &\qquad c_3 = -c_1\frac{1}{2\times 3} &&= -c_1\frac{1}{3!} \\ x^2: &\qquad c_4 = -c_2\frac{1}{3\times 4} &&= c_0\frac{1}{4!} \\ x^3: &\qquad c_5 = -c_3\frac{1}{4\times 5} &&= c_1\frac{1}{5!} \end{aligned} \tag{4.14}$$

Notice that the even c_n are all related back to c_0. All the odd c_n are related back to c_1. Only the c_0 and c_1 coefficients are not determined. Inserting the c_n coefficients back into the power series expansion and collecting in c_0 and c_1:

$$y(x) = c_0\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots\right] + c_1\left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right] \ .$$

By inspection of the pattern we deduce that the terms multiplying c_0 go as $(-1)^k x^{2k}/(2k)!$ (for $k = 0, 1, 2, \ldots$) and those multiplying c_1 are $(-1)^k x^{2k+1}/(2k+1)!$. We then recognise that we have

$$y(x) = c_0\cos(x) + c_1\sin(x) \ .$$

Notice that this is the general solution with the undetermined coefficients c_0 and c_1 acting as the two required arbitrary constants.

4.3.3 Method 2 – Recurrence Relation

The other method, arguably more elegant, is to work directly with the summations without expanding out. The power series and its second derivative are

$$y(x) \quad = \quad \sum_{n=0}^{\infty} c_n x^n \qquad , \qquad y''(x) = \sum_{\mathbf{n=2}}^{\infty} c_n(n)(n-1)x^{n-2} \ .$$

Our goal is to insert these into the ODE $y'' + y = 0$, and then combine the summations so that the general term involves a single power of x. (Doing this will yield a recurrence relation.) However at the moment the summands involve different powers of x. Also the lower limits are different. We can fix this by using a shifted index $m = n - 2$ (thus also $n = m + 2$) for y'' and then relabel $m \to n$:

$$y''(x) = \sum_{n=0}^{\infty} c_{n+2}(n+2)(n+1)x^n \ ,$$

Now we can proceed:

$$\begin{aligned} \sum_{n=0}^{\infty} c_{n+2}(n+2)(n+1)x^n + \sum_{n=0}^{\infty} c_n x^n &= 0 \\ \therefore \qquad \sum_{n=0}^{\infty} \left[c_{n+2}(n+2)(n+1) + c_n\right] x^n &= 0 \ . \end{aligned}$$

We require the overall coefficient of each and every power of x to vanish. This is the only way to guarantee that the LHS equals zero for *any* x. Thus we immediately see

$$\begin{aligned} c_{n+2}(n+2)(n+1) + c_n &= 0 \\ \therefore \qquad c_{n+2} &= -\frac{1}{(n+2)(n+1)}c_n \qquad (\text{for } n = 0, 1, 2, \dots) \end{aligned}$$

which is known as a ***recurrence relation***. This automatically recovers all the columns in the table that was previously manually built in the preceding method. For instance setting $n = 0$ in the above recurrence relation yields the first column, summed and set equal to zero. Evaluating this recurrence relation repeatedly starting with $n = 0$ and then increasing n allows one to generate exactly the same coefficients as in (4.14).

Notice that this recurrence relation separately links c_n together for even n and odd n. For even $n = 2k$:

$$c_{2(k+1)} = -\frac{1}{(2k+2)(2k+1)}c_{2k} \tag{4.15}$$

where $k = 0, 1, 2, \dots$. We again deduce by inspection that $c_{2k} = \frac{(-1)^k}{(2k)!}c_0$.

Proof by induction

Mathematically, it is a perhaps bit of a leap of faith to deduce a general expression, like $c_{2k} = c_0(-1)^k/(2k)!$, just going by a few terms in a sequence. How can we be sure it really does hold for all k? Proof by induction is a mathematically rigorous way to confirm the deduced expression does hold out to $k \to \infty$.

Assume that

$$c_{2k} = \frac{(-1)^k}{(2k)!}c_0 \tag{4.16}$$

is true. Then applying the above recurrence relation (4.15);

$$\begin{aligned} c_{2(k+1)} &= -\frac{1}{(2k+2)(2k+1)}\frac{(-1)^k}{(2k)!}c_0 = \frac{(-1)^{k+1}}{(2k+2)!}c_0 \\ \therefore \qquad c_{2(k+1)} &= \frac{(-1)^{k+1}}{[2(k+1)]!}c_0 \end{aligned}$$

which shows that our assumed expression is then also true for $k + 1$. We now check that our original assumption is true by considering (4.16) for $k = 0$:

$$c_0 = \frac{(-1)^0}{0!}c_0 = c_0 \ .$$

It is. Since (4.16) is true for the first value of k, and we showed using the recurrence relation that if (4.16) is true for any k it is also true for the next value $k + 1$

$$c_{2k} = \frac{(-1)^k}{(2k)!}c_0 \qquad (\text{for all integer } k \geq 0) \qquad \textbf{QED} \ .$$

We can similarly prove by induction (try it) that $c_{2k+1} = \frac{(-1)^k}{(2k+1)!}c_1$.

(*In this course, it is okay to deduce the general expression for coefficients of a power series by inspection. You are not expected to prove by induction unless explicitly asked to.)*

Thus (relabelling $k \to n$)

$$y(x) = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} + c_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \quad ,$$

which we again note is $y(x) = c_0 \cos(x) + c_1 \sin(x)$.

Radius of convergence

Let us check the infinite series multiplying c_0. We can speed things up by using the recurrence relation (4.15), rather than working with the general form of the coefficients of the power series.

$$\begin{aligned} r_n &= \left| \frac{t_{n+1}}{t_n} \right| = \left| \frac{c_{2(n+1)} x^{2(n+1)}}{c_{2n} x^{2n}} \right| = \left| \frac{-x^2}{(2n+2)(2n+1)} \right| \\ \therefore \quad r_n &= \frac{x^2}{4n^2 \left(1 + \frac{1}{n}\right)\left(1 + \frac{1}{2n}\right)} \\ \therefore \quad r &= \lim_{n \to \infty} r_n = \lim_{n \to \infty} \frac{x^2}{4n^2} \\ &= 0 \ . \end{aligned}$$

Thus for any x the series converges absolutely, thus converges. Therefore the radius of convergence is $R \to \infty$. This accords with the known behaviour of $\cos(x)$. Similar arguments apply for the series multiplying c_1.

We are in the lucky position that the power series solution has yielded known (transcendental) functions. Thus the above is all rather synthetic! However, it is quite possible that for a different ODE, the power series solution will not be a known function, in which case obtaining a general expression for the c_n coefficients and understanding the radius of convergence is essential!

4.4 Solving Legendre's Equation

In solving PDEs we often first seek a *separable* solution (see later in the course). e.g. in problems involving $\nabla^2 u$ in spherical polars, we look for solutions of the form

$$u(r,\theta,\phi) = R(r)T(\theta)F(\phi).$$

When we do this, we frequently encounter this equation

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left[\sin\theta\frac{dT}{d\theta}\right] + \left[k - \frac{m^2}{\sin^2\theta}\right]T = 0$$

where k is a *separation constant* to be determined and m is an integer. We will see where this comes from when we consider PDEs.

If we let $x = \cos\theta$, and $y = T$, this reduces to the Associated Legendre Equation

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + \left[k - \frac{m^2}{1-x^2}\right]y = 0$$

which will be covered in section 9.1. For the special case $m = 0$, this is the ***Legendre Equation***

$$\boxed{(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + ky = 0 \ .} \tag{4.17}$$

Since $x = \cos\theta$, the range is $-1 \le x \le 1$.

We are going to obtain a power series solution about the point $x_0 = 0$, but first we should check whether this is an ordinary point or not. To do this, the equation needs to be written in standard form. Doing this reveals, $p(x) = -2x/(1-x^2)$ and $q(x) = k/(1-x^2)$. Both p and q are analytic at $x = 0$, so $x_0 = 0$ is an ordinary point, and we can make a series expansion about $x_0 = 0$. We anticipate some problems at $x = \pm 1$, where p and q are not analytic.

Important - For ODEs such as Legendre's equation, where $a_2(x)$ is a function of x rather than a constant, the power series solution procedure itself requires the ODE to be in the original form $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, e.g., equation (4.17) here. The procedure does not readily work if the ODE is in standard form, as the $p(x) = a_1(x)/a_2(x)$ and $q(x) = a_0(x)/a_2(x)$ 'coefficients' would need to be expanded into power series themselves, which complicates things.

Substitute the series expansion into the (original) equation (note the limits on the sums):

$$\begin{aligned}
(1-x^2)\frac{d^2}{dx^2}\sum_{n=0}^{\infty} c_n x^n - 2x\frac{d}{dx}\sum_{n=0}^{\infty} c_n x^n + k\sum_{n=0}^{\infty} c_n x^n &= 0 \\
(1-x^2)\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - 2x\sum_{n=1}^{\infty} n c_n x^{n-1} + k\sum_{n=0}^{\infty} c_n x^n &= 0 \\
\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=2}^{\infty} c_n n(n-1)x^n - 2\sum_{n=1}^{\infty} c_n n x^n + k\sum_{n=0}^{\infty} c_n x^n &= 0
\end{aligned} \tag{4.18}$$

In the first term, let $m = n-2$, and then relabel $m \to n$ (this makes all the powers x^n):

$$\sum_{n=0}^{\infty} c_{n+2}(n+2)(n+1)x^n - \sum_{n=2}^{\infty} c_n n(n-1)x^n - 2\sum_{n=1}^{\infty} c_n n x^n + k\sum_{n=0}^{\infty} c_n x^n = 0 \ .$$

Now we can **extend** the second and third sums to **start from $n = 0$**, since **the terms we add are all zero**:

$$\sum_{n=0}^{\infty} c_{n+2}(n+2)(n+1)x^n - \sum_{n=0}^{\infty} c_n n(n-1)x^n - 2\sum_{n=0}^{\infty} c_n n x^n + k\sum_{n=0}^{\infty} c_n x^n = 0 \ .$$

Combining summations:

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1)c_{n+2} - n(n-1)c_n - 2nc_n + kc_n \right] x^n = 0 \ .$$

Since the solution must work for any value of x (within the range of convergence), the only way this can happen is if the coefficients of x^n are all zero, which gives (for all $n \geq 0$):

$$(n+2)(n+1)c_{n+2} - n(n-1)c_n - 2nc_n + kc_n = 0$$

which gives the ***recurrence relation***

$$\boxed{c_{n+2} = \frac{[n(n+1)-k]}{(n+2)(n+1)} c_n \qquad \text{for } n \geq 0 \ .} \tag{4.19}$$

Repeatedly applying the recurrence relation, the coefficients can be determined. The first few are:

$$\begin{aligned}
x^0 \ (n=0): &\qquad c_2 = c_0 \frac{0-k}{2\cdot 1} &&= -c_0 \frac{k}{2!} \\
x^1 \ (n=1): &\qquad c_3 = c_1 \frac{1\cdot 2-k}{2\cdot 3} &&= c_1 \frac{2-k}{3!} \\
x^2 \ (n=2): &\qquad c_4 = c_2 \frac{2\cdot 3-k}{3\cdot 4} &&= -c_0 \frac{k(6-k)}{4!} \\
x^3 \ (n=3): &\qquad c_5 = c_3 \frac{3\cdot 4-k}{4\cdot 5} &&= c_1 \frac{(2-k)(12-k)}{5!}
\end{aligned} \tag{4.20}$$

Note that the odd terms are linked, and the even terms are linked, but they aren't related to each other, so we need to specify both c_0 and c_1, and the solution splits into two series [$y(x) = c_0 y_0(x) + c_1 y_1(x)$] with two arbitrary coefficients:

$$y(x) = c_0 \left[1 - \frac{k}{2!}x^2 - \frac{k(6-k)}{4!}x^4 + \ldots \right] + c_1 \left[x + \frac{(2-k)}{3!}x^3 + \frac{(2-k)(12-k)}{5!}x^5 + \ldots \right] \tag{4.21}$$

The terms in square brackets, which are y_0 and y_1 respectively, are quite complicated, but note that y_0 is an even function of x and y_1 is odd.

4.4.1 Dealing with $x = \pm 1$

A problem is that for general k, the series in equation (4.21) both diverge when $x = \pm 1$. Each series has

$$r_n = \left| \frac{[n(n+1)-k]}{(n+1)(n+2)} x^2 \right| = \left| \frac{\left[(1+1/n) - k/n^2\right]}{(1+1/n)(1+2/n)} x^2 \right|$$

so $r = \lim_{n\to\infty} r_n = |x^2|$ so the radius of convergence is $R = 1$, and there is no guarantee that the series converge when $x = \pm 1$. In fact they do not. [4] Since $x = \pm 1$ are in the physical range ($\theta = 0, \pi$) this is potentially a problem.

In order to get solutions that do not diverge at $x = \pm 1$, we need either to set the coefficients c_0 or c_1 to zero, *or*, we need a series to stop at some point. In general, we demand that one series is not infinite, and the coefficient of the other is zero.

You may notice that for certain values of k, one or other series is truncated. This happens when the numerator of the recursion relation vanishes. These values are

$$k = 0, 2, 6, 12, 20, 30, \ldots$$

or more compactly

$$k = \ell(\ell + 1) \tag{4.22}$$

where $\ell = 0, 1, 2\ldots$. As usual, c_0 and c_1 are set by boundary conditions.

The solution diverges (and is thus unphysical) unless $k = \ell(\ell + 1)$ (which will truncate one series) and one coefficient is zero. Specifically (setting $c_0 = 1$ for even ℓ, and $c_1 = 1$ for odd ℓ):

- $\ell = 0 \Rightarrow k = 0 \Rightarrow y_0 = 1$. y_1 diverges unless we set $c_1 = 0$.
- $\ell = 1 \Rightarrow k = 2 \Rightarrow y_1 = x$. y_0 diverges unless we set $c_0 = 0$.
- $\ell = 2 \Rightarrow k = 6 \Rightarrow y_0 = (1 - 3x^2)$. y_1 diverges unless we set $c_1 = 0$.
- etc.

[4] This can be verified using Gauss' test for convergence.

4.5 Legendre Polynomials

The truncated power series is a polynomial. Thus we see that we get *polynomial* solutions, $P_\ell(x)$. It is customary to normalise them (unusually) by setting $P_\ell(1) = 1$, so

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= (3x^2 - 1)/2 \\ P_3(x) &= (5x^3 - 3x)/2 \end{aligned} \tag{4.23}$$

These are shown in figure 4.1. In general,

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \frac{d^\ell}{dx^\ell} \left[(x^2 - 1)^\ell \right]. \tag{4.24}$$

This is ***Rodriguez' formula***, which you do not need to remember.

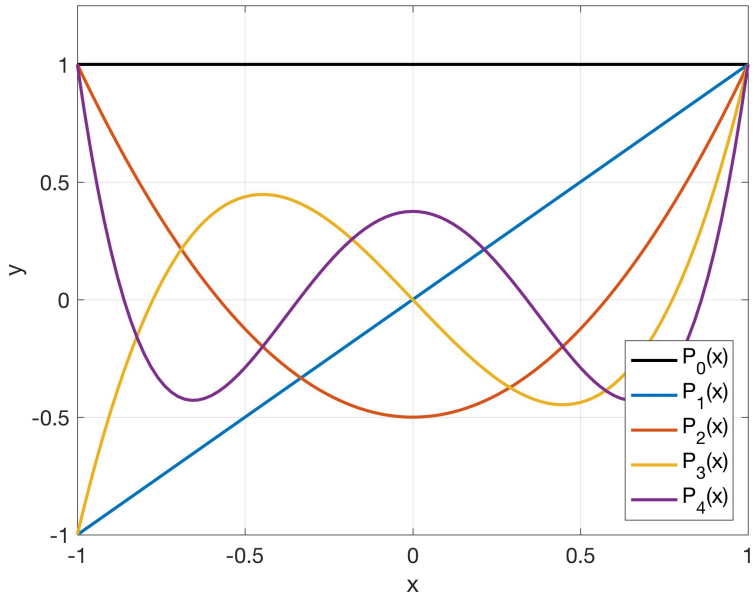


Figure 4.1: The first 5 Legendre Polynomials, $P_0(x)$, $P_1(x)$, $P_2(x)$, $P_3(x)$ and $P_4(x)$.

4.5.1 Orthogonality

The Legendre polynomials form a set of ***orthogonal functions***, in the sense that

$$\boxed{\int_{-1}^{1} dx\, P_\ell(x) P_n(x) = 0 \qquad \text{if } \ell \neq n \ .} \tag{4.25}$$

In principle, you could manually verify that the first four Legendre polynomials, shown in equation (4.23), indeed satisfy (4.25). Legendre polynomial orthogonality will be proved later in section 5.4. [5]

[5] An alternative, *non-examinable* proof can be found in appendix D if you cannot wait!

4.5.2 Normalisation

The normalisation of the Legendre polynomials is unusual. Most orthogonal functions are also *normalised* so that the integral is unity if $\ell = n$, but Legendre polynomials are not normalised in this way, but for historical reasons the amplitude is set differently ($P_\ell(1) = 1$). This is reflected in the (rather arbitrary) appearance of $2/(2\ell + 1)$ on the right hand side of the orthogonality relation (not proven here):

$$\int_{-1}^{1} P_\ell(x) P_n(x) dx = \frac{2}{2\ell + 1} \delta_{\ell n}. \tag{4.26}$$

4.5.3 Complete Set

In fact the Legendre polynomials form a *complete set*, meaning that any function $f(x)$ defined on $-1 \leq x \leq 1$ that satisfies the Dirichlet conditions (i.e. the same conditions that apply to Fourier series) can be expressed as a sum

$$\boxed{f(x) = \sum_{\ell=0}^{\infty} a_\ell P_\ell(x)} \tag{4.27}$$

and we get the coefficients from the orthogonality

$$\int_{-1}^{1} f(x) P_n(x) dx = \sum_{\ell=0}^{\infty} a_\ell \int_{-1}^{1} P_n(x) P_\ell(x) dx = \sum_{\ell=0}^{\infty} a_\ell \frac{2}{2\ell + 1} \delta_{\ell n} = \frac{2}{2n + 1} a_n$$

$$\therefore \qquad \boxed{a_\ell = \frac{2\ell + 1}{2} \int_{-1}^{1} f(x) P_\ell(x) dx \ .} \tag{4.28}$$

Other special functions share this completeness property, and it will be covered in a little more detail in section 5.3.4.

Figure 4.2 shows an example of reconstructing a step function (black curve) via Legendre polynomials. The approximation becomes better, the more polynomials are used in the reconstruction.

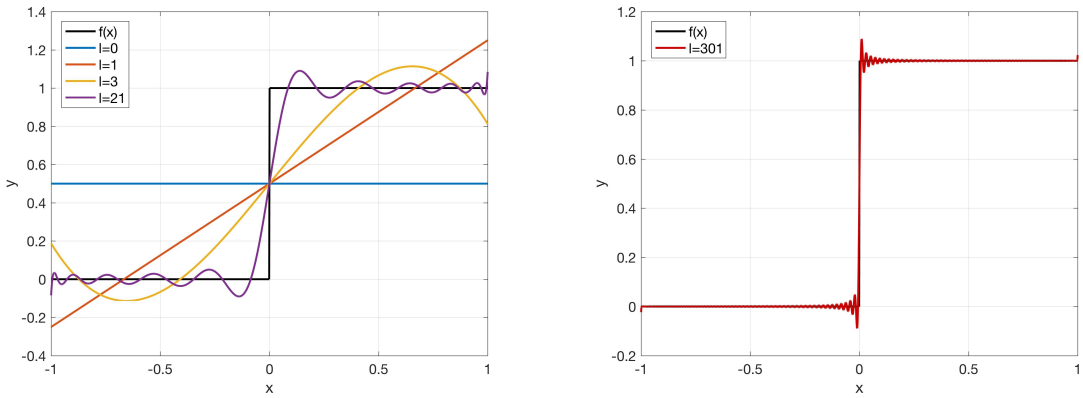


Figure 4.2: Reconstruction of a step function (black curve) via a Legendre polynomial series. In the legends, ℓ indicates the maximum value of ℓ used in the series (4.27).

4.6 Power Series Solution about Regular Singular Points (Frobenius' Method)

The expansion $y(x) = \sum_{n=0}^{\infty} c_n(x-x_0)^n$ works if x_0 is an **ordinary point**, but what happens if not? There are some important equations in physics that have points that are not ordinary, e.g. Bessel's equation

$$\boxed{x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - m^2)y = 0} \tag{4.29}$$

which in standard form is

$$\frac{d^2y}{dx^2} + \frac{1}{x}\frac{dy}{dx} + \left(1 - \frac{m^2}{x^2}\right)y = 0 \tag{4.30}$$

so $p(x)$ and $q(x)$ are not analytic at $x = 0$, and $x = 0$ is a ***singular point***. We can still find solutions provided the singularity is 'not too severe' (to be defined). Note that since the parameter m appears as m^2, we consider only $m \geq 0$.

4.6.1 Regular singular points

If **$(x - x_0)p(x)$** and **$(x - x_0)^2 q(x)$ are analytic at x_0**, then x_0 is called a ***regular singular point***. Note that $x_0 = 0$ is a regular singular point for Bessel's equation. We can make expansions about regular singular points using ***Frobenius' method***, which is a generalised power series.

Theorem. *If x_0 is a regular singular point, there is at least one solution of the equation in the form of a generalised power series*

$$\begin{aligned} y(x) &= (x - x_0)^r\left[c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + \ldots\right] \\ &= \sum_{n=0}^{\infty} c_n(x - x_0)^{n+r} \end{aligned} \tag{4.31}$$

for some r, to be determined.

It will be important later to note that $c_0 \neq 0$. Why? Well, if it were zero, we can absorb a factor of $(x - x_0)$ into $(x - x_0)^r$:

$$y(x) = (x - x_0)^r\left[0 + c_1(x - x_0) + c_2(x - x_0)^2 + \ldots\right] = (x - x_0)^{r+1}\left[c_1 + c_2(x - x_0) + \ldots\right]$$

so if we relabel c_1 to c_0, we have a series starting with $c_0 \neq 0$, and can simply redefine $r + 1$ as a new index r. The condition $c_0 \neq 0$ leads to the ***indicial equation*** for r (i.e. the equation for the *index* r), which is quadratic. Usually there are two solutions and hence two series. If the roots for r differ by an integer, we have to be careful. We will see this shortly!

4.6.2 Bessel's equation

The goal to so use Frobenius' method to find solutions to Bessel's equation (4.29), taking the expansion about $x_0 = 0$. First, it should be verified that this point is a regular singular point, which requires that Bessel's equation be written in standard form (4.30). This reveals that $p(x) = 1/x$ and $q(x) = 1 - m^2/x^2$, so we have

$$(x - x_0)p(x) = (x)\frac{1}{x} = 1 \qquad \text{and} \qquad (x - x_0)^2 q(x) = (x)^2\left(1 - \frac{m^2}{x^2}\right) = x^2 - m^2$$

which implies that they are analytic at $x_0 = 0$, so that $x_0 = 0$ is indeed a regular singular point. Hence Frobenius' method is guaranteed to yield at least one solution and is worth using.

It best *not* to apply the method of Frobenius to Bessel's equation in standard form, but rather in the original form, equation (4.29). Relabelling $m \to p$ (to avoid confusing this parameter with a summation index) this is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - p^2)y = 0 \ .$$

We substitute

$$y(x) = \sum_{n=0}^{\infty} c_n x^{n+r} \quad , \qquad y'(x) = \sum_{\mathbf{n=0}}^{\infty} c_n(n+r)x^{n+r-1} \quad , \qquad y''(x) = \sum_{\mathbf{n=0}}^{\infty} c_n(n+r)(n+r-1)x^{n+r-2}$$

into Bessel's equation, noting that *the limits are all* $n = 0$ since the $n = 0$ and $n = 1$ terms do not necessarily differentiate to zero. If r is not an integer then the leading term does not vanish upon differentiating, e.g., if $r = 1/2$ then $d[c_0x^r]/dx = d[c_0x^{1/2}]/dx = c_0/(2x^{1/2})$. Thus we find

$$x^2\sum_{n=0}^{\infty} c_n(n+r)(n+r-1)x^{n+r-2} + x\sum_{n=0}^{\infty} c_n(n+r)x^{n+r-1} + (x^2-p^2)\sum_{n=0}^{\infty} c_n x^{n+r} = 0 \ .$$

Absorb the x^2 or x^1 pre-factors into the sums:

$$\sum_{n=0}^{\infty} c_n(n+r)(n+r-1)x^{n+r} + \sum_{n=0}^{\infty} c_n(n+r)x^{n+r} - p^2\sum_{n=0}^{\infty} c_n x^{n+r} + \sum_{n=0}^{\infty} c_n x^{n+r+2} = 0 \ .$$

Letting $m = n+2$ in the final sum: (and noting that $n = m-2$ and $n = 0$ occurs for $m = 2$)

$$\ldots + \ldots + \ldots + \sum_{m=2}^{\infty} c_{m-2}x^{m+r} = 0 \ .$$

Finally, relabel $m \to n$, collect in powers of x and split off the $n = 0$ and $n = 1$ terms of the first three sums:

$$c_0\Big[r(r-1) + r - p^2\Big]x^r + c_1\Big[(r+1)r + (r+1) - p^2\Big]x^{r+1} + \sum_{n=2}^{\infty}\Big[c_n\Big[(n+r)^2 - p^2\Big] + c_{n-2}\Big]x^{n+r} = 0 \ . \tag{4.32}$$

As before, we require all coefficients in front of each power of x to vanish. The first two terms (for $n = 0$ and $n = 1$) are the 'orphan terms', which yield the indicial equation for determining r. The general term inside the summation gives the recurrence relation that generates the coefficients of the power series solution.

$\underline{n = 0}$:
After simplification, $r(r-1) + r - p^2 = 0$ yields the first ***indicial equation***

$$\boxed{r^2 - p^2 = 0 \qquad \Rightarrow \qquad r = \pm p \ .} \tag{4.33}$$

where p is regarded as being positive This is because substituting $p \to -p$ in $r^2 - p^2 = 0$ yields the same indicial equation and the same roots $r = \pm p$ again. Also remember that c_0 cannot be zero.

$\underline{n = 1}$:

$$c_1\left[(r+1)^2 - p^2\right] = 0$$

thus either

$$\boxed{c_1 = 0 \qquad \text{or} \qquad (r+1)^2 - p^2 = 0} \tag{4.34}$$

We will consider the second possibility a little later. For now we take $c_1 = 0$.

$\underline{n \geq 2}$:

$$\boxed{c_n = \frac{-c_{n-2}}{(r+n)^2 - p^2} = \frac{-c_{n-2}}{n^2 + 2rn}} \tag{4.35}$$

where the last equality comes from $r^2 = p^2$, i.e., the indicial equation. Using the recursion relation to evaluate the c_n coefficients and inserting in the generalised power series yields

$$y(x) = c_0\, x^r \left[1 - \frac{x^2}{2(2+2r)} + \frac{x^4}{2\times 4(2+2r)(4+2r)} - \dots\right] . \tag{4.36}$$

(*You do not need to remember this formula, but in principle should be able to derive it.*)

Remembering that $c_1 = 0$, the recurrence relation implies that all $c_{\text{odd}} = 0$. Thus there is no second series. However, the two roots $r = \pm p$ will *usually* yield the two independent solutions.

Radius of convergence

Finally, having found the series solution, we should determine its radius of convergence. Using the ratio test, and noting that every other term in the power series is zero,

$$\begin{aligned} r &= \lim_{k\to\infty}\left|\frac{t_{2k}}{t_{2(k-1)}}\right| = \lim_{k\to\infty}\left|\frac{c_{2k}x^{2k+r}}{c_{2(k-1)}x^{2(k-1)+r}}\right| \\ &= \lim_{k\to\infty}\left|\frac{c_{2k}}{c_{2(k-1)}}x^2\right| = \lim_{k\to\infty}\left|-\frac{1}{(2k)^2 + 2r(2k)}x^2\right| = \lim_{k\to\infty}\frac{x^2}{4(k^2+rk)} \\ \therefore \quad r &= 0 \qquad \forall x\ . \end{aligned}$$

Thus the series converges absolutely (and therefore converges) for any x, hence the radius of convergence is $R \to \infty$.

4.6.3 Bessel functions

Let us call the solution for the $r = p$ case $y_1(x) = y_p(x)$ and denote the $r = -p$ case as $y_2(x) = y_{-p}(x)$. If p is *not* an integer then it turns out (not proven in this course) that these two solutions are linearly independent (i.e. their Wronskian determinant is non-zero), so that the general solution is

$$y_{\text{GS}}(x) = c_1 y_p(x) + c_2 y_{-p}(x) \qquad (\text{for } p \neq 0, 1, 2, \dots)$$

where c_1 and c_2 are (the usual) arbitrary constants.

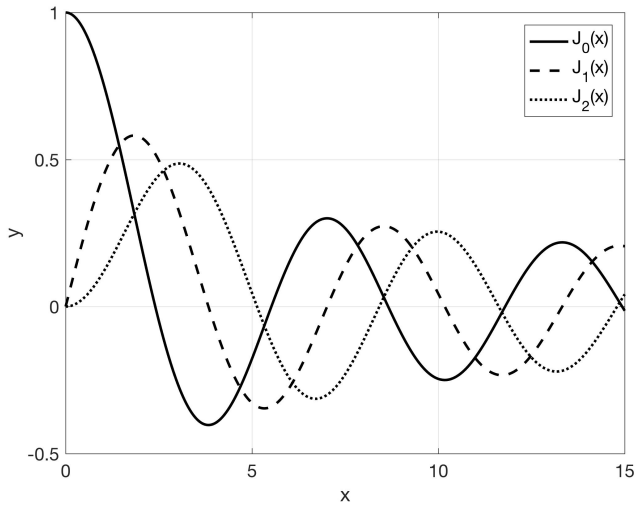


Figure 4.3: Bessel functions $J_0(x)$, $J_1(x)$ and $J_2(x)$.

It is customary to multiply the power series (4.36) by $[2\Gamma(1+r)]^{-1}$ where $\Gamma(x)$ is the ***gamma function*** which is a generalisation of the factorial function to non-integer and/or negative arguments.[6] Thus for the $r = p$ case it turns out (not proven here) that

$$J_p(x) = \left(\frac{x}{2}\right)^p \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\Gamma(p+n+1)} \left(\frac{x}{2}\right)^{2n} \quad . \tag{4.37}$$

(*You do not need to remember this formula!*)

These special functions J_p and J_{-p} are known as ***Bessel Functions*** *of the First Kind (of order p)* and the GS is

$$y_{GS}(x) = aJ_p(x) + bJ_{-p}(x) \qquad (\text{for } \; p \neq 0, 1, 2, \dots) \tag{4.38}$$

where a and b are arbitrary constants.

These Bessel functions might look horrendous, but their properties are as well studied as those of cos and sin and available in maths textbooks and handbooks. Figure 4.3 shows $J_0(x)$, $J_1(x)$, $J_2(x)$. Figure 4.4 shows $J_{1/3}(x)$ and $J_{-1/3}(x)$. All are approximately decaying oscillatory functions. However, the period of the oscillations varies as x increases. We will see in section 9.2 (of the PDE part of the course) that when solving the PDE $\nabla^2 y = 0$ in systems with cylindrical symmetry and with $y = 0$ on a circular boundary $r = R$, that $J_0(r)$ will be of great importance (being an eigenfunction)!

4.6.4 Bessel's equation of integer order

When $p \to n = 0, 1, 2\ldots$, it is possible to show (not proven here) that $J_{-n}(x) = (-1)^n J_n(x)$ so that it is apparent that they are linearly dependent. In principle, the formal method of determining the second, independent solution (covered back in section 3.6) can be used. Procedures such as this yield *Bessel*

[6] **Non-examinable** – For positive integers $x \to n$ with $n = 0, 1, 2, \ldots$ the gamma function is simply $\Gamma(n+1) = n!$. Also, $\Gamma(x+1) = x\Gamma(x)$, which is consistent with the behaviour of the factorial function. For $x = 1/2$, $\Gamma(1/2) = \sqrt{\pi}$. For negative integers $n = -1, -2, \ldots$ it turns out that $\Gamma(n)$ is infinite.

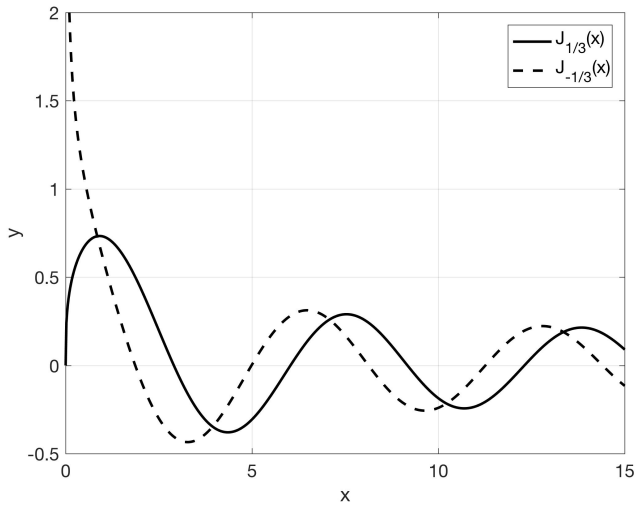


Figure 4.4: Bessel functions $J_{1/3}(x)$ and $J_{-1/3}(x)$. These are an example of the two independent solutions (basis functions) for Bessel's equation; in this case for order 1/3.

Functions of the Second Kind, denoted by $Y_p(x)$, and allow the GS to be completed;

$$y_{\text{GS}} = aJ_p(x) + bY_p(x) \qquad (\text{for } p = 0, 1, 2, \ldots) \ . \tag{4.39}$$

This form of the GS is essential for integer p: Equation (4.38) does not work. However, $Y_p(x)$ is also defined for non-integer p and is still linearly independent to $J_p(x)$. Hence (4.39) also applies for non-integer p. (More details about $Y_p(x)$ can be found in *non-examinable* appendix E.)

4.6.5 Bessel's equation of order $p = 1/2$.

Let us go back to the second indicial equation (4.34); the one obtained from $n = 1$. What if $\left[(r+1)^2 - p^2\right] = 0$? Since $r^2 = p^2$, this requires $2r+1 = 0$ or $r = -1/2$, so this is a special case of the equation when $p = 1/2$. So we look for solutions Bessel's equation with $p = 1/2$.

The $r = +1/2$ solution – From (4.35) the recurrence relation becomes

$$c_n = \frac{-c_{n-2}}{n(n+1)} \tag{4.40}$$

so

$$y(x) = c_0 x^{1/2}\left[1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \ldots\right]$$

which we can write in terms of something more familiar by dividing and multiplying by x:

$$y(x) = \frac{c_0}{x^{1/2}}\left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right] = \frac{c_0 \sin x}{x^{1/2}}. \tag{4.41}$$

There is no second series, since $c_1 = 0$ (only for $r = -1/2$ is c_1 undetermined).

The $r = -1/2$ solution – It has a different recurrence relation

$$c_n = \frac{-c_{n-2}}{n(n-1)} \ . \tag{4.42}$$

The c_1 coefficient is undetermined [see equation (4.34)], so

$$y(x) = c_0 x^{-1/2}\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots\right] + c_1 x^{-1/2}\left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right]$$

which we can write in terms of something more familiar again, as

$$\boxed{y(x) = c_0 \frac{\cos x}{x^{1/2}} + c_1 \frac{\sin x}{x^{1/2}} = y_{\text{GS}}(x) \ .} \tag{4.43}$$

The second term just duplicates the solution we found for $r = 1/2$, so that solution is already present here, and this is why we identify (4.43) as the general solution.

Things like this happen when the roots differ by an integer, as they do here ($r = -1/2, +1/2$).

Chapter 5

Boundary-Value and Eigenvalue Problems

So far we have mainly focussed on initial value problems. For linear 2nd-order ODEs, these involve initial conditions; values of y and dy/dx at a specific point x_0. We saw that the two independent solutions in the complementary function allow these to be satisfied when suitable constants, say c_1 and c_2, are found. The two solutions give the general solution $y_{GS} = c_1 y_1 + c_2 y_2$ the freedom to fit any initial conditions.

We now move on to consider ***boundary value problems*** (BVP) involving *linear 2nd-order ODEs*, where we seek solutions to the ODE that have prescribed values at two *distinct* points, e.g., $y = y_a$ at $x = a$ and $y = y_b$ at $x = b$, which are the boundary conditions (BCs). In this chapter we will see that such BCs endow 2nd-order linear ODEs with profound properties that are extremely useful (even indispensable) for deep understanding of the behaviour of physical systems.

Firstly, the combination of the ODE and BCs can yield an ***eigenvalue problem***. This happens when the *problem is homogeneous* : That is, when both the ODE and BCs are homogeneous. In such problems there is an adjustable parameter in the ODE – the ***eigenvalue*** – which must be fine-tuned to meet the BCs. Often these eigenvalues are distinct and each value yields a different solution; an ***eigenfunction***. This is the mathematical basis of quantisation in Quantum Mechanics. Furthermore, for many of the most common ODE + BC combinations in physics, it turns out that virtually any function $f(x)$ can be built from a linear combination of the (usually infinite) set of orthogonal eigenfunctions that naturally arise for the homogeneous BVP. Fourier series is one example of this. The Legendre series seen in the last chapter is another, where the Legendre polynomials are the eigenfunctions. Many other examples exist; Laguerre, Hermite and Chebyschev polynomials, and Bessel functions. The linearity of the ODE ensures that a function constructed using a Fourier series, or Legendre Series (or whatever) satisfies the problem too, and thus is a solution.

This chapter is quite general. It will show that these desirable properties arise for ***self-adjoint problems***. We will see that a particular (but quite general) form of ODE with special BCs (i.e. homogeneous BCs) forms a self-adjoint problem. This particular combination of ODE and BCs are known as ***Sturm-Liouville problems*** and are prevalent when solving PDEs arising in physics. Thus this chapter develops the concepts seen so far into a mathematical framework that will be applied to solving PDEs in the next part of the course.

5.1 Boundary Value Problems

We consider here second-order linear ODEs with inhomogeneous boundary conditions (BCs), specified at two different values of x, say $y(a) = y_a$ and $y(b) = y_b$, where y_a and y_b are known constants.

Starting with the general solution

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

we use the BCs to determine the constants c_1, c_2 that allow the GS to satisfy those BCs:

$$\begin{aligned} c_1 y_1(a) + c_2 y_2(a) &= y_a \\ c_1 y_1(b) + c_2 y_2(b) &= y_b \end{aligned}$$

i.e.

$$\begin{pmatrix} y_1(a) & y_2(a) \\ y_1(b) & y_2(b) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_a \\ y_b \end{pmatrix} \qquad (5.1)$$
$$\mathbf{M}\mathbf{c} = \mathbf{b} \ .$$

Let us define the 2×2 matrix as $\mathbf{M}$. Geometrically, the GS evaluated at each boundary represents a straight line in the (c_1, c_2) plane. Solvability hinges on the crossing of these lines.

5.1.1 Inhomogeneous BCs

If $\det(\mathbf{M}) \neq 0$ then $\mathbf{M}^{-1}$ exists, and we have a unique solution for c_1, c_2 by premultiplying by $\mathbf{M}^{-1}$. (We did something very similar in section 3.5 on linear independence.)

If $\det(\mathbf{M}) = 0$, there may be no solutions, or infinitely many. Geometrically, the two lines in the (c_1, c_2) plane have the same gradient if $\det(\mathbf{M}) = 0$, so they either coincide nowhere or everywhere.

5.1.2 Homogeneous BCs

The simplest case of homogeneous boundary conditions is $y_a = y_b = 0$, which can be considered as a special case of inhomogeneous conditions. There are other types of homogeneous BCs as will be discussed shortly in sections 5.2 (pages 47 and 49) and 5.4.1. Considering the case $y_a = y_b = 0$; if $\det(\mathbf{M}) \neq 0$, we have a unique solution again, which is the trivial one $c_1 = c_2 = 0$. This is rarely of any physical interest. Much more interesting is if $\det(\mathbf{M}) = 0$, in which case there are non-trivial solutions. To make $\det(\mathbf{M})$ precisely zero usually requires the ***fine-tuning*** of a parameter. Typically, if $\hat{L}$ contains an ***adjustable parameter*** λ (e.g. k in Legendre's equation, m (or p) in Bessel's equation), $\mathbf{M}$ can be singular if λ is chosen correctly, and usually λ is an ***eigenvalue*** of the problem, and the solution(s) y_1 or y_2 that meets the BCs is the eigenfunction associated with that particular eigenvalue λ. Furthermore, typically there are an infinite set of eigenvalues, and the particular form of $y_1(x)$ for each eigenvalue will be different. The same goes for y_2.[1] Hence we might denote this by, e.g., $y_{1,\lambda}(x)$, $y_{2,\lambda}(x)$ to acknowledge that the functions vary with λ.

The maths here is behind *all sorts* of quantisation problems (waves on violin string as well as QM).

[1] It may be that it is always only y_1 or only y_2 that fits the BCs and provides all the eigenfunctions. Or perhaps a specific linear combination does. In other cases (e.g., Legendre polynomials), y_1 provides some of the eigenfunctions (e.g. even polynomials) whiles y_2 provides the rest (e.g. odd polynomials.)

5.1.3 Example – SHO equation

Let us consider an equation we are comfortable with

$$\begin{aligned} y'' + k^2 y &= 0 \qquad (0 \le x \le b) \\ \text{with BCs:} \qquad y(0) &= y(b) = 0 \ . \end{aligned}$$

The basis functions are $y_1 = \cos(kx)$, $y_2 = \sin(kx)$. Hence

$$\det(\mathbf{M}) = \begin{vmatrix} 1 & 0 \\ \cos(kb) & \sin(kb) \end{vmatrix} = \sin(kb) \ . \tag{5.2}$$

Hence for non-trivial solutions:

$$\det(\mathbf{M}) = 0 \quad \Rightarrow \quad \sin(kb) = 0 \quad \Rightarrow \quad k = \frac{n\pi}{b} \quad (n \in \mathbb{Z}) \ .$$

Hence we have found the value of the adjustable parameter (k) that ensures the homogeneous BCs are met.

5.2 Eigenvalue Problems

You have seen eigenvalue problems (EVPs) in connection with solving equations like $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ in the year 1 'Linear Algebra' course. There you learnt how to determine the eigenvalues λ and associated eigenvectors $\mathbf{v}$ of the matrix $\mathbf{M}$.

Now we define eigenproblems in terms of *linear* differential operators (instead of matrices) and functions (instead of vectors). There are very strong parallels. This arises because both systems possess *linearity*. (Recall that matrix-vector multiplication is a linear operation too, i.e., $\mathbf{M}(\alpha\mathbf{v}+\beta\mathbf{u}) = \alpha\mathbf{M}\mathbf{v}+\beta\mathbf{M}\mathbf{u}$.)

Consider the eigenproblem

$$\boxed{\hat{L}[y(x)] = \lambda h(x) y(x) \qquad \text{for} \quad a \le x \le b} \tag{5.3}$$

where

- $\hat{L}$ is a linear operator, [2]
- λ is an eigenvalue,
- $h(x)$ is a function,
- $y(x)$ is an eigenfunction,

and the problem is subject to ***homogeneous boundary conditions***

- normally $y(a) = y(b) = 0$,
- or $y(a) - y(b) = 0$ and $y'(a) - y'(b) = 0$, which are ***periodic BCs***,
- or most generally, for some constants d_i;

$$\boxed{d_1 y(a) + d_2 y'(a) = 0 \qquad \text{and} \qquad d_3 y(b) + d_4 y'(b) = 0 \ .} \tag{5.4}$$

The boundary conditions are an *important* and *essential* part of the specification of the problem. Usually they are set by *physical considerations*.

Self-adjoint form – When the linear operator is ***self-adjoint*** (to be defined in section 5.3), then equation (5.3) takes the special form

$$\boxed{\hat{L}_{SA}[y(x)] = \lambda \rho(x) y(x) \qquad \text{for} \quad a \le x \le b} \tag{5.5}$$

where

- $\rho(x)$ is a *real* valued '***weight function***' which is strictly positive ($\rho > 0$).

[2] A word on notation. $\hat{L}(y)$, $\hat{L}[y(x)]$ and $\hat{L}y(x)$ will be used interchangeably going forward. The choice will be a matter of aesthetics; striking a balance between clarity/precision and cluttering the equation with brackets and parentheses!

The weight function is of critical importance. We will shortly see, in section 5.3, that it determines what it means for eigenfunctions to be orthogonal. We will then see in section 5.4, a form of a linear operator that (together with suitable boundary conditions) is guaranteed to be self-adjoint. That section also shows how multiplying equation (5.3) through by a suitable integrating factor $\mu(x)$ can convert $\hat{L} \to \hat{L}_{SA}$ and $h(x) \to \rho(x)$. (This does not change the actual eigenfunctions, but does reveal the weight function.)

Comments – Equation (5.3) is just a linear, homogeneous ODE. The non-derivative term involving the adjustable parameter has just been moved to the RHS. The general BCs (5.4) applies to ***regular*** *eigenvalue problems*, where both boundaries are ordinary points of the ODE. Later, in section 5.4, we will see the form of homogeneous BCs for ***singular*** *EVPs* where $x = a$ and/or $x = b$ are singular points of the ODE.

Normalisation of eigenfunctions – We *usually* choose

$$\boxed{\int_a^b |y(x)|^2 \rho(x) dx = 1 \ .} \tag{5.6}$$

Example 1 – Legendre's equation and Legendre polynomials

We saw that Legendre's equation

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + ky = 0 \qquad \text{on} \qquad -1 \le x \le 1 \ ,$$

can be made to yield solutions that do not diverge at $x = \pm 1$ by tuning the adjustable parameter k to be $k = \ell(\ell+1)$, where ℓ is a positive integer which yields discrete values of k. Rearranging Legendre's equation to

$$\begin{aligned} (1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} &= -ky \\ \hat{L}(y) &= -ky \end{aligned}$$

allows us to identify the linear operator $\hat{L} = (1-x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx}$, weight function $\rho = 1$ and eigenvalues $\lambda_\ell = -k = -\ell(\ell+1)$. The eigenfunctions are then the Legendre polynomials, i.e., $y_\ell(x) = P_\ell(x)$ which satisfy the BCs; y_ℓ and y'_ℓ are bounded (finite) at the boundaries. (These BCs will be discussed in section 5.4.) We know that $P_\ell(1) = 1$ and $P_\ell(-1) = (-1)^\ell$. It can also be shown that $P'_\ell(\pm 1) = \pm\frac{1}{2}\ell(\ell+1)P_\ell(\pm 1)$ by considering Legendre's equation at $x = \pm 1$ (where the second derivative term vanishes): Hence $P'_\ell(\pm 1)$ is also bounded. It is apparent that there are an infinite number of eigenvalues λ_ℓ and eigenfunctions y_ℓ. (Note: Recall that the normalisation of P_ℓ is unusual.)

Example 2 – SHO equation

A familiar example of an eigenproblem is the simple harmonic oscillator equation with homogeneous BCs;

$$y'' = \lambda y \qquad \text{with, e.g.,} \qquad y(0) = y(1) = 0 \ .$$

Let us explore the how different values of λ may or may not satisfy this BVP.

Case: $\lambda = 0$ – Since $y'' = 0$ yields $y = c_1 x + c_2$, application of the BC $y(0) = 0$ requires $c_2 = 0$. Then $y(1) = 0$ yields $c_1 = 0$. Only the trivial solution $y(x) = 0$ is possible in this case.

Case: $\lambda > 0$ – The general solution is then

$$y(x) = c_1 e^{\sqrt{\lambda}x} + c_2 e^{-\sqrt{\lambda}x} \ .$$

Application of the BCs

$$\begin{aligned} y(0) = c_1 \qquad + c_2 \qquad &= 0 \\ y(1) = c_1 e^{\sqrt{\lambda}} \quad + c_2 e^{-\sqrt{\lambda}} \quad &= 0 \end{aligned}$$

yields $c_1 = -c_2$ and $c_1(e^{\sqrt{\lambda}} - e^{-\sqrt{\lambda}}) = 0$ which then implies $c_1 = 0$ and $c_2 = 0$. Again, only the trivial solution is possible in this case.

Case: $\lambda < 0$ – The general solution is then

$$y(x) = c_1 \cos(\sqrt{-\lambda}x) + c_2 \sin(\sqrt{-\lambda}x) \ ,$$

Application of the BCs

$$\begin{aligned} y(0) = c_1 \qquad\qquad &= 0 \\ y(1) = c_2 \sin(\sqrt{-\lambda}) \quad &= 0 \end{aligned}$$

means that the parameter $\boxed{\sqrt{-\lambda} = n\pi}$ is required for a non-trivial solution.

Thus we identify: linear operator $\hat{L} = \frac{d^2}{dx^2}$, weight function $\rho = 1$, eigenvalues $\lambda_n = -n^2\pi^2$ and eigenfunctions $y_n = a_n \sin(n\pi x)$ for $n = 1, 2, 3, \ldots$. Again there are an infinite number of eigenvalues λ_n and eigenfunctions y_n.

You should be able to find the normalisation factor a_n using equation (5.6).

Real meaning of 'homogeneous'

Fundamentally, a ***homogeneous problem*** is one where the solution can be scaled (i.e. multiplied by a scalar) and still solves the problem. This means $y(x) \to \bar{y}(x) = \alpha y(x)$ must still (i) solve the ODE and (ii) meet the BCs. The general solution $y = c_1 y_1(x) + c_2 y_2$ to a homogeneous linear ODE satisfies the first requirement. We will shortly see that eigenfunctions are just specialised instances of the GS with the arbitrary constants restricted. Thus eigenfunctions also satisfy the first requirement. We can show that our previous definition of homogeneous BCs, equation (5.4), automatically meets the second requirement, hence deserves to be designated as *homogeneous* BCs. Inserting $\bar{y}(x)$ into (5.4)

$$\begin{aligned} d_1\bar{y}(a) + d_2\bar{y}'(a) &= d_1\alpha y(a) + d_2 \left.\frac{d(\alpha y)}{dx}\right|_{x=a} \\ &= \alpha\left[d_1 y(a) + d_2 y'(a)\right] \\ &= \alpha \cdot 0 \ = \ 0 \ . \end{aligned}$$

The same argument holds at the other boundary.

In example 2 – a regular EVP – it is readily seen that multiplying $y_n = a_n \sin(n\pi x)$ by α still (i) solves the ODE, (ii) meets the BCs $y(0) = y(1) = 0$.

In example 1 – a singular EVP – multiplying $y_\ell = P_\ell(x)$ by α still (i) solves Legendre's equation, and (ii) meets the BCs; $y_\ell(x)$, $y'_\ell(x)$ are bounded (finite) at the boundaries.

5.3 Self-Adjoint Problems

An eigenproblem is said to be ***self-adjoint*** if, for any pair of sufficiently differentiable functions $u(x)$ and $v(x)$ that satisfy the BCs given previously by equation (5.4), the following holds;

$$\boxed{\int_a^b u^*(x)\hat{L}v(x)dx = \int_a^b [\hat{L}u(x)]^* v(x)dx \ .} \tag{5.7}$$

Note that the BCs are an essential part of the specification. Here we explicitly allow the functions to be complex, so the analysis applies to QM. Note that an operator $\hat{L}$ that satisfies (5.7) is also referred to as ***Hermitian***.

5.3.1 (*) Self-adjoint operators

[**(*) Non-examinable**, but worth being aware of!]

Consider the general linear operator $\hat{L}(y) = a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y$. We take the functions a_2, a_1 and a_0 to be real.[3] The *adjoint* of this operator is defined by

$$\hat{L}^\dagger(y) \equiv \frac{d^2}{dx^2}[a_2(x)y] - \frac{d}{dx}[a_1(x)y] + a_0(x)y \ . \tag{5.8}$$

When $\hat{L}^\dagger(y) = \hat{L}(y)$ the operator is ***self-adjoint***.

Consider the LHS of (5.7). Integrating the $a_2(x)d^2v/dx^2$ term by parts twice and the $a_1(x)dv/dx$ term by parts once, it can be shown that

$$\int_a^b u^*(x)\hat{L}v(x)dx = \int_a^b [\hat{L}^\dagger u(x)]^* v(x)dx + \left[u^* a_2 \frac{dv}{dx} - \frac{d}{dx}(u^* a_2)v + u^* a_1 v\right]_a^b \ .$$

If the operator is self-adjoint then

$$\int_a^b u^*(x)\hat{L}v(x)dx = \int_a^b [\hat{L}u(x)]^* v(x)dx + \left[u^* a_2 \frac{dv}{dx} - \frac{du^*}{dx}a_2 v - u^* \cancel{\frac{da_2}{dx}} v + u^* \cancel{a_1 v}\right]_a^b \ .$$

The cancelled terms arise because $da_2/dx = a_1$ is required for an operator to be self-adjoint. At this point the *problem* is not necessarily self-adjoint. But if u and v satisfy the BCs (5.4), then it can be shown that the boundary terms [...] vanish individually at $x = a$ and $x = b$ and the problem is self-adjoint, i.e., equation (5.7) holds. Then the operator is referred to as an Hermitian operator. This is a more stringent requirement than being self-adjoint, since BCs must be satisfied too.

[**... end of Non-examinable matter.**]

[3] Generalisation to complex a_2, a_1 and a_0 functions is possible, but places more restrictions on the BCs required to make the boundary terms, arising from integration by parts, vanish.

5.3.2 Eigenvalues of self-adjoint problems are real

Proof: Put $u = v = y$ (an eigenfunction with eigenvalue λ) into the LHS of (5.7):

$$\int_a^b y^*(x)\hat{L}y(x)dx = \lambda \int_a^b y^*(x)\rho(x)y(x)dx = \lambda \int_a^b \rho(x)|y(x)|^2 dx \ .$$

Doing the same with the RHS of (5.7) and remembering that the weight function is real ($\rho^* = \rho$):

$$\int_a^b [\hat{L}y(x)]^* y(x)dx = \lambda^* \int_a^b y^*(x)\rho(x)y(x)dx = \lambda^* \int_a^b \rho(x)|y(x)|^2 dx \ .$$

Since the operator is self-adjoint, the LHSs of the last two equations are equal, so the RHSs are too. The integral cannot vanish (since $\rho(x) > 0$), therefore $\boxed{\lambda = \lambda^*}$ and we see that the eigenvalues are real. **[QED]**

5.3.3 Eigenfunctions with different eigenvalues are orthogonal

Proof: put $u = y_\alpha$ and $v = y_\beta$,[4] which are eigenfunctions with (real!) eigenvalues λ_α and λ_β, respectively:

$$\int_a^b y_\alpha^*(x)\hat{L}y_\beta(x)dx = \lambda_\beta \int_a^b y_\alpha^*(x)\rho(x)y_\beta(x)dx \tag{5.9}$$

$$\int_a^b [\hat{L}y_\alpha(x)]^* y_\beta(x)dx = \lambda_\alpha \int_a^b y_\alpha^*(x)\rho(x)y_\beta(x)dx \ . \tag{5.10}$$

$\rho^*(x) = \rho(x)$ (i.e. it is real) has been used in the RHS of (5.10). Again, for self-adjoint operators, the LHSs are equal, so

$$(\lambda_\beta - \lambda_\alpha)\int_a^b y_\alpha^*(x)\rho(x)y_\beta(x)dx = 0$$

and since the eigenvalues are different, we require the orthogonality of the eigenfunctions, defined by

$$\int_a^b y_\alpha^*(x)\rho(x)y_\beta(x)dx = 0 \qquad \text{for } \alpha \neq \beta \ .$$

If the eigenfunctions are normalised, according to equation (5.6), then

$$\boxed{\int_a^b y_\alpha^*(x)\rho(x)y_\beta(x)dx = \delta_{\alpha\beta} \ .} \tag{5.11}$$

Such eigenfunctions are said to be ***orthonormal***. Note that one of the functions appears with a complex conjugate, and that the weight function $\rho(x)$ is part of the definition of orthogonality. For the examples we have seen so far, the functions have been real and $\rho(x) = 1$.

For problems that have some equal eigenvalues, we can construct linear combinations of the 'degenerate' eigenfunctions that are orthogonal. The procedure is called *Gram-Schmidt orthogonalisation*. It will not be covered here. (However, it is very similar to that used to construct orthogonal vectors, which you may have seen.)

[4] Here α and β are integer indices that label the eigenvalues and eigenvectors.

5.3.4 Completeness of eigenfunctions

This implies that any function, say $f(x)$, defined on the interval $a \le x \le b$ on which the eigenfunctions exist, and satisfying the ***Dirichlet conditions***[5] (which you saw in the Fourier course), can be expanded in terms of those eigenfunctions:

$$\boxed{f(x) = \sum_{n=0}^{\infty} c_n y_n(x) \ .} \tag{5.12}$$

(This will not be proven here.) ***Complete*** means that

$$\lim_{N\to\infty} \int_a^b \left| f(x) - \sum_{n=0}^{N} c_n y_n(x) \right|^2 dx = 0 \ . \tag{5.13}$$

Note that this definition means that the eigenfunction expansion does not necessarily have to converge to $f(x)$ at every point on the interval $a \le x \le b$. This is often referred to as ***convergence in the mean***. This allows the eigenfunction expansion to overshoot in fine spikes (of non-vanishing amplitude as $N \to \infty$) where $f(x)$ has a (finite) discontinuity; the Gibbs phenomenon. You saw this in the Fourier course. It also occurs with Legendre polynomial expansions, as shown in figure 5.1 for the case of a step function.

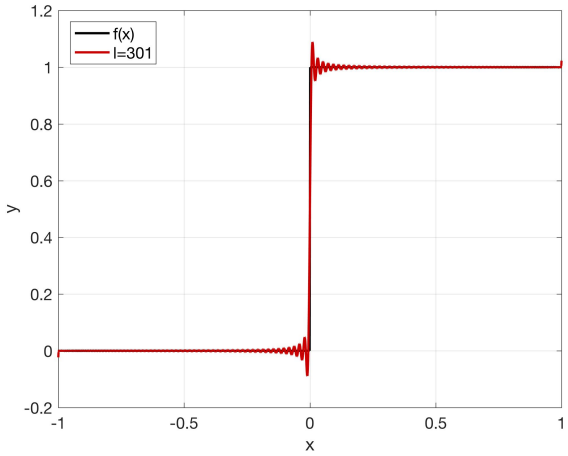


Figure 5.1: Gibbs phenomenon for Legendre expansion out to $\ell = 301$ of the step function $f(x) = \{0, 1\}$ for $\{x < 0, x \ge 0\}$. This is an example of convergence in the mean, yet $\sum_{n=0}^{\infty} c_n y_n(x)$ is not equal to $f(x)$ everywhere.

The expansion coefficients c_n can be found via

$$c_n = \int_a^b \rho(x) y_n^*(x) f(x) dx \ , \tag{5.14}$$

when the eigenfunctions are normalised. This follows by performing the integral on the RHS above on the expansion of $f(x)$, eqn. (5.12), and then using the orthonormal property (5.11). Note that the requirement $\rho(x) > 0$ throughout the range of the problem, specified after (5.3), is necessary for the set of eigenfunctions to be complete.[6]

[5] The Dirichlet conditions are: (1) $f(x)$ is absolutely integrable: $\int_a^b |f(x)| dx$ is finite. (2) $f(x)$ is single valued. (3) The function has a finite number of extrema on the interval $a \le x \le b$. (4) $f(x)$ can have a finite number of *finite* discontinuities.

[6] It also ensures that $\int_a^b u^*(x)\rho(x)v(x)dx$ behaves as a valid inner product; e.g. $\int_a^b \rho(x)|u(x)|^2 dx = \langle u|u\rangle = 0$ iff $u(x) = 0$

5.4 Sturm-Liouville problems

Sturm-Liouville problems are eigenvalue problems involving a linear operator $\hat{L}$ in a particular form, together with special BCs. The key point is that **Sturm-Liouville problems are self-adjoint**. Therefore, they inherit all the properties above: real eigenvalues and orthogonal eigenfunctions, which are also complete.

The specific form of $\hat{L}(y)$ needed is $\hat{L}(y) = [p(x)y']' + q(x)y$ which results in the ODE

$$\boxed{\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + q(x)y(x) = \lambda\rho(x)y(x)} \tag{5.15}$$

where

- $p(x)$, $p'(x)$, $q(x)$ [7] and $\rho(x)$ are continuous and real,
- $\rho(x) > 0$ and $p(x) > 0$ (except that $p = 0$ is allowed at the ends of the interval).

The required BCs are that, for any solutions u and v of (5.15),

$$\boxed{\left[u^* pv'\right]_{x=a}^{x=b} = 0 \ .} \tag{5.16}$$

Proof – Eigenvalue problems in Sturm-Liouville form are self-adjoint.

$$\begin{aligned}\int_a^b u^*\hat{L}v\,dx &= \int_a^b u^*\left\{\frac{d}{dx}\left[p(x)\frac{dv}{dx}\right] + q(x)v(x)\right\}dx \\ &= \cancel{\left[u^* pv'\right]_a^b} - \int_a^b (u^*)'pv'dx + \int_a^b u^* qvdx \ , \end{aligned} \tag{5.17}$$

where integration by parts has been used on the 2nd derivative term. The term in square brackets is zero because of the required boundary conditions (5.16). Applying integration by parts once more on the first integral in (5.17), using the [complex conjugate of the] BCs again, and noting that p and q are real results in:

$$\begin{aligned}\int_a^b u^*\hat{L}v\,dx &= -\cancel{\left[(u^*)'pv\right]_a^b} + \int_a^b [p(u^*)']'vdx + \int_a^b u^* qvdx \\ &= \int_a^b [(pu')' + qu]^* vdx = \int_a^b [\hat{L}u]^* vdx \ . \end{aligned} \tag{5.18}$$

[QED]

The required BCs, eq. (5.16) are needed for the proof. Many physical problems satisfy these (e.g. in spherically-symmetric quantum problems, $a = 0$ and $b \to \infty$, u may be the wave function, which often vanishes as $r \to \infty$, and p involves r^2, which vanishes at $r = 0$, so the BC condition is satisfied).

everywhere. Ultimately, $\rho(x) > 0$ is necessary for eigenfunctions to form a linear vector space! (More detail is beyond the scope of this course!)

[7] A word on notation: $p(x)$ and $q(x)$ here are different to the coefficient of y' and y, respectively, in the ODE written in standard form! The S-L form of the linear operator expands to $\hat{L}(y) = p(x)y'' + p'(x)y(x) + q(x)y$ and illustrates this!

5.4.1 Regular and Singular Problems

The beauty of equation (5.16) is that it covers both ***regular*** and ***singular*** *eigenvalue problems*. It extends the definition of homogeneous BC seen earlier (5.4) to include singular problems where $p(a) = 0$ or $p(b) = 0$, because one or both of the boundaries are singular points of the ODE in the problem. Solutions $u(x)$ and $v(x)$ of the ODE which are *finite* at a singular boundary (e.g. $x = a$), and have a finite first-derivative there too – i.e. $u(a)$, $u'(a)$ and $v(a)$, $v'(a)$ are bounded – will ensure that $\left[u^* p v'\right]_{x=a}$ vanishes. Some of the most prominent and useful EVPs in physics are singular; Legendre's equation on $[-1,1]$, Bessel's equation on $[0,\infty)$, etc. Hence singular EVPs are important.

Note that (5.4) does not apply at singular points, or at least is too restrictive. Remember that for the problem to be homogeneous, the fundamental requirement is that multiplying a solution by a scalar does not stop it being a solution. Inserting $\alpha u(x)$ and $\beta v(x)$ into (5.16) will not interfere with meeting the BC at a singular point, since $\left[(\alpha u)^* p(\beta v)\right]_{x=a} = \alpha^*\beta\left[u^* p v'\right]_{x=a} = \alpha^*\beta \cdot 0 = 0$.

Example – Legendre's equation

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = -\ell(\ell+1)y \qquad (-1 \le x \le 1)$$

can be written as

$$\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right] = -\ell(\ell+1)y \ .$$

Thus we identify $p(x) = (1-x^2)$, $q(x) = 0$, $\rho(x) = 1$ and $\lambda = -\ell(\ell+1)$. We then note that p, p', q, ρ meet the requirements for an ODE in Sturm-Liouville form. Also note that $a = -1$ and $b = +1$, thus $p(a) = p(b) = 0$. Therefore we have *singular boundaries* (i.e. boundaries lie at singular points of the ODE) which means that the Sturm-Liouville BCs (5.16) are satisfied as long as the eigenfunctions and their derivatives are finite there. Remembering back to section 4.4.1, adjusting the eigenvalues to $\lambda = -k = -\ell(\ell+1)$ truncated the infinite series which diverges at the boundaries. The truncated series – Legendre polynomials P_ℓ – are finite at the boundaries and so is their derivative. So we can verify that the S-L BCs are indeed satisfied. Hence we know that we have a singular Sturm-Liouville problem. That means that the eigenfunctions are guaranteed to be orthogonal & complete, and the eigenvalues real. This accords with what we already saw in section 4.5.1, before we knew about self-adjoint EVPs!

The Legendre polynomials do <u>not</u> satisfy (5.4); the homogeneous BC that apply to *regular* EVPs. There is no universal choice of d_3 and d_4 that satisfies $d_3 P_\ell(1) + d_4 P'_\ell(1) = 0$ for any ℓ. At $x = 1$, Legendre's equation collapses to $-2y' + \ell(\ell+1)y = 0$. Hence $P'_\ell(1) = \frac{1}{2}\ell(\ell+1)P_\ell(1)$. This means that if, e.g., d_3 and d_4 are tuned to make $P_1(x)$ satisfy $d_3 P_1(1) + d_4 P'_1(1) = 0$, then P_2, P_3, ...will not! But $\alpha P_\ell(x)$ (where α is a scalar) solves the ODE (5.15) and satisfies BCs (5.16), hence it is a valid solution of the singular homogeneous problem.

5.4.2 Transforming into Sturm-Liouville form

If an eigenvalue equation is not in Sturm-Liouville form, it is sometimes possible to get it into Sturm-Liouville form by using an ***integrating factor***. This will also reveal the weight function ρ.

Say that the eigenvalue equation is

$$y'' + f(x)y' + g(x)y = \lambda h(x)y \ . \tag{5.19}$$

The trick is to apply the IF to $z \equiv dy/dx$ rather than y. Looking only at the first and second derivatives

of y, we want to make them into a perfect derivative. This can be achieved using the same integrating factor technique used for (linear) first order ODEs and covered back in section 2.1.2. So, the key part of the equation (in standard form) to concentrate on is

$$y'' + f(x)y' = z' + f(x)z \ .$$

If we multiply by the integrating factor [i.e. equation (2.12)]

$$\mu(x) \equiv \exp\left[\int^x f(s)ds\right] \tag{5.20}$$

then we get

$$\mu(x)\left[y'' + f(x)y'\right] \ = \ \frac{d}{dx}\left[\mu(x)z\right] \ = \ \frac{d}{dx}\left[\mu(x)\frac{dy}{dx}\right] \ .$$

So multiplying the *entire* ODE by the IF yields

$$\frac{d}{dx}\left[\mu(x)\frac{dy}{dx}\right] + \mu(x)g(x)y \ = \ \lambda\,\mu(x)h(x)\,y \ . \tag{5.21}$$

This is now in Sturm-Liouville form. We immediately identify that $p(x)$ is in fact the IF $\mu(x)$, and that the weight function $\rho(x) = \mu(x)h(x)$.

Note that conversion of the ODE in an EVP to Sturm-Liouville form does not change the solutions. In other words, the eigenfunctions and eigenvalues are the same for equations (5.19) and (5.21), assuming the same BCs are used. However, conversion to Sturm-Liouville form does reveal the weight function needed to use the eigenfunctions as an orthogonal set of functions (and therefore to act like a 'generalised' Fourier series).

Example

Consider

$$(1-x^2)\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + ky = 0 \qquad (-1 \le x \le 1)$$

It is similar to Legendre's equation, but with −4 instead of −2 in the y' term. We can convert to standard form and write this as

$$z' - \frac{4x}{1-x^2}z \ = \ -\frac{k}{1-x^2}y \ .$$

The integrating factor is

$$\begin{aligned}\mu(x) \ &= \ \exp\left[\int^x -\frac{4s}{1-s^2}\,ds\right] \\ &= \ \exp\left[2\ln(1-x^2) + c\right] \ = \ e^c \exp\left\{\ln[(1-x^2)^2]\right\} \\ &= \ e^c(1-x^2)^2 \ .\end{aligned}$$

Hence multiplying through by $\mu(x)$ yields

$$\begin{aligned}(1-x^2)^2\,z' - 4x(1-x^2)z \ &= \ -k(1-x^2)y \\ \therefore \qquad \frac{d}{dx}\left[(1-x^2)^2 z\right] \ = \ \frac{d}{dx}\left[(1-x^2)^2\frac{dy}{dx}\right] \ &= \ -k(1-x^2)y \ .\end{aligned}$$

This is now in S-L form. We identify $p(x) = (1-x^2)^2$, $q(x) = 0$ and $\rho(x) = (1-x^2)$. The eigenvalue is $\lambda = -k$. It is a singular S-L problem since $p(-1) = 0$ and $p(1) = 0$.

5.5 Summary

Since this chapter is quite technical and deeply interconnected, it is worth summarising the main points and logic:

- A boundary value problem is comprised of an ODE and a boundary condition specified at two separate points. We limited ourselves to linear, 2nd-order ODEs. (Even that yielded a cornucopia of riches!)
- The prescription becomes increasingly precise moving from "boundary value problem" to "eigenvalue problem" to "self-adjoint problem" to "Sturm-Liouville problem". (So S-L is a subset of EVP, etc.)
- Homogeneous problems are particularly interesting and profound. A problem is homogeneous if given a solution, we can multiply it by a scalar (i.e. a constant or number) and that still solves the same problem; that is, it still (i) solves the ODE and still (ii) meets the BCs. Linear, homogeneous ODEs satisfy the 1st requirement. Homogeneous BCs, i.e. (5.4) & (5.16), meet the 2nd requirement.
- Fine tuning of an adjustable parameter in the ODE is needed for its solution to meet homogeneous BCs. (There will normally be an infinite set of values that work.)
- Eigenvalue problems (5.3) are homogeneous BVPs. The term with the adjustable parameter is put on the RHS. The remaining terms on the LHS effectively define the linear operator $\hat{L}$. The values of the adjustable parameter required yield the eigenvalues. The realisations of the power-series solution with that set of parameters are the eigenfunctions.
- Furthermore, if the eigenproblem is self-adjoint (5.7) – which restricts the form of $\hat{L}$ and requires that the BCs are homogeneous (5.4) – then the eigenfunctions are guaranteed to be orthogonal and complete! Furthermore eigenvalues are guaranteed to be real. (Which is all jolly useful for physics!)
- If $\hat{L}$ has Sturm-Liouville form (5.15), then $\hat{L}$ together with homogeneous BCs (5.16) is guaranteed to be self-adjoint. Hence S-L problems inherit the 'nice' properties of self-adjoint problems, noted above.
- Eigenvalue problems can be regular (when both boundaries are ordinary points of the ODE) or singular (when either boundary coincides with a singular point of the ODE). Some of the most important 'problems' in physics are singular, e.g., Legendre and Bessel!

This all underpins solving PDEs!

Part II

Part Two – PDEs

Chapter 6

Introduction to Partial Differential Equations

In this second part of the course, we consider solutions of *linear* partial differential equations (PDEs). The knowledge of ODEs and their solutions covered in the first part will be extensively used here. We now consider differential equations with two, or even three, independent variables. The two independent variables might be x, t or x, y or r, ϕ. We will also see PDEs depending on x, y, t, or r, θ, ϕ, etc. The solution techniques to be covered later in chapter 7 and 8 can readily be extended to more than three independent variables.

This short chapter introduces three common PDEs in physics that will be attacked. It will also introduce some terminology on the classification of PDEs and the boundary conditions that apply to them.

6.1 Key Equations in Physics

The course concentrates on 3, specific, linear PDEs:

(i) The Wave Equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \tag{6.1}$$

where $u(x, t)$ could be the displacement of a string, density of air, strength of E or B field, etc., and c is the ***phase speed*** of the wave, e.g., speed of sound, speed of light, etc. To generalise to 3D, replace $\partial^2 u/\partial x^2$ by $\nabla^2 u$. The wave equation is classified as a ***hyperbolic*** PDE.

(ii) The Diffusion Equation

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} \ , \tag{6.2}$$

where D is the ***diffusion coefficient***. To generalise to 3D, replace $\partial^2 u/\partial x^2$ by $\nabla^2 u$. If $u(x, t)$ is temperature T then this is known as the ***heat equation***. It describes the diffusion of thermal energy through matter. It is formed by the equation for conservation of thermal energy

$$c_v \frac{\partial T}{\partial t} + \nabla \cdot \mathbf{q} = 0 \tag{6.3}$$

where c_v is the heat capacity (per unit volume) and $\mathbf{q}$ is the the heat flow (i.e. an thermal energy flux with dimensions of power / area) which is given by Fourier's law of thermal conduction

$$\mathbf{q} = -\kappa \nabla T \tag{6.4}$$

where κ is the thermal conductivity (which can be experimentally measured or determined from microscopic physics theories). This basically says that heat flows down temperature gradients. If the thermal conductivity is independent of position, then combining these two equations gives $\partial_t T = \frac{\kappa}{c_v} \nabla^2 T$ where we can identify the thermal diffusion coefficient $D_T = \kappa / c_v$. In other contexts, $u(x,t)$ could be number density n, mass density ρ_m, charge density ρ, etc. There is a corresponding flux of the quantity down its gradient.

The diffusion equation is classified as a ***parabolic*** PDE.

(ii) Laplace and Poisson Equations

$$\nabla^2 u = 0 \qquad \text{Laplace}$$

$$\nabla^2 u = -\frac{\rho}{\epsilon_0} \qquad \text{Poisson}$$

and $u(\mathbf{r}, t)$ is the electrostatic potential, and ρ is the charge density. These equations are classified as ***elliptic*** PDEs.

For derivations, see the handout on blackboard. Derivations of the physics equations are not examinable in this course, but you **are** expected to be familiar with these equations.

There are other hyperbolic PDEs than the wave equation, other parabolic PDEs than the diffusion equation, etc. The classification can be broadly understood by considering the *domain of influence* and *domain of dependence* of a given point x, t (or $\mathbf{r}, t$). Hyperbolic PDEs share the same essential characteristics: propagation of information at finite speed. Think of light cones. A point cannot influence something too far away at a given time later. Parabolic-type PDEs share the same essential characteristic of describing a vanishingly small but non-zero influence spreading out infinitely fast, infinitely far. Think of a Gaussian shape curve/function spreading out with time while keeping its area fixed. The domain of influence for x_0, t_0 is $-\infty < x < \infty$ for $t > t_0$. The domain of dependence on the state of the system at earlier times $t < t_0$ is $-\infty < x < \infty$ too. Elliptic PDEs share the characteristic of the solution being intimately coupled throughout the domain (x, y). The solution at a point x_0, y_0 affects and is affected by the solution at all other points.

The procedure for classification of PDEs into hyperbolic, parabolic and elliptic is covered in the year 3, Computational Physics course. Using numerical methods implemented on a computer (in a language like, e.g., Python, C++ or FORTRAN), is a way to approximately solve nonlinear PDEs (of first degree). However the methods for treating each class of PDEs are different, so it is essential to know the flavour of your PDE before attempting a numerical solution!

6.2 Classification of BCs

(*) Non-examinable – This is just to help you understand the terminology when you see it in the future. Most will not be seen again in this course. 'Periodic' is the exception.

u defined on boundaries	: Dirichlet conditions,
∇u defined on boundaries	: Neumann conditions,
both u and ∇u defined on boundaries	: Cauchy conditions.
u or ∇u applied on diff. parts of boundary	: mixed conditions.
$u(x_r) = u(x_l + L) = u(x_l)$ and $u'(x_r) = u'(x_l)$	: Periodic BC (e.g. in x).
$u(-x) = u(x)$	: Reflective BC (e.g. about $x_l = 0$).

where, e.g., x_l and x_r denote the left and right edges of the domain. For elliptic problems in 2D, the 'boundary' can, in principle, be a closed curve of arbitrary shape. In 3D, it is a closed surface. For hyperbolic/parabolic initial-value problems, Cauchy conditions are applied only at $t = t_0$. For $t \to \infty$, parabolic problems normally insist that $u \to 0$, while hyperbolic problems might use an *open boundary* where u is free to do what it wants. ∇u means the gradient normal to the boundary curve/surface. If the boundary is aligned with a pair of coordinate axes, e.g., x and y, then the condition on ∇u simply becomes a condition on, e.g., $\partial u/\partial z$. A reflective BC is equivalent to $\partial u/\partial x = 0$ at a boundary, so is a special case of a Neumann condition.

Chapter 7

Separation of Variables

This chapter considers the main technique for solving linear PDEs, whereby the equation is separated into ODEs in each independent variable. Having already learnt how to solve second-order ODEs with variable coefficients and then eigenvalue problems (in one independent variable), this is the next key step towards being able to tackle important differential equations found in physics!

Each linear ODE that arises can be solved using one of the techniques from Part 1. However, these ODEs are linked through ***separation constants***. The spatial boundary conditions are, as ever, crucial. Together with the ODE, they yield boundary value problems, e.g., x or r. In fact, the separation constant will play the role of an eigenvalue, so we will really be dealing with eigenvalue problems in spatial dimensions. The fundamental separable solutions that will be obtained are, therefore, eigenfunctions of the PDE, which are also referred to as ***normal modes*** (or just *modes*). For PDEs involving time, each mode can have a different time dependence, e.g., frequency or decay rate.

Usually, the initial conditions, e.g., $u(\mathbf{r}, t = 0)$ can be expanded into these normal modes, thanks to the Sturm-Liouville form of 'problem' in each spatial dimension.

Solutions to a PDE (that fit the imposed BCs) that are not separable are perfectly possible. Interestingly, we will see that non-separable solutions can be built out of the separable ones, i.e., the normal modes.

This chapter will only consider PDEs in terms of Cartesian coordinates, Since the separated ODEs are the easiest to solve, they yield the most straightforward examples of the method of separation of variables. In Cartesian coordinates, the eigenfunctions involved are $\sin(k_n x)$ and $\cos(k_n x)$ in a 1D case, with k_n tuned to meet the BCs. Thus, ***Fourier series*** will be involved. The method will be applied to spherical polar and cylindrical polar coordinates in chapter 9. This chapter will also only consider systems of finite spatial extent. Extension to infinitely long systems is covered in chapter 8.

This is perhaps a lot to take in. Come back and read this again once you have finished this section! (Hopefully, it will make much more sense!)

7.1 Method

This is a general technique that is useful for a wide range of PDEs, including the three we concentrate on. It works for **linear**, **homogeneous** PDEs **without mixed partial derivatives**. Nonlinear PDEs, e.g., the Navier-Stokes equation, cannot be solved this way. It is not guaranteed to work with PDEs involving mixed derivatives, though in some particular cases,it turns out to be possible to separate them into ODEs. Luckily, many, if not most, important PDEs arising in physics do not involve mixed partial derivatives. The method cannot be applied directly to inhomogeneous PDEs, i.e. ones with a 'source' term not involving the dependent variable. However, the eigenfunctions for the associated homogeneous equation (found using the separation of variables method) are involved in the solution to the full inhomogeneous equation. (See appendix H.)

The individual solutions found by the separation of variables method are **not** general: Most solutions are not separable. However, they form a ***basis*** in terms of which the general solution may be expanded. Note that the linear sum of separable solutions is not in general separable. This will be demonstrated later on.

The method of separation of variables is as follows. The outline here is for two independent variables x (space) and t (time). (The extension to three or more will be illustrated in section 7.4.) We assume a linear, homogeneous PDE (in two independent variables) without mixed partial derivatives. Such a PDE can always be written as

$$\hat{L}_x[u(x,t)] = \hat{L}_t[u(x,t)] \tag{7.1}$$

where the linear differential operator $\hat{L}_x$ only involves partial derivatives with respect to x and $\hat{L}_t$ only involves partial derivatives with respect to t. For example, in the case of the 1D diffusion equation $\hat{L}_x = D\frac{\partial^2}{\partial x^2}$ and $\hat{L}_t = \frac{\partial}{\partial t}$. We then seek ***separable solutions*** of the form

$$\boxed{u(x,t) = X(x)T(t)} \tag{7.2}$$

where $X(x)$ is some function of x (to be determined) and $T(t)$ is some function of t. In other words, the dependence on each independent variable factorizes into a separate function for each.

Step 1 – Separate variables

This is achieved by substituting the form XT into the equation, dividing through by $u = XT$ and then collecting all terms involving x on one side and terms involving t on the other. For each side to hold for arbitrary x and t, each must be equal to a constant; the ***separation constant***:

$$\boxed{\frac{1}{X(x)}\hat{L}_x(X) = \frac{1}{T(t)}\hat{L}_t(T) = S = \text{const}} \tag{7.3}$$

where S is the separation constant. If one cannot arrange for one side to depend on a single dependent variable only, and the other side to depend on the *other* independent variable (or variables) only, then the PDE is not separable and the method fails.

Step 2 – Solve the ODEs to determine the normal modes

Equation (7.3) implies there are two ODEs to be solved:

$$\hat{L}_x(X) = SX \ , \tag{7.4}$$
$$\hat{L}_t(T) = ST \ . \tag{7.5}$$

Typically, homogeneous BCs are applied in x. Thus (7.4) then forms an eigenvalue problem for $X(x)$ where S is the eigenvalue. For finite-length spatial domains, there will be an infinite set of discrete values; $S \longrightarrow S_n$ where $n = 1, 2, \ldots$. Each eigenvalue has an associated eigenfunction $X_n(x)$. For spatial domains of infinite extent, it will be a continuous eigenvalue. (See later.) Having determined S, the second ODE, (7.5) can be solved, with the separation constant (i.e. the eigenvalue) providing a 'link' between x and t. Thus picking a specific eigenvalue sets the ODE for T, e.g., $\hat{L}_t(T) - S_n T = 0$, and results in a specific $T(t)$ solution, which we denote as $T_n(x)$. Together, this yields an infinite set of separable solutions

$$\boxed{u_n(x,t) = X_n(x)T_n(t) \qquad \text{for integer } n \ .} \tag{7.6}$$

These are called ***eigenstates***, ***eigenfunctions*** or ***normal modes***.

Step 3 – Find the particular solution that satisfies the initial conditions

Since $u_n(x,t)$ satisfies the PDE and the homogeneous BCs, so does their linear superposition

$$\boxed{u(x,t) = \sum_n c_n u_n(x,t)} \tag{7.7}$$

where the sum is an infinite sum. This is the ***general solution***.

The initial conditions (on the line $t = 0$) are normally inhomogeneous; $u(x, t = 0) = u_0(x)$. The remaining task is then to determine the coefficients c_n to satisfy the initial conditions. Since the spatial eigenfunctions $X_n(x)$ are usually orthogonal and complete, c_n can be determined by, e.g., Fourier analysis of $u_0(x)$ or the equivalent for whatever orthogonal basis function applies (e.g. Legendre polynomials, Bessel functions, …).

Note that equation (7.7) only works because the spatial BCs are homogeneous: It does *not* work if they are inhomogeneous. Sometimes, the spatial BCs are homogeneous on some boundaries and not others, e.g., u is fixed at fixed a value at one end (e.g. $x = a$ for all t). Special consideration will then be needed.

More general solutions are not separable

Although each normal mode is a separable solution, more general solutions do not have to be separable; they will not have the form $u(x,t) = \chi(x)\tau(t)$. This is because the **sum** of normal modes is, in general, **not separable**. This is easy to see just considering two modes:

$$\begin{aligned} u(x,t) &= u_1(x,t) + u_2(x,t) = X_1(x)T_1(t) + X_2(x)T_2(t) = (X_1 + X_2)(T_1 + T_2) - \{X_1T_2 + X_2T_1\} \\ &= \chi(x)\tau(t) + f(x,t) \ . \end{aligned} \tag{7.8}$$

It cannot be arranged into the form $u(x,t) = \chi(x)\tau(t)$. While the first term on the RHS is separated (into sums of the eigenfunctions in x & t), the 'cross term' $f(x,t) = -(X_1T_2 + X_2T_1)$ then arises and spoils the

overall separation.

7.1.1 Recap – Fourier Series

The Fourier series on an interval $[-L, L]$, in terms of trigonometric functions, is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right] \tag{7.9}$$

where the mode amplitudes are determined via [1]

$$\begin{aligned} a_n &= \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx \\ b_n &= \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx \ . \end{aligned} \tag{7.10}$$

The Fourier modes are orthogonal

$$\begin{aligned} \frac{1}{L}\int_{-L}^{L}\cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx &= \delta_{mn} \\ \frac{1}{L}\int_{-L}^{L}\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx &= \delta_{mn} \\ \frac{1}{L}\int_{-L}^{L}\cos\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx &= 0 \ . \end{aligned} \tag{7.11}$$

Given the above mode orthogonality, the expression 7.10 can easily be shown, e.g.,

$$\begin{aligned} \int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx &= \int_{0}^{L}\sum_{m=1}^{\infty}\left[a_m\cos\left(\frac{m\pi x}{L}\right) + b_m\sin\left(\frac{m\pi x}{L}\right)\right]\sin\left(\frac{n\pi x}{L}\right)dx \\ &= \sum_{m=1}^{\infty}\left\{a_m\int_{-L}^{L}\cos\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)dx + b_m\int_{-L}^{L}\sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)dx\right\} \\ &= \sum_{m=1}^{\infty}\left\{a + m\cdot 0 + b_m\left(L\,\delta_{nm}\right)\right\} = L\,b_n \ . \end{aligned}$$

In terms of complex exponentials:

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} c_n\, e^{in\pi x/L} \\ c_n &= \frac{1}{2L}\int_{-L}^{L} f(x)\, e^{-in\pi x/L}dx \ . \end{aligned} \tag{7.12}$$

[1] Watch out in what follows! b_n and a_n may be switched, so that a_n is associated with the amplitude of sin modes, etc. Sorry!

7.2 Wave Equation in 1D (finite domain)

Problem: Solve the wave equation (6.1) on $0 \le x \le L$ with homogeneous boundary conditions $u(0,t) = 0$, $u(L,t) = 0$, with initial conditions $u(x,0) = u_0(x)$ and $\partial u/\partial t = v_0(x)$ when $t = 0$, for some functions $u_0(x)$ and $v_0(x)$.

Step 1: Separate variables – Substitute $u(x,t) = X(x)T(t)$ into wave equation:

$$\begin{aligned} c^2 \frac{\partial^2}{\partial x^2}[X(x)T(t)] &= \frac{\partial^2}{\partial t^2}[X(x)T(t)] \ , \\ \therefore \qquad c^2 T(t)\frac{d^2X}{dx^2} &= X(x)\frac{d^2T}{dt^2} \ . \end{aligned}$$

We do not use partial derivatives here because X is only a function of x and T depends only on t, thus, they are not necessary. Collect terms

$$\frac{1}{X}\frac{d^2X}{dx^2} = \frac{1}{c^2T}\frac{d^2T}{dt^2} \ .$$

Let them both be equal to $F(x,t)$, say. But terms on LHS do not depend on t, so $\partial F/\partial t = 0$, and terms on the RHS don't depend on x, so $\partial F/\partial x = 0$. F is equal to something that is neither a function of x nor of t, so is equal to a constant, which we will call S:

$$\frac{1}{X}\frac{d^2X}{dx^2} = \frac{1}{c^2T}\frac{d^2T}{dt^2} = S \ . \tag{7.13}$$

Step 2: Solve the ODEs for the normal modes – First solve the equation for $X(x)$

$$\frac{d^2X}{dx^2} = SX \tag{7.14}$$

subject to homogeneous BCs:

$$\begin{aligned} u(0,t) &= X(0)T(t) = 0 \qquad \forall\, t \\ u(L,t) &= X(L)T(t) = 0 \qquad \forall\, t \ . \end{aligned} \tag{7.15}$$

We ignore the trivial solution $T(t) = 0$ which just gives $u = 0$ everywhere for all time, so we need $X(0) = X(L) = 0$. This is the same (eigenvalue) problem that was covered in section 5.2. The eigenvalues and eigenfunctions are

$$S_n = -\frac{n^2\pi^2}{L^2} \quad , \qquad X_n(x) = \sin\left(\frac{n\pi x}{L}\right) \qquad (n = 1, 2, \ldots) \ . \tag{7.16}$$

We do not bother normalising the eigenfunctions at this stage, for reasons that will become clear. The function $T_n(t)$ that accompanies $X_n(x)$ satisfies

$$\frac{d^2T_n}{dt^2} = S_n c^2 T_n = -\frac{n^2\pi^2c^2}{L^2}T_n \ . \tag{7.17}$$

Hence

$$T_n(t) = A_n \sin\left(\frac{n\pi ct}{L}\right) + B_n \cos\left(\frac{n\pi ct}{L}\right) \tag{7.18}$$

for constants A_n, B_n. Thus the normal modes are

$$u_n(x,t) = X_n(x)T_n(t) = \sin\left(\frac{n\pi x}{L}\right)\left[A_n \sin\left(\frac{n\pi ct}{L}\right) + B_n \cos\left(\frac{n\pi ct}{L}\right)\right] \ . \tag{7.19}$$

Step 3: Form the general solution and impose the initial conditions – The GS is the sum of the normal modes. Because u_n above already carry their arbitrary amplitudes through A_n and B_n we write

$$u(x,t) \quad = \quad \sum_{n=1}^{\infty} u_n(x,t) \ . \tag{7.20}$$

This satisfies the homogeneous BCs at $x = (0, L)$ and the PDE.

Now the trickier part. We need to choose the constants so that the solution satisfies the initial conditions (which are inhomogeneous BCs in t);

$$u(x,0) = u_0(x) \quad , \qquad \frac{\partial}{\partial t}u(x,t)\bigg|_{t=0} = v_0(x) \ .$$

Since the $\sin(n\pi ct/L)$ terms vanish at $t = 0$, and the $\cos(n\pi ct/L)$ terms equal 1,

$$u(x,0) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) = u_0(x) \tag{7.21}$$

which we can solve using orthogonality of the sin functions on the range, i.e., ***Fourier analysis***:

$$B_n = \frac{2}{L}\int_0^L u_0(x)\sin\left(\frac{n\pi x}{L}\right)dx \ . \tag{7.22}$$

In fact, since an integer number of *half*-periods of the sin functions fit into the range $[0, L]$, equation (7.21) is a ***half-range Fourier series***, rather than the usual Fourier series, recapped in section 7.1.1. The expression for B_n is easily obtained in exactly the same way as for regular Fourier series (which was shown in 7.1.1), noting that $\sin(n\pi x)$ (over $0 \le x \le L$) is also orthogonal. For completeness, orthogonality of $\sin(n\pi x)$ in half-range Fourier series and the origin of the factor $2/L$ will be shown in section 7.2.1.

Similarly, differentiating (7.20) w.r.t. t, and noting that

$$\frac{\partial u_n(x,t)}{\partial t} = \sin\left(\frac{n\pi x}{L}\right)\left[A_n \frac{n\pi c}{L}\cos\left(\frac{n\pi ct}{L}\right) - B_n \frac{n\pi c}{L}\sin\left(\frac{n\pi ct}{L}\right)\right]$$

and then using similar arguments that when $t = 0$ the sin terms vanish and the cos terms are unity, we find

$$\frac{\partial}{\partial t}u(x,t)\bigg|_{t=0} = \sum_{n=1}^{\infty} A_n \frac{n\pi c}{L}\sin\left(\frac{n\pi x}{L}\right) = v_0(x) \tag{7.23}$$

giving

$$A_n \frac{n\pi c}{L} = \frac{2}{L}\int_0^L v_0(x)\sin\left(\frac{n\pi x}{L}\right)dx \ . \tag{7.24}$$

So the constants are coefficients in a Fourier expansion of the initial displacement and velocity. Substituting A_n and B_n into the general solution (7.20) yields the particular solution that satisfies the initial conditions.

7.2.1 Orthogonality of half-range Fourier modes

From the Y1 Fourier course, you know that $\int_0^L \sin(2\pi n x/L)\sin(2\pi m x/L)dx = 0$ when $m \neq n$ (where $n, m \in \mathbb{Z}$). Here integer numbers of whole periods fit in the domain. But what about $\int_0^L \sin(\pi n x/L)\sin(\pi m x/L)dx$, where we have integer number of half-periods? It turns out that they are orthogonal too, which can be shown using the same method as you used for standard $\sin(2\pi n x/L)$ Fourier modes. Consider

$$I = \int_0^L \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx \ .$$

Using the trig identity $2\sin\theta\sin\phi = \cos(\theta-\phi) - \cos(\theta+\phi)$ the above integral becomes

$$\begin{aligned} I &= \frac{1}{2}\int_0^L \cos\left(\frac{(n-m)\pi x}{L}\right)dx - \frac{1}{2}\int_0^L \cos\left(\frac{(n+m)\pi x}{L}\right)dx \\ &= I_{n-m} + I_{n+m} \ . \end{aligned}$$

For $m \neq n$

$$\begin{aligned} I_{n-m} &= \frac{1}{2}\frac{L}{(n-m)\pi}\left[\sin\left(\frac{(n-m)\pi x}{L}\right)\right]_0^L \\ &= \frac{1}{2}\frac{L}{(n-m)\pi}\left[\sin(0) - \sin((n-m)\pi)\right] = \frac{1}{2}\frac{L}{(n-m)\pi}\left[0-0\right] = 0 \ . \end{aligned}$$

The second integral $I_{n+m} = 0$ too by the same arguments. For $m = n$

$$I = \frac{1}{2}\int_0^L 1\cdot dx - \frac{1}{2}\int_0^L \cos\left(\frac{2n\pi x}{L}\right)dx = \frac{L}{2} - 0 \ ,$$

where in the second integral we use the result that integrating over a whole number periods of cos (or sin) yields zero. Putting this all together, we have

$$\boxed{\frac{2}{L}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx = \delta_{nm}} \tag{7.25}$$

where δ_{nm} is the Kronecker delta function.

In fact, using the machinery of section 5.4, we could shortcut all this but noting again that the eigen-problem arising from the ODE for $X(x)$ and the homogenous boundary conditions is in Sturm-Liouville form. Thus the eigenfunctions are *guaranteed* to be orthogonal! Of course when $n = m$, one still has to do the integral to actually work out the normalisation constant, which turned out to be $2/L$. But even that can be shortcut by noting that for $m = n$ the integral is $I = \int_0^L \sin^2(n\pi x/L)dx$ and noticing that the average of $\sin^2$ (or $\cos^2$) over an integer number of half-periods is $\frac{1}{2}$ and then multiplying by the integration range L to get $L/2$.

7.3 Laplace's Equation in 2D

Let us now try to solve Laplace's equation in 2D:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \ . \tag{7.26}$$

We consider the rectangular spatial domain $0 \leq x \leq a$ and $0 \leq y \leq b$. We chose homogeneous boundary conditions at the boundaries for y; the solution is zero at the lower and upper boundaries. However, we apply ***inhomogeneous boundary conditions*** in x; the solution on the left boundary is still zero, however at the right boundary it is specified by a function $f(y)$. The system is depicted in figure 7.1. The BCs in full are;

- $u(x, y = 0) = 0 \ , \quad u(x, y = b) = 0 \qquad (0 \leq x \leq a)$
- $u(x = 0, y) = 0 \ , \quad u(x = a, y) = f(y) \qquad (0 \leq y \leq b)$

Specifying homogeneous BCs in x as well would simply yield $u(x, y) = 0$ throughout the interior! In that case, it would take sources (i.e. a non-zero right hand side in equation(7.26)) to get a non-trivial solution.

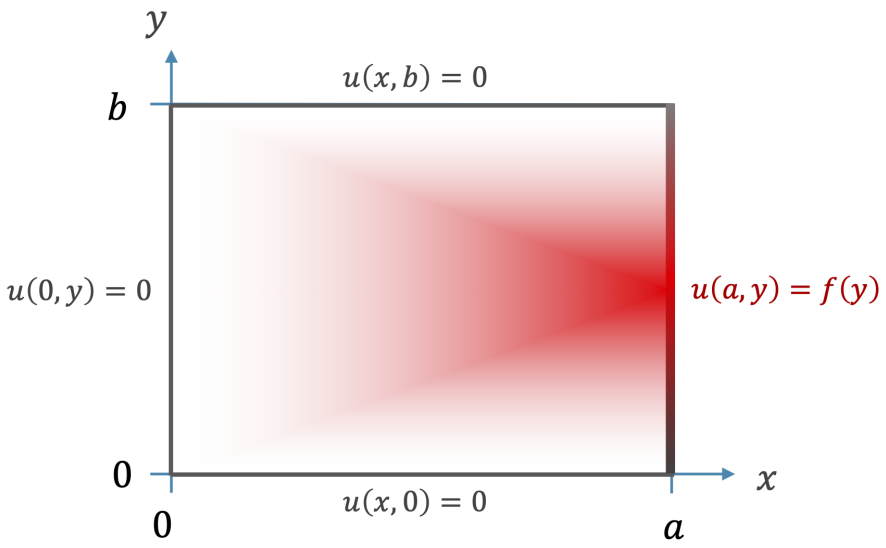


Figure 7.1: Rectangular domain used for Laplace's equation, showing the inhomogeneous BC on the right boundary. The colours illustrate what the solution could be like.

Physically, this problem could describe the ***steady state*** temperature distribution inside a block of material, when the boundaries are held at a fixed temperature. Alternatively, it applies to the electrostatic potential inside a 2D box when the left, lower and upper sides are grounded, and the right-hand side is an insulator with a distribution of surface charge.

Separation of Variables

We try separating variables, $u(x,y) = X(x)Y(y)$, to find

$$\frac{1}{Y}\frac{d^2Y}{dy^2} = -\frac{1}{X}\frac{d^2X}{dx^2} = -k^2 \tag{7.27}$$

where we have written the separation constant as $-k^2$. So the separated ODEs are

$$\frac{d^2Y}{dy^2} = -k^2Y \quad , \qquad \frac{d^2X}{dx^2} = +k^2X \ . \tag{7.28}$$

We choose to write the separation constant as $-k^2$, because we anticipate that the problem in the y direction – governed by the differential operator $\hat{L} = \frac{d^2}{dy^2}$ and subject to homogeneous BCs – will be the same eigenvalue problem that arose for $X(x)$ in the 1D wave equation example seen in 7.2. For now let us assume that $k^2 > 0$, but we should revisit this to check it makes sense.

In terms of the separated variables, the boundary conditions become

- $Y(0) = Y(b) = 0$
- $X(0) = 0$

Determination of the Normal Modes

The solution to $\frac{d^2Y}{dy^2} + k^2Y = 0$, along with $Y(0) = Y(b) = 0$, are the eigenfunctions,

$$Y_n(y) = \sin(k_n y) \quad , \qquad k = k_n = \frac{n\pi}{b} \ . \tag{7.29}$$

What happens if $k^2 = 0$? In that case the general solution for Y is $Y(y) = A + By$ and the only way to meet the homogeneous BCs in y is for $Y(y) = 0$. Then $u = XY = 0$ (no matter what X is) and the BC at $x = a$ cannot be met. The same logic applies for $k^2 < 0$. Let $\gamma^2 = -k^2$, so that $\gamma^2 > 0$. Then the general solution is $Y(y) = Ae^{\gamma y} + Be^{-\gamma y}$, which requires $B = -A$ to meet the lower BC, but then $Y(b) = 2A\sinh(\gamma b)$ which requires $A = 0$.

Back to considering the case $k^2 > 0$: It follows that the general solution for $X(x)$ is

$$X(x) \ = \ Ce^{kx} + De^{-kx} \tag{7.30}$$

and the left BC implies

$$X(0) = C + D = 0 \qquad \Longrightarrow \qquad D = -C \qquad \Longrightarrow \qquad X(x) = 2D\sinh(kx)$$

so that

$$X_n(x) \ = \ \sinh(k_n x) \tag{7.31}$$

where the parameter k is the eigenvalue determined by the eigenvalue problem in y, hence we denote it as k_n. Thus the x part of the problem is linked to the eigenvalue problem in y, and we denote this by placing a subscript n on X. Putting it all together yields the normal modes

$$\boxed{u_n(x,y) \ = \ \sinh(k_n x)\sin(k_n y) \quad , \qquad k_n = \frac{n\pi}{b}} \tag{7.32}$$

i.e. the eigenmodes of the problem. Note that we drop the factor of $2D$ since the coefficient c_n in the general solution serves as the arbitrary constant. Figure 7.2 depicts the normal mode with $n = 3$. You can see that it indeed conforms to the homogeneous boundary conditions on the lower, left and upper sides of the domain.

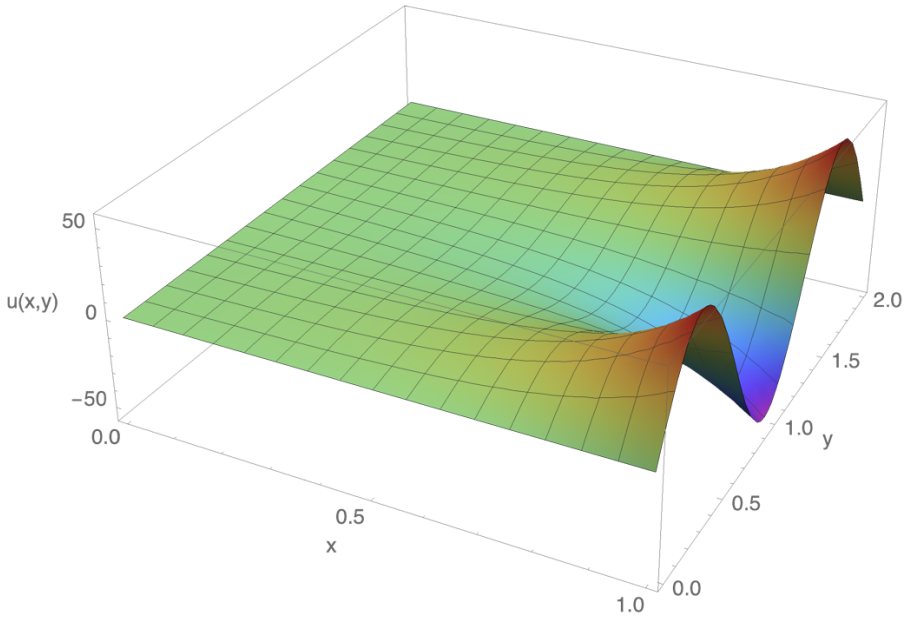


Figure 7.2: Normal mode for $n = 3$, in the case $a = 1$, $b = 2$.

General solution

The general solution is the superposition of the separable solutions (normal modes), each with arbitrary amplitude c_n;

$$u(x,y) = \sum_{n=1}^{\infty} c_n u_n(x,y) = \sum_{n=1}^{\infty} c_n \sinh\left(\frac{n\pi x}{b}\right)\sin\left(\frac{n\pi y}{b}\right) .$$

All that remains is to determine the mode amplitudes c_n.

Particular solution

This is where the inhomogeneous boundary condition $u(a,y) = f(y)$ comes into play. Since the problem is in Cartesian coordinates, the mode coefficients c_n are found by Fourier analysis. They are related to the ***half-range Fourier (sine) series*** coefficients of $f(y)$. [2] By setting the general solution at $x = a$ equal to $f(y)$, multiplying by the basis function $\sin(k_m y)$, integrating over y and using the orthogonality of the basis functions, it is possible to show that

$$\boxed{c_m = \frac{1}{\sinh(k_m a)}\frac{2}{b}\int_0^b f(y)\sin(k_m y)\,dy \ .} \tag{7.33}$$

[2] The half-range Fourier (sine) series is where the basis functions are $\sin(n\pi x/L)$, so that an integer number n of half-wavelengths fit into the 'half' range $0 \le x \le L$. This is what we also used in section 7.2.

Dealing with Additional Inhomogeneous BCs

Now imagine that we want to find the solution to $\nabla^2 U(x,y) = 0$ for the composite inhomogeneous BCs formed by $U(x=a,y) = f(y)$ (the previous one) and $U(x,y=b) = g(x)$ (the new, additional one). The BC is still $U = 0$ along the lower and left sides of the domain. How do we find the solution?

This can be done by finding the solution to the new inhomogeneous BC ***in isolation***, and then adding this on to our previous solution. We need to solve $\nabla^2 v = 0$ subject to the new BCs

- $v(x,0) = 0$, $v(x,b) = g(x)$,
- $v(0,y) = 0$, and now $v(a,y) = 0$.

It is **crucial** that the previous BC at $x = a$ is removed and $v(a,y) = 0$ is used instead. This means that we now apply homogeneous BCs on the x boundaries. It has the effect of switching the eigenvalue problem that we previously had in y to the x direction instead. Thus, the normal modes for this (new) part of the composite problem can be inferred by simply interchanging x and y in equation (7.32) and replacing b with a (i.e. the eigenmodes are determined by the x boundaries which are a distance a apart):

$$v_n(x,y) = \sinh(l_n y)\sin(l_n x) \quad , \qquad l_n = \frac{n\pi}{a} \ . \tag{7.34}$$

Thus the solution to the new inhomogeneous BC (in isolation) is then

$$v(x,y) = \sum_{n=1}^{\infty} d_n v_n(x,y) \qquad \text{with} \qquad d_n = \frac{1}{\sinh(l_n b)} \frac{2}{a} \int_0^a g(x)\sin(l_n x)\, dx \ . \tag{7.35}$$

The solution to the full problem is then

$$\boxed{U(x,y) = u(x,y) + v(x,y)} \tag{7.36}$$

The new part $v(x,y)$ vanishes on $x = a$, so does not interfere with the previous part $u(x,y)$ meeting the right-hand inhomogeneous BC. Similarly, the old part $u(x,y)$ vanishes on $y = b$, so does not interfere with the new part $v(x,y)$ meeting the upper inhomogeneous BC. Thus $\nabla^2 U = 0$ (which is guaranteed since $\nabla^2 = \frac{d^2}{dx^2} + \frac{d^2}{dy^2}$ is a linear differential operator, and $\nabla^2 u = 0$ and $\nabla^2 v = 0$) and

- $U(x,0) = 0$, $U(x,b) = g(x)$,
- $U(0,y) = 0$, $U(a,y) = f(y)$

as required to solve the composite problem.

In principle, we could extend this further to deal with inhomogeneous BCs on every side of the domain!

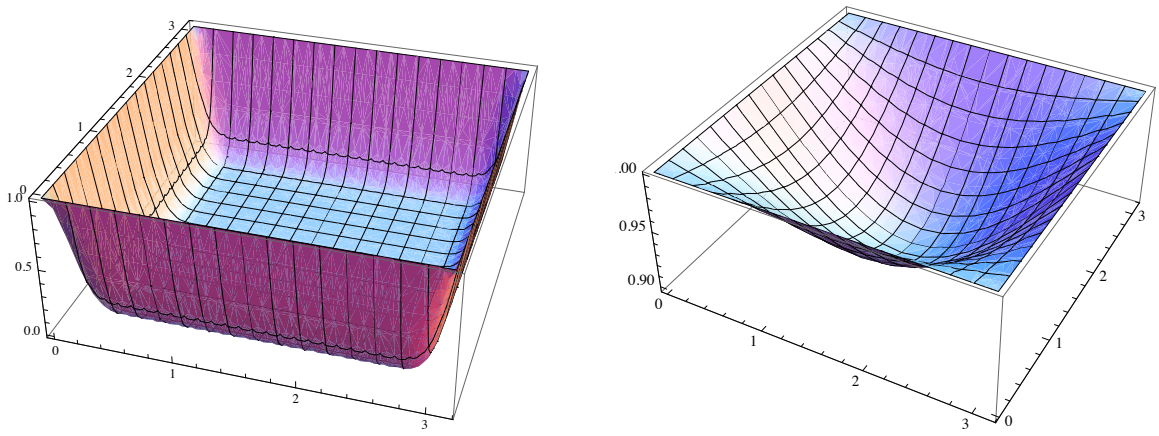


Figure 7.3: Temperature at an early time $t = 0.01$ (left) and a late time $t = 1$ (right), for $T_0 = 1$, $\kappa = 1$ and $L = \pi$.

7.4 Diffusion Equation in 2D

As another example of solving equations via separation of variables, let us consider a more complicated situation, where we have

- 2 space dimensions (and time),
- inhomogeneous spatial BCs.

Consider an infinite square column of side L which is at zero temperature. At $t = 0$ it is immersed in a heat bath at fixed temperature T_0. This means that initially $u(x, y, t = 0) = 0$ on the interior $0 < x < L$ and $0 < y < L$, and $u(x, y, t \geq 0) = T_0$ on the boundaries $x = 0$, $x = L$, $y = 0$ and $y = L$. The heat bath clamps the surface temperature to T_0 and intuitively, it will take time for this temperature to diffuse into the interior. We ignore z as by symmetry there is no heat flowing along this direction. Note that these BCs are inhomogeneous. Note also that, at $t = 0$, the value of u an infinitesimal distance inside the boundary is zero.

We need to solve the heat equation

$$\nabla^2 u = \frac{1}{\kappa}\frac{\partial u}{\partial t} \ . \tag{7.37}$$

Figure 7.3 shows the solution (7.49) to the problem at an early time and a later time. The boundary temperature diffuses in and brings the internal temperature up to T_0. The centre takes the longest to equilibrate. Below is how we arrive at this solution.

Important: First we note that $u =$ constant is a solution to this, so we can always add a constant to any solution we find by separation of variables. So let us now make $U = u - T_0$, to render the boundary condition for $U(x, y)$ homogeneous. Initially, $U(x, y, 0) = -T_0$ *inside* and $U(x, y, t \geq 0) = 0$ on the four boundaries ($x = 0$, $x = L$, $y = 0$ and $y = L$). We will solve for $U(x, y, t)$ and then add T_0 to get u.

Let us look for separable solutions,

$$U(x, y, t) = X(x)Y(y)T(t) \tag{7.38}$$

giving

$$YT\frac{d^2X}{dx^2} + XT\frac{d^2Y}{dy^2} - \frac{XY}{\kappa}\frac{dT}{dt} = 0$$

Dividing by XYT,

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} - \frac{1}{\kappa T}\frac{dT}{dt} = 0. \tag{7.39}$$

First separation

Important: Now we have to argue quite carefully. The first *two* terms *together* are not functions of t, and the third term is not a function of x or y, so they are both equal to a constant:

$$\boxed{\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} = \frac{1}{\kappa T}\frac{dT}{dt} = \text{constant} = -s^2 \ .} \tag{7.40}$$

Note that we *cannot* say

$$\frac{1}{X}\frac{d^2X}{dx^2} = \frac{1}{Y}\frac{d^2Y}{dy^2} = \frac{1}{\kappa T}\frac{dT}{dt} = \text{constant} \qquad \textbf{WRONG!} \tag{7.41}$$

which looks tempting, but is *wrong*. **Make sure you understand the logic here.** The logic given leads us to equation (7.40), but there is no way to go further to (7.41): It simply does not follow. Instead, we first find the solution for T and then afterward explicitly separate X and Y in a second stage of separation.

Proceeding

$$\frac{1}{\kappa T}\frac{dT}{dt} = -s^2$$

for some constant s, from which we find

$$T(t) \propto e^{-s^2\kappa t} \ .$$

Second separation

We next find the equation for X, by going back to Eq. (7.40) and separating the x and y dependence. Since

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} = -s^2.$$

we *can* now do a second stage of separation, with a second separation constant:

$$\boxed{\frac{1}{X}\frac{d^2X}{dx^2} = -\frac{1}{Y}\frac{d^2Y}{dy^2} - s^2 = -k_x^2 = \text{constant}} \tag{7.42}$$

The LHS is not a function of y, the Y term is not a function of x, hence they must be equal to another constant. So we introduce a **second separation constant**. (We will justify that it is negative later.)

$$\frac{d^2X}{dx^2} = -k_x^2 X \qquad \Rightarrow \qquad X(x) = Ae^{ik_x x} + Be^{-ik_x x}$$

and similarly for Y:

$$\frac{d^2Y}{dy^2} = -(s^2 - k_x^2)Y \qquad \Rightarrow \qquad Y(y) = Ce^{ik_y y} + De^{-ik_y y}$$

where

$$k_y^2 \equiv s^2 - k_x^2 \ . \tag{7.43}$$

These terms we calculate here must be zero on the boundaries at $x = \{0, L\}$, $y = \{0, L\}$. Hence the solutions for X and Y must be sinusoidal, with the correct period

$$k_x = \frac{m\pi}{L} \quad , \qquad k_y = \frac{n\pi}{L}$$

where m and n are integers. Thus

$$X(x) \propto \sin\left(\frac{m\pi x}{L}\right) \quad , \qquad Y(y) \propto \sin\left(\frac{n\pi y}{L}\right) \ . \tag{7.44}$$

Note that to make u zero on the boundary, we must have k_x^2 and k_y^2 positive, otherwise k_x is imaginary and $X(x)$ will be a exponentially growing/decaying function that does not satisfy the BCs (and similarly for k_y.) This justifies writing the separation constants as we did.

So a separable solution (which is the normal modes) is

$$\boxed{U_{mn}(x,y,t) = \sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi y}{L}\right)e^{-s_{mn}^2 \kappa t}} \tag{7.45}$$

where

$$s_{mn}^2 = \frac{\pi^2}{L^2}(m^2 + n^2) \ . \tag{7.46}$$

This comes from Equation 7.43, which reads

$$-\left(\frac{m^2\pi^2}{L^2}\right) - \left(\frac{n^2\pi^2}{L^2}\right) + s_{mn}^2 = 0 \ .$$

General solution

Now we can add the separable solutions (normal modes),

$$U(x,y,t) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} C_{mn} U_{mn}(x,y,t) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} C_{mn} \sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi y}{L}\right)e^{-s_{mn}^2 \kappa t} \ . \tag{7.47}$$

All that remains is to determine the mode amplitudes C_{mn}.

Particular solution

We use the initial condition that inside the volume $u = 0$ when $t = 0$ (when the exponential term is unity). So for $U(x,y,0) = U_0(x,y)$ we have $U_0(x,y) = -T_0$ for $0 < x < L$ and $0 < y < L$. However, as noted above, right on the boundary $x = 0, L$, $y = 0, L$ we have $U_0 = 0$. Therefore (changing to index variables p and q)

$$\sum_{p,q=1}^{\infty} C_{qp} \sin\left(\frac{p\pi x}{L}\right)\sin\left(\frac{q\pi y}{L}\right) = -T_0 \ .$$

But note that this looks very much like a Fourier Series, and we can use the same trick of the orthogonality of the sin functions. Multiply by $\sin(m\pi x/L)$ and integrate with respect to x, giving 0 for $p \neq m$,

and $L/2$ if $p = m$. Similarly for y. Therefore

$$\begin{aligned} C_{mn}\left(\frac{L}{2}\right)^2 &= -T_0 \int_0^L \sin(m\pi x/L)dx \int_0^L \sin(n\pi y/L)dy \\ &= -T_0 \left[-\frac{L}{m\pi}\cos\left(\frac{m\pi x}{L}\right)\right]_0^L \left[-\frac{L}{n\pi}\cos\left(\frac{n\pi y}{L}\right)\right]_0^L . \end{aligned} \tag{7.48}$$

The cosines are zero if m or n are even. If m, n are both odd, the right hand side is $4L^2/(mn\pi^2)$, from which we get

$$C_{mn} = \begin{cases} -16T_0/(\pi^2 mn) & m, n \quad \text{both odd} \\ 0 & \text{otherwise} \end{cases} . \tag{7.49}$$

Finally remembering $u = T_0 + U$ the particular solution is

$$u(x, y, t) = T_0 \left\{ 1 - \frac{16}{\pi^2} \sum_{m,n \text{ odd}} \frac{1}{mn} \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi y}{L}\right) \exp\left[-(m^2 + n^2)\frac{\kappa\pi^2}{L^2} t\right] \right\} . \tag{7.50}$$

Chapter 8

Solving PDEs with Fourier Methods

The ***Fourier transform*** (FT) is one example of an ***integral transform***, a general technique for solving differential equations. Fourier methods can be especially useful if the coefficients in the equation are constants.

You already saw an example of Fourier transforms applied to solving a PDE in the Fourier Maths course. In the preceding chapter on separation of variables, Fourier analysis arose when we formed the general solution from normal modes and when the initial conditions were used to find the (eigenvalue/wavenumber dependent) constants in the GS to arrive at the particular solution. (See sections 7.2, 7.3 and 7.4.) This chapter sets out to formalise this.

The PDE will be Fourier transformed from the outset, in contrast to the separation of variables approach where it was effectively used later on. Transformation of a PDE (e.g. from position x to wave number k) often leads to simpler equations (algebraic or ODE typically) for the Fourier transform of the unknown function. This is because spatial derivatives turn into factors of ik. Fourier transforming with respect to x is especially useful if the coefficients in the equation are independent of x. These simpler equations are then solved in k space, and the answer is transformed back to x to give the required solution.

It should also be noted that the FT method works best for infinite systems. Applying it to a finite system essentially reduces the problem into a Fourier series of modes with discrete wavelengths (as seen in chapter 7), rather than k being continuous.

Note: For a PDE which depends on t and x, we can choose to Fourier transform from x to k, resulting in equations which are still functions of t, or we can transform from t to ω, or we can transform both. Actually, when dealing with PDEs that are initial value problems in time and boundary value problems in space (as is very common), it is often better to perform a *Laplace transform* in time than a Fourier transform in the time variable.[1] (Not covered in this course.)

The definitions of the Fourier transform and inverse Fourier transform that will be used are, respectively,

$$\boxed{\mathcal{F}[f(x)] = \int_{-\infty}^{\infty} f(x)e^{-ikx}\,dx = \tilde{f}(k)\ ,} \tag{8.1}$$

$$\boxed{\mathcal{F}^{-1}[\tilde{f}(k)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{f}(k)e^{+ikx}\,dk = f(x)\ .} \tag{8.2}$$

[1] The Laplace transform uses the transform kernel $\exp(-st)$ rather than $\exp(i\omega t)$, so that $\mathcal{L}[f(t)] = \int_0^\infty f(t)\exp(-st)dt = \tilde{f}(s)$. This lower limit of 0 is better suited to dealing with situations where there is no knowledge of the system, or $f(t) = 0$, for $t < 0$.

This is one of many definitions used. [2] The two functions $f(x)$ and $\tilde{f}(k)$ form a transform pair and can be considered as different representations of the same thing. To acknowledge this, we can write

$$f(x) \rightleftharpoons \tilde{f}(k) \ . \tag{8.3}$$

This course will focus on applying Fourier transforms in one spatial dimension. However, be aware that the technique works just as well and elegantly in 2 or 3 dimensions. Then a function of $\mathbf{r}$ transforms into a function of $\mathbf{k}$, e.g., $f(x, y, z) \rightleftharpoons \tilde{f}(k_x, k_y, k_z)$.

Fourier transform of a derivative

As we shall see, FTs are very useful in solving PDEs. This is because the FT of a derivative of a function is very similar to the FT of the original function:

$$\begin{aligned} \mathcal{F}\left[\frac{df}{dx}\right] &= \int_{-\infty}^{\infty} \frac{df(x)}{dx} e^{-ikx}\, dx \\ &= \left[e^{-ikx} f(x)\right]_{-\infty}^{\infty} + ik \int_{-\infty}^{\infty} f(x) e^{-ikx}\, dx \ . \end{aligned}$$

We have done this integral by parts, setting $u = e^{-ikx}$ and $dv/dx = f'$. Now, we *require*

$$f(x) \to 0 \quad \text{as} \quad |x| \to \infty$$

so that the first term vanishes. For any physical problem, it is likely that $f(x)$ will vanish at $\pm\infty$ otherwise, for instance, the wave function would not be normalisable, or there would be infinite (thermal) energy or mass in the system.

The remaining term is proportional to the FT of the original function $f(x)$, so that

$$\boxed{\mathcal{F}\left[\frac{df}{dx}\right] = (ik)\tilde{f}(k) \ ,} \tag{8.4}$$

and for higher-order derivatives, repeated integration by parts yields

$$\boxed{\mathcal{F}\left[\frac{d^p f}{dx^p}\right] = (ik)^p \tilde{f}(k) \ .} \tag{8.5}$$

[2] There are other conventions. Sometimes $\mathcal{F}$ and $\mathcal{F}^{-1}$ each have a factor of $\frac{1}{\sqrt{2\pi}}$. Sometimes, the kernel $e^{\pm 2\pi ikx}$ is chosen, in which case there is no normalisation factor in the definitions of $\mathcal{F}$ and $\mathcal{F}$ (i.e. $1/2\pi$ is absent).

8.1 Diffusion Equation in 1D

As an example, we will consider the (linear) diffusion equation for an infinite system

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} \qquad (-\infty < x < \infty\,,\;\; t \geq 0)$$

with initial condition

$$u(x, t = 0) = u_0(x)$$

and solve it using Fourier transform methods. The diffusion coefficient D is assumed to be independent of position. This is important; otherwise, the FT method is difficult to work with.

Step 1: FT the PDE to convert to an ODE

This will give an ODE, in terms of time, that describes the evolution of each Fourier mode $\exp(+ikx)$. To Fourier transform the PDE, we do

$$\mathcal{F}\left[\frac{1}{D}\frac{\partial u}{\partial t}\right] = \mathcal{F}\left[\frac{\partial^2 u}{\partial x^2}\right] .$$

On the LHS, the time derivative can be pulled outside the integral over x that performs the FT. On the RHS, we use (8.5) with $p = 2$. This yields

$$\frac{1}{D}\frac{\partial \tilde{u}(k,t)}{\partial t} = (ik)^2 \tilde{u}(k,t) \;\; .$$

This is true for each value of k (remember k is a continuous variable). Fortunately, in this case, the evolution of $\tilde{u}(k,t)$ at a particular value of k does not depend on $\tilde{u}$ at a different value of k. In other words, the Fourier modes are uncoupled and evolve independently.[3] So we can regard k as a parameter and $\frac{\partial}{\partial t}$ behaves the same as $\frac{d}{dt}$, which yields the **ODE**

$$\frac{d\ln|\tilde{u}|}{dt} = -k^2 D \;\; , \tag{8.6}$$

where we have noted that $d(\ln|\tilde{u}|) = d\tilde{u}/\tilde{u}$.

Step 2: Solve for the temporal evolution of each mode

The solution to the above ODE is $\ln|\tilde{u}(k,t)| = -k^2 Dt + \text{constant}$. Note that the constant can be different for different values of k.

$$\tilde{u}(k,t) = \tilde{u}_0(k)\, e^{-Dk^2 t} \;\; , \tag{8.7}$$

where $\tilde{u}_0(k) \equiv \tilde{u}(k, t = 0)$ is the initial 'spectrum' of Fourier modes, to be determined by the initial conditions. The interpretation of equation (8.7) is that diffusion erodes the spectrum at a rate that increases with wave number.[4]

[3] Uncoupled modes occur when the coefficients are constants, e.g., D in this example. It isn't true if they are functions of x. FTs could still be useful, but it involves more complicated equations (such as convolutions).

[4] Note that equation (8.6) tells us that the $|\tilde{u}(k,t)|$ decays exponentially to zero. Negative values of $\tilde{u}$ stay negative while they decay; positive values stay positive. Equation (8.7) does this regardless of the sign of $\tilde{u}$. Therefore, it is valid for positive and negative $\tilde{u}$.

Step 3: Apply the initial conditions to find the particular solution

The ***general solution*** is the superposition of the Fourier modes $\exp(+ikx)$. It is the inverse Fourier transform of the (time-dependent) spectrum;

$$u(x,t) = \mathcal{F}^{-1}\left[\tilde{u}(k,t)\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{u}_0(k)\, e^{ikx - Dk^2 t}\, dk \ . \tag{8.8}$$

We get the initial spectrum by FT of the initial profile $n_0(x) \equiv n(x, t=0)$:

$$\tilde{u}_0(k) = \mathcal{F}\left[u_0(x)\right] \ . \tag{8.9}$$

Example — A Gaussian

A drop of ink falls into an infinitely large water tank (assumed 1-dimensional) at $x = 0$, $t = 0$. We want to find the density of ink as a function of space and time, i.e., determine the solution $u(x,t)$. Initially, all the ink (S particles) is concentrated at one point:

$$u(x, t=0) = S\,\delta(x) \ ,$$

where $\delta(x)$ is the Dirac delta function. Using the 'sifting property' of the Dirac delta function[5], we can find the corresponding initial spectrum of Fourier modes:

$$\begin{aligned} \tilde{u}_0(k) &= \mathcal{F}\left[u(x,t=0)\right] = \int_{-\infty}^{\infty} S\delta(x)\, e^{-ikx}\, dx \\ &= S \ . \end{aligned}$$

Inserting this into the general solution (8.8) we get:

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S\, e^{ikx - Dk^2 t}\, dk \ . \tag{8.10}$$

To proceed, we consider the argument of the exponential function in the integrand, and complete the square:

$$\begin{aligned} \alpha &= -Dk^2 t + ikx = \left(-Dk^2 t + ikx + \frac{x^2}{4Dt}\right) - \frac{x^2}{4Dt} \\ &= -\left(k\sqrt{Dt} - i\frac{x}{2\sqrt{Dt}}\right)^2 - \frac{x^2}{4Dt} \\ &= -z^2 - \frac{x^2}{4Dt} \ . \end{aligned}$$

Since we have defined $z \equiv k\sqrt{Dt} - i\frac{x}{2\sqrt{Dt}}$, $dk = dz/\sqrt{Dt}$ and equation (8.10) becomes

$$u(x,t) = \frac{1}{2\pi}\frac{S}{\sqrt{Dt}}\exp\left(-\frac{x^2}{4Dt}\right)\int_{-\infty}^{\infty} e^{-z^2}\, dz \ .$$

[5] Sifting property: $\int_a^b \delta(x - x_0) f(x) dx = f(x_0)$ if x_0 lies in the range (a, b)

Using the standard result for the 'Gaussian integral', $\int_{-\infty}^{\infty} e^{-z^2}\,dz = \sqrt{\pi}$, results in

$$\therefore \qquad \boxed{u(x,t) = \frac{S}{\sqrt{2\pi}\sqrt{2Dt}} \exp\left(-\frac{x^2}{4Dt}\right)\ .} \tag{8.11}$$

Comparing this with the usual expression for a Gaussian,

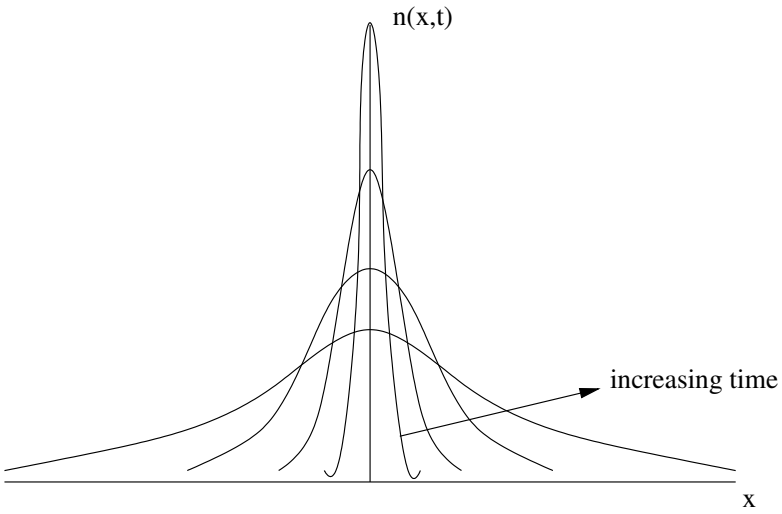


Figure 8.1: Variation of concentration with distance x at various diffusion times. (The turnups are a mistake in the plotting package).

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \tag{8.12}$$

we identify the width σ with $\sqrt{2Dt}$. So, the ink spreads out with concentration described by a normalised Gaussian centred on the origin with width $\sigma = \sqrt{2Dt}$. The solution is sketched for various t in Fig. 8.1.

The important features of the solution (8.11) are:

- Normalised: there are always S particles in total at every value of t.
- Centred on the origin: where we placed the initial drop.
- Width $\sigma = \sqrt{2Dt}$: gets broader as time increases
  - $\sigma \propto \sqrt{t}$: characteristic of random walk ("stochastic") process.
  - $\sigma \propto \sqrt{D}$: if we increase the diffusion constant D, the ink spreads out more quickly.

Equation (8.11), the response to an initial impulse/spike, is known as a ***Green's Function***. The Green's function is a general concept that is very useful for solving inhomogeneous PDEs (i.e. ones with driving terms) and will be considered later in chapter 10.

8.2 Wave Equation in 1D – D'Alembert's Solution

One is used to thinking of solutions to the wave equation being sinusoidal, but they don't have to be. We can use Fourier Transforms to show this rather elegantly, applying a partial FT ($x \to k$, but keeping t as is).

The wave equation is

$$c^2 \frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial^2 u(x,t)}{\partial t^2}$$

where c is the wave speed. We Fourier Transform w.r.t. x to get $\tilde{u}(k,t)$ (note the arguments), remembering that the FT of $\partial^2/\partial x^2$ is $-k^2$:

$$-c^2 k^2 \tilde{u}(k,t) = \frac{\partial^2 \tilde{u}(k,t)}{\partial t^2} \quad .$$

This is a harmonic equation for $\tilde{u}(k,t)$, with solution

$$\tilde{u}(k,t) = Ae^{ikct} + Be^{-ikct}$$

A crucial point is that each Fourier mode, e^{ikx}, has its own constants, thus A and B are functions of k. (This is analogous to each of the eigenmodes y_n possessing its own arbitrary constant c_n in the general solution, as seen in chapters 5 and 7.) Therefore, let us write A and B as arbitrary functions as $\tilde{f}(k)$ and $\tilde{g}(k)$, respectively, so that

$$\tilde{u}(k,t) = \tilde{f}(k)e^{ikct} + \tilde{g}(k)e^{-ikct} \quad . \tag{8.13}$$

We now invert the transform, to give

$$\begin{aligned} u(x,t) &= \mathcal{F}^{-1}[\,\tilde{u}(k,t)\,] \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{f}(k)e^{ik(x+ct)}\,dk + \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{g}(k)e^{ik(x-ct)}\,dk \quad . \end{aligned}$$

Hence

$$\boxed{u(x,t) = f(x+ct) + g(x-ct)} \tag{8.14}$$

and f and g are arbitrary functions, i.e., anything that satisfies the Dirichlet conditions. This is ***D'Alembert's solution*** of the wave equation.[6] This follows since by inspection of the complex exponential 'kernel'. In the first integral, it serves to transform the independent variable from k to the combination $x+ct$, and the second integral from k to $x-ct$. Physically, $f(x+ct)$ represents an arbitrary wave disturbance (i.e. a pulse) that propagates steadily at speed c in the negative x-direction, preserving its shape. Similarly, $g(x-ct)$ represents a pulse that moves in the positive x-direction at constant speed c.

Another way to understand D'Alembert's solution is as follows. The 'oscillating' $\exp(\pm ikct)$ factors in equation (8.13) change the phase of the associated plane wave $\exp(ikx)$, which shifts the peaks at the phase speed $v = \omega/k = (ck)/k = c$. All modes have the same phase speed (for this dispersionless case), so the pulse that they combine to form also shifts with them, retaining its shape.

Note that generally, $f(x+ct)$, etc., will not be separable in x and t. For instance, a Gaussian shape $f(x+ct) = \exp[-(x+ct)^2]$ cannot be factorised into parts in only x and t. Even a pure travelling mode

$$f(x+ct) = \cos(x+ct) = \cos(x)\cos(ct) - \sin(x)\sin(ct)$$

[6] Technically, it is the general solution before the initial conditions $u(x, t=0)$ and $\partial u/\partial t\big|_{t=0}$ are specified.

is not separable but rather the superposition of two separable solutions. One of the few functions that is separable is the complex exponential!

8.3 Fourier Methods in 2D and 3D

Although this course focuses on performing Fourier transforms in one spatial dimension, it will be useful to learn more about FT in other dimensions for future reference. For the 2D finite problem seen in section 7.4 the solution involved a 2D Fourier series. The normal modes, given in equation (7.45) had both x and y dependence going as $\sin(m\pi x/L)\sin(n\pi y/L)$. The general solution, eqn (7.47), involved a double sum; one over n (the y dependence) and one over m (i.e. x). The 2D Fourier transform involves a double integral, one over x, involving basis functions $\exp(ik_x x)$ and one over y involving basis functions $\exp(ik_y y)$. (The basis function in integral transforms is also referred to the kernel (of the integral transform)). The basis functions are multiplied so that one can write the combined kernel for 2D or 3D Fourier transforms as $\exp(i\mathbf{k}\cdot\mathbf{r})$ where $\mathbf{r}$ is the position vector and $\mathbf{k}$ is the wave vector. So, the set of basis functions involved in 2D or 3D FTs are all possible plane waves; one of every wavelength $2\pi/|\mathbf{k}|$ and every direction $\hat{\mathbf{k}}$.

For future reference, the Fourier transform and inverse FT are

$$\tilde{f}(\mathbf{k}) = \mathcal{F}[f(\mathbf{r})] = \int_{-\infty}^{\infty} f(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}}\, d\mathbf{r} \tag{8.15}$$

$$f(\mathbf{r}) = \mathcal{F}^{-1}[f(\mathbf{r})] = \frac{1}{(2\pi)^d} \int_{-\infty}^{\infty} \tilde{f}(\mathbf{k}) e^{+i\mathbf{k}\cdot\mathbf{r}}\, d\mathbf{k} \ , \tag{8.16}$$

where the power $d = \{1,2,3\}$ selects the number of spatial dimensions and $\int_{-\infty}^{\infty} \ldots d\mathbf{r} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \ldots dx\, dy$ in 2D and $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \ldots dx\, dy\, dz$ in 3D. Notation: $f(\mathbf{r})$ means $f(x,y,z)$, and $\tilde{f}(\mathbf{k})$ means $\tilde{f}(k_x,k_y,k_z)$.

For *separable functions*, e.g.,

$$f(\mathbf{r}) = u(x)v(y)w(z) \ , \tag{8.17}$$

the multi-dimensional FT factorizes into a 1D Fourier transform in x, y and z, e.g.,

$$\mathcal{F}[f(\mathbf{r})] = \mathcal{F}_x[u(x)]\,\mathcal{F}_y[v(y)]\,\mathcal{F}_z[w(z)] \ , \tag{8.18}$$

where $\mathcal{F}_y[\ldots] = \int_{-\infty}^{\infty} \ldots e^{ik_y y}\, dy$, etc.

Chapter 9

The Laplacian in Spherical and Cylindrical Polar Coordinates

In this chapter, we consider how to deal with the operator ∇^2 in spherical-polar and cylindrical polar coordinate systems. We will limit ourselves to considering Laplace's equation

$$\nabla^2 u(\mathbf{r}) = 0 \ .$$

However, $\nabla^2 u$ also appears in the wave equation (a hyperbolic PDE) and diffusion equation (a parabolic PDE). The techniques about to be seen are quite general and apply to such PDEs too.

The first section deals with spherical-polar coordinate systems. It will draw together many elements from the course; separation of variables, ODEs that are solved using power series methods (though we will just quote the result here), Legendre-polynomials and orthogonal sets of eigenfunctions. We will first separate the Laplacian in this coordinate system and then obtain the separable solutions (i.e. normal modes, or eigenfunctions). A key result will be ***spherical harmonics*** which are basis functions over the angles θ, ϕ and are prevalent throughout physics: angular momentum electron wavefunction eigenstates in hydrogen; dipole, quadrupole, multipole EM fields arising from charge distributions; oscillation modes of the sun, etc. We then proceed to work out the electric field of a spherical conducting sphere in an initially uniform electric field.

The next section does much the same but for cylindrical coordinates. We will see how Bessel's equation arises during separation of variables in this case. It turns out that the general solution involves a sum of Bessel functions in r. An appreciation of how this can be used to satisfy homogeneous boundary conditions, will be given. (Spoiler: the expansion in terms of Bessel functions J_p is *not* in their order p, but involves the zeros of $J_0(x)$.)

9.1 Spherical Coordinates

If the symmetry of the problem suggests the use of spherical coordinates, then we seek separable solutions for $u(r, \theta, \phi)$, where θ is the angle from the z-axis (i.e. the polar axis) and ϕ is the angle in the x-y plane, as illustrated in figure 9.1.

We first have to write ∇^2 in spherical polar coordinates. In this coordinate system, Laplace's equation is

$$\nabla^2 u = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial u}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 u}{\partial \phi^2} = 0 \ . \tag{9.1}$$

(*You do not need to remember this equation.*)

Try separable solutions, $u(r,\theta,\phi) = R(r)\Theta(\theta)\Phi(\phi)$. Substitute into Laplace's equation:

$$\frac{\Theta\Phi}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{R\Phi}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{R\Theta}{r^2\sin^2\theta}\frac{d^2\Phi}{d\phi^2} = 0 \ .$$

As usual, divide by $R\Theta\Phi$; also multiply by r^2:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2} = 0 \ .$$

We also change variable from θ to $\mu = \cos\theta$ for convenience. This transforms the derivatives as follows; $\frac{d}{d\theta} = \frac{d\mu}{d\theta}\frac{d}{d\mu} = -\sin\theta\frac{d}{d\mu}$. The result of $\theta \to \mu$ is then:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{\Theta}\frac{d}{d\mu}\left[(1-\mu^2)\frac{d\Theta}{d\mu}\right] + \frac{1}{\Phi(1-\mu^2)}\frac{d^2\Phi}{d\phi^2} = 0 \ . \tag{9.2}$$

First separation – Let us separate out the simplest term first; the one in ϕ:

$$\boxed{\frac{(1-\mu^2)}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{(1-\mu^2)}{\Theta}\frac{d}{d\mu}\left[(1-\mu^2)\frac{d\Theta}{d\mu}\right] = -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = \text{constant} = m^2} \tag{9.3}$$

where we have introduced a first separation constant m^2. The logic behind this is: The LHS is not a function of ϕ and the Φ term is not a function of r,θ.

The **boundary condition** on Φ is $\Phi(0) = \Phi(2\pi)$, i.e. periodic, which requires trig (or complex exponential) solutions. These require $m^2 \geq 0$.

$$\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = -m^2 \qquad \Rightarrow \qquad \Phi(\phi) = Ae^{im\phi} + Be^{-im\phi} \ .$$

$\Phi(0) = \Phi(2\pi)$ requires $m = 0, 1, 2, \ldots$. Actually, either of these solutions gives rise to a separable solution to the equation so we can split them up, and write the possible solutions for Φ as

$$\boxed{\Phi(\phi) = e^{im\phi} \quad , \qquad m = 0, \pm1, \pm2, \ldots} \tag{9.4}$$

i.e. m = any integer. We have not bothered to normalise Φ. Note that for $m = 0$, $\Phi(\phi) = a\phi + b$, where a, b are constants. However, this does not fit the periodic BCs unless $a = 0$.

Second separation – To get Θ, we divide Eq. (9.3) by $(1-\mu^2)$, and separate r and θ variables:

$$-\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) = \frac{1}{\Theta}\frac{d}{d\mu}\left[(1-\mu^2)\frac{d\Theta}{d\mu}\right] - \frac{m^2}{1-\mu^2} = -k \tag{9.5}$$

where we introduce a second separation constant $-k$. Hence Θ satisfies the ***associated Legendre equation***:

$$\frac{d}{d\mu}\left[(1-\mu^2)\frac{d\Theta}{d\mu}\right] + \left(k - \frac{m^2}{1-\mu^2}\right)\Theta = 0 \ . \tag{9.6}$$

Compare with Legendre's equation seen in section 4.4. Series expansion shows that these have finite

solutions only for

$$k = \ell(\ell+1) \tag{9.7}$$

for ℓ = positive integer . (We proved this earlier in section 4.4 for $m = 0$, but not in the general case.) The solutions are ***associated Legendre polynomials***:

$$\boxed{\Theta(\theta) = P_\ell^m(\mu)\ .} \tag{9.8}$$

Important: Note that m here is an index that together with ℓ labels the polynomial. (It is not a power!) Nonsingular solutions also require

$$0 \le |m| \le \ell \tag{9.9}$$

for convergence. [1] (Not proven here.)

The first few $P_\ell^m(x)$ are

$$\begin{aligned} &P_0^0(x) = 1 \\ &P_1^0(x) = x\ , && P_1^1(x) = (1-x^2)^{1/2} \\ &P_2^0(x) = \frac{1}{2}(3x^2-1)\ , && P_2^1(x) = 3x(1-x^2)^{1/2}\ , \qquad P_2^2(x) = 3(1-x^2) \end{aligned} \tag{9.10}$$

Note that $m = 0$ yields the 'ordinary' Legendre polynomials; $P_\ell^0(x) = P_\ell(x)$. Also note the triangular pattern of the above table, continues for $\ell > 2$. Remembering that $x = \mu = \cos\theta$, we see that the $\sqrt{1-x^2}$ factors appearing for $m > 0$ are simply $\sin\theta$.

For R,

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) = \ell(\ell+1)R\ . \tag{9.11}$$

Expanding out the derivative, one gets $r^2R'' + 2rR' - \ell(\ell+1)R = 0$ which has the form of Euler's differential equation (see (2.17) in section 2.1.4). Thus we try $R \propto r^\alpha$. (There are alternative solution strategies. [2]) Then $\alpha(\alpha+1) = \ell(\ell+1)$, with solutions $\alpha = \ell$ and $\alpha = -(\ell+1)$, so

$$\boxed{R(r) = Ar^\ell + Br^{-(\ell+1)}\ .} \tag{9.12}$$

If the range of the solution includes $r = 0$, we would set $B = 0$ to avoid divergences.

Hence the separable solutions of Laplace's equation, i.e., the normal modes, are

$$\boxed{u_{\ell m}(r,\theta,\phi) = \left[A_{\ell m}r^\ell + B_{\ell m}r^{-(\ell+1)}\right]P_\ell^m(\cos\theta)e^{im\phi} \qquad (\ell = 0,1,2,\dots\ ,\ -\ell \le m \le \ell)\ .} \tag{9.13}$$

Note that $A_{\ell m}$ and $B_{\ell m}$ are coefficients; each combination of ℓ, m has its own pair of these constants. The general solution is then

$$u(r,\theta,\phi) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} u_{\ell m}(r,\theta,\phi)\ . \tag{9.14}$$

Note that the expansion coefficients have been built into the normal modes here.

[1] There is actually another set of solutions, Q_ℓ^m, but these diverge at $\mu = \pm 1$, so we do not consider them.

[2] Alternatively, let $x \equiv \ln r$ and solve the simple resulting equation, or, if you prefer, use the Frobenius method, where $c_0 \neq 0$ gives the solution given for $n = 0$, and you can show that $c_n = 0$ for $n > 0$; this is not the quickest way.

9.1.1 Spherical Harmonics

Spherical harmonics are the angular part of the normal modes (9.13) of the Laplacian operator, in spherical polar coordinates. They are

$$\boxed{Y_\ell^m(\theta,\phi) = (-1)^m \left[\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}\right]^{1/2} P_\ell^m(\cos\theta) e^{im\phi}} \tag{9.15}$$

for $\ell = 0, 1, 2, \ldots$ and $-\ell \le m \le \ell$. The complicated factor serves to normalise them.
(*You do not need to memorise the details of this formula.*)
They are shown in Fig. 9.1. They are eigenfunctions of the angular part of the ∇^2 differential operator. They form a complete set of basis functions that can be used to expand any function over the full solid angle. (This is utilised in (9.14).) They obey the orthogonormality relation

$$\boxed{\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} [Y_\ell^m(\theta,\phi)]^* \, Y_{\ell'}^{m'}(\theta,\phi)\sin\theta \, d\theta d\phi = \delta_{\ell\ell'}\delta_{mm'} \ .} \tag{9.16}$$

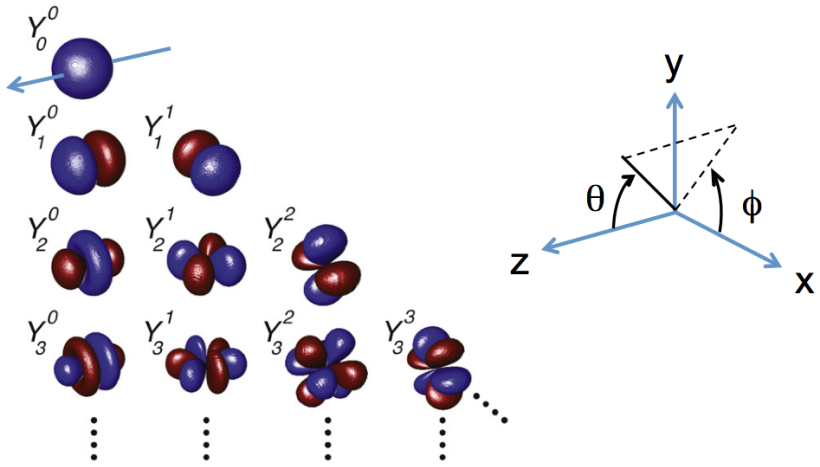


Figure 9.1: Polar plot of spherical harmonics $Y_{\ell m}(\theta,\phi)$. The rows depict increasing l, then columns increasing m. Note that $\mathcal{R}e\{Y_l^m\}$ is plotted. Blue and red correspond to positive and negative values of $\mathcal{R}e\{Y_l^m\}$, respectively. (Adapted from Tzoufras, et al., J. Comput. Phys **230** (2011).)

9.1.2 Example: Conducting Sphere in an Applied E-field

Consider an uncharged conducting sphere of radius a, placed in an initially uniform electric field $\mathbf{E} = \mathbf{E}_0$. The charges can flow freely on the surface and will reconfigure to ensure that the E-field is normal to the surface. We wish to determine the electrical potential $u(r,\theta,\phi)$ for $r \ge a$. Fig. 9.2 shows the electric field $\mathbf{E} = -\nabla u$ corresponding to the potential solution, when the sphere is present. This solution is obtained below.

There are no charges outside the sphere, so Laplace's equation holds there. Let the initial field *before*

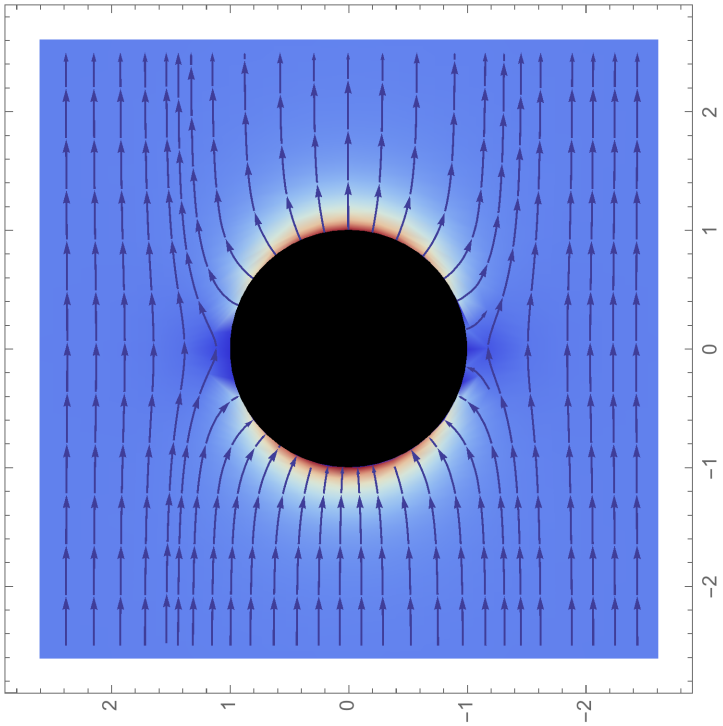


Figure 9.2: Electric field lines of an uncharged sphere placed in an initially uniform electric field.

the sphere is inserted be in the direction of the polar axis (z):

$$u_0 = -Ez = -Er\cos\theta \tag{9.17}$$

so that $\mathbf{E}_0 = -\nabla u_0$ has uniform magnitude and points in the polar direction $\hat{\mathbf{z}}$ everywhere (in the absence of the sphere).

Boundary conditions: These are

$$\begin{aligned} u &\to -Er\cos\theta && (r\to\infty)\\ u &= \text{constant} && (r=a) \end{aligned} \tag{9.18}$$

The first is that the solution is such that $\mathbf{E} = -\nabla u$ approaches the unperturbed, applied field far from the sphere. The second is that the potential is constant on the surface of the sphere.

Symmetry: To solve the problem, we note that it is axisymmetric (i.e. no ϕ dependence), so we will have $m = 0$, and the associated Legendre polynomials are then the ordinary Legendre polynomials, $P_\ell(\mu)$. The general axisymmetric solution is then [from (9.14)]

$$u(r,\theta,\phi) = u(r,\theta) = \sum_{\ell=0}^{\infty}\left[A_\ell\, r^\ell + B_\ell\, r^{-(\ell+1)}\right]P_\ell(\cos\theta). \tag{9.19}$$

Using the BC for large r allows us to compute the A_ℓ coefficients. The B_ℓ terms disappear at large r, since they are proportional to $r^{-(\ell+1)}$, so we cannot determine these directly from the large-r behaviour. To find the A_ℓ, we remind ourselves that $P_0(x) = 1$, $P_1(x) = x$ and so $\cos\theta = P_1(\cos\theta)$. Hence at large r the BC looks like

$$u_0 = -Er\,P_1(\cos\theta)$$

in terms of Legendre polynomials. Therefore only the $\ell = 1$ mode exists

$$u \to A_\ell r^\ell P_\ell(\cos\theta) \qquad \text{with } \ell = 1 \text{ only} \qquad (\text{for } r \to \infty) \tag{9.20}$$

Matching with the preceding equation, we see that $\boxed{A_1 = -E}$ and all other $A_\ell = 0$. Note that formally, we could find A_ℓ using the orthogonality of the spherical harmonics,[3] i.e.,

$$A_\ell = \frac{2\pi\, h_\ell^0}{r^\ell} \int_{\theta=0}^{\pi} Y_\ell^0(\theta) u_0(r,\theta,\phi) \sin\theta \, d\theta \ ,$$

where $m = 0$ has been used, in which case $\int \ldots d\phi$ just yields 2π, and $h_\ell^m = (-1)^m[\ldots]^{1/2}$ is the normalisation constant in the definition of spherical harmonics (9.15). Indeed, this would need to be used for more complicated BCs (for large r).

To obtain B_ℓ we use the BC for the potential on the sphere. Since $u(a)$ is constant, there can be no θ dependence of the solution when $r = a$. However, the $A_1 r\cos\theta$ term just found above, does not fit this requirement. This can be fixed by adding in $B_1 r^{-2}\cos\theta$, in such a way that it cancels the $A_1 r\cos\theta$ 'component' precisely at $r = a$. Luckily, $B_1 r^{-2}\cos\theta$ will not interfere with satisfying the BC at $r \to \infty$. Since P_ℓ *does* have θ dependence except for $\ell = 0$, we need to suppress contributions from $\ell \neq 0$ precisely at $r = a$. This means that each coefficient of P_ℓ (evaluated at $r = a$) in the general solution, equation (9.19), must be zero for $\ell > 0$:

$$A_\ell\, a^\ell + B_\ell\, a^{-(\ell+1)} = 0 \qquad (\text{for } \ell > 0) \ . \tag{9.21}$$

Here we have used the fact that since the Legendre polynomials are orthogonal (and thus linearly independent), there is no way to add them up to cancel to zero for all θ. Since $A_\ell = 0$ for $\ell \neq 1$ (deduced previously from the BC at $r \to \infty$), we deduce from equation (9.21) that $B_\ell = 0$ for $\ell \neq 1$ and for $\ell = 1$, we find $B_1 = -a^3 A_1$ hence $\boxed{B_1 = Ea^3}$. $\ell = 0$ just gives a constant contribution to the potential, which we can ignore. Hence the final answer is

$$\boxed{u(r,\theta,\phi) = -Er\left(1 - \frac{a^3}{r^3}\right)\cos\theta \qquad (r \geq a) \ .} \tag{9.22}$$

Physically, the term in a^3/r^3 corresponds to the dipole E-field caused by the polarisation of the sphere in response to the applied field. The redistribution of charge on the surface of the sphere ($r = a$) results in a dipole moment.[4]

[3] Alternatively, with this axisymmetric problem just multiply by $P_\ell(\cos\theta)$ and use the Legendre polynomial orthogonality, eqn. (4.26).

[4] Conceptually, a less 'symmetrical' distribution of charge would result in a quadrupole correction to u, involving Y_2^m and falling off faster than $1/r^3$, and even higher order 'multipole' corrections involving Y_l^m for $l > 2$.

9.2 Cylindrical Symmetry

If the symmetry of the problem requires cylindrical polar coordinates, then we seek separable solutions for $u(r,\phi,z)$. The Laplacian in cylindrical polar coordinates has the form

$$\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} \ . \tag{9.23}$$

(*You do not need to remember this equation.*)

In a similar fashion to the preceding section, we try separable solutions, $u(r,\phi,z) = R(r)\Phi(\phi)Z(z)$. Substituting into Laplace's equation ($\nabla^2 u = 0$), expanding the outer r derivative, then dividing through by u yields

$$\frac{1}{R}\left[\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr}\right] + \frac{1}{r^2\Phi}\frac{d^2\Phi}{d\phi^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} = 0 \qquad .$$

First separation – Let us separate out the simplest term first; the one in z:

$$\boxed{h(r,\phi) = \frac{1}{Z}\frac{d^2Z}{dz^2} = \text{constant} = k^2 \ ,} \tag{9.24}$$

assuming $k^2 \geq 0$.[5]

$h(r,\phi)$ denotes the other terms in the Laplacian. The solution for $Z(z)$ is

$$Z(z) = Ae^{kz} + Be^{-kz} \ . \tag{9.25}$$

Note that one part of this general solution for Z diverges for $z \to \infty$ and the other for $z \to -\infty$.

Second separation – Equation (9.24) now becomes

$$\frac{1}{R}\left[\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr}\right] + \frac{1}{r^2\Phi}\frac{d^2\Phi}{d\phi^2} = -k^2$$

which can be separated into

$$-\frac{r^2}{R}\left[\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr}\right] - k^2r^2 = \frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = -m^2 \ , \tag{9.26}$$

where m here is the second separation constant. The ODE and BCs for Φ are the same as in the spherical case and the solutions are given again by $\Phi(\phi) = \exp(im\phi)$ for $m \in \mathbb{Z}$.

The ODE for $R(r)$ is then

$$r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} + (k^2r^2 - m^2)R = 0 \ . \tag{9.27}$$

Finally, changing to a scaled radial variable $x = kr$ yields Bessel's equation of order m

$$\boxed{x^2\frac{d^2R}{dx^2} + x\frac{dR}{dx} + (x^2 - m^2)R = 0 \ ,} \tag{9.28}$$

[5] A positive, rather than negative, separation constant is chosen in (9.24), in anticipation of the $u \to 0$ as $z \to \infty$ BC that will be used shortly.

with general solution

$$R(x) = CJ_m(x) + DY_m(x) \tag{9.29}$$

(seen before in section 4.6.4) since m is an integer. J_m and Y_m are Bessel functions of the first and second kind, of order m, respectively. (Also remember that $J_{-m}(x) = (-1)^m J_m(x)$ for integer m and similarly for $Y_m(x)$. Hence we know how to deal with, e.g., $J_{-2}(x)$.)

Separable solution – Putting everything together, the separable solution is

$$u_{mk}(r,\phi,z) = [CJ_m(kr) + DY_m(kr)]e^{im\phi}[Ae^{kz} + Be^{-kz}] \ . \tag{9.30}$$

Although this is complicated, parts will vanish when insisting the physically sensible condition that u_{mk} is finite in certain limits. For bounded solutions at $r = 0$, D must vanish since $Y_m(kr)$ diverges at $r = 0$. For a semi-infinite system in $0 \le z < \infty$, A must vanish. In this case, the separable solution becomes

$$u_{mk}(r,\phi,z) = J_m(kr)e^{im\phi}e^{-kz} \ . \tag{9.31}$$

As will be shown in the example below, application of homogeneous BCs in the r direction will set the eigenvalues k which will make this a normal mode.

9.2.1 Example: Semi-infinite rod

Consider a semi-infinite rod with $0 \le z < \infty$ and radius a, as depicted in figure 9.3. We look for the equilibrium temperature profile $u(r,\phi,z)$ inside the rod ($0 \le r \le a$) for boundary conditions

$$\begin{aligned} \lim_{z\to\infty} u(z) &\to 0 \\ u(r = a) &= 0 \\ u(r,\phi,z = 0) &= f(r) \end{aligned} \tag{9.32}$$

i.e. zero temperature on the curved surface and an arbitrary temperature that is only dependent on r (and not ϕ) at the end $z = 0$. The separable solution above, eqn. (9.31), is appropriate to this problem.

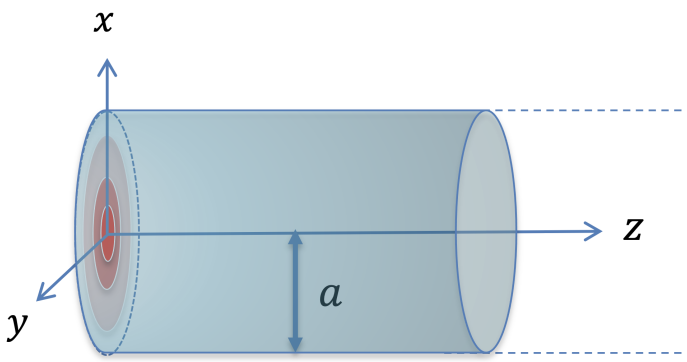


Figure 9.3: Illustration of the semi-infinite rod problem. A radial heating profile $f(r)$ is applied at the base $z = 0$. The surface at $r = a$ is held at zero value. The profile as $z \to \infty$ tends to zero.

Symmetry: Immediately, we deduce that $m = 0$ is the only azimuthmal mode permitted, given the rotational symmetry about the axis. Thus only $J_0(kr)$, the zeroth order Bessel function, is going to be present in the solution.

Homogeneous BC at $r = a$: The ***zeros of the Bessel functions*** will be key to solving this problem. Referring back to Fig. 4.3, we note that $J_0(x)$ goes to zero at specific points; call them x_n for $n = 1, 2, 3, \ldots$. The first few values are $x_n = \{2.40, 5.52, 8.65, \ldots\}$. They are not quite equidistant.

To ensure that at $J_0(kr) = 0$ at $r = a$, so that $u(a) = 0$ and satisfies the BC on the exterior surface, requires

$$ka = x_n \quad \Rightarrow \quad \boxed{k = x_n/a \equiv k_n \qquad (\text{for } n = 1, 2, \ldots) \ .} \tag{9.33}$$

Thus only discrete values of the k separation constant are allowed, in this case. Thus the *normal modes* are

$$\boxed{u_n(r, z) = J_0(k_n r) e^{-k_n z} \ .} \tag{9.34}$$

and the *general solution* for this problem is

$$\boxed{u(r, z) = \sum_{n=1}^{\infty} c_n J_0(k_n r) e^{-k_n z}} \tag{9.35}$$

which is a ***Bessel series expansion***. It turns out that Bessel functions that fall to zero at the boundary $r = a$ are orthogonal [6]

$$\boxed{\int_0^a J_0(k_n r) J_0(k_j r)\, r\, dr = \frac{a^2 [J_1(k_n a)]^2}{2}\, \delta_{nj} \ .} \tag{9.36}$$

This will not be rigorously proven here, but we note that it stems from the fact that Bessel's equation together with the homogeneous BC at $r = a$ forms a (singular) Sturm-Liouville problem (with weight function $\rho(r) = r$, or $\rho(x) = x$ in terms of x). The k_n values that ensure the BCs are met are the eigenvalues. The eigenfunctions $y_n(x)$ are $y_n(x) = J_0(k_n r)$. We know that for Sturm-Liouville problems, eigenfunctions with different eigenvalues are orthogonal.

(*You do <u>not</u> need to memorise details of equation (9.36), like the precise form of the RHS, but you should know in general terms how orthogonality of Bessel functions works and be able to apply it.*)

Finally, the coefficients c_n in the general solution can be evaluated using the orthogonality property (9.36) yielding

$$c_n = \frac{2}{a^2 [J_1(k_n a)]^2} \int_0^a J_0(k_n r) f(r)\, r\, dr \ . \tag{9.37}$$

Inserting these c_n (which can in principle be computed/calculated for a specific $f(r)$) into the general solution (9.35) means that $u(r, z)$ has been determined and the problem solved.

An extension of this to the non-axisymmetric case can be found in non-examinable appendix F.

[6] In fact (9.36) can be generalised to $J_0 \to J_p$ in the integral and $J_1 \to J_{p+1}$ in the factor infront, for $p \in \mathbb{R}$.

Chapter 10

Green's Functions for Solving Inhomogeneous Equations

Green's functions (GFs) are a way to solve inhomogeneous ODEs and PDEs, i.e., ones with a driving term. The method is developed for linear DEs, e.g.,

$$\hat{L}[y(x)] = f(x) \qquad \text{and} \qquad \hat{L}[u(\mathbf{r},t)] = h(\mathbf{r},t)$$

and results in a solution that is an integral over the driving term $f(x)$, or $h(\mathbf{r},t)$, multiplied by the Green's function $G(x,z)$ (or $G(\mathbf{r},t;\mathbf{x},\tau)$, etc.) So it gives an *integral solution*.

Green's functions are well-suited to dealing with boundary value problems. They can also be used for initial value problems. We will consider second (and first) order DEs, though the GF method works for higher orders too. Note that the boundary conditions (or initial conditions) are built into the GFs. The GF will vary for different BCs and ICs for a given DE.

Green's functions are physically intuitive. They represent the response of the DE to a ***impulse*** driving force $\delta(x-z)$ located at $x = z$ (or, e.g., $\delta(\mathbf{r}-\mathbf{z})\delta(t-\tau)$ located at $\mathbf{r} = \mathbf{x}$, occurring at $t = \tau$). In fact, you have already used GFs in your Electricity and Magnetism course, when calculating the electrical potential from an arrangement of point sources. The Green's function there is $1/r$ potential from the source (i.e. location of the charge $\mathbf{z}$) to the observation point (i.e. $\mathbf{x}$). This is the GF for Laplace's equation in 3D. Green's functions are known for other key physics equations – wave, diffusion – in 1D, 2D & 3D. They are given in non-examinable appendix I.

We will only consider driving terms that do not involve the function being solved for, though Green's functions can be used when, e.g., $f(x) = w(x)y(x)$ which results in an ***integral equation*** that can be solved using an iteration technique. This is beyond the scope of this course. This occurs in physics when using *perturbation theory* to solve for quantum mechanical scattering of a particle.

There are two different formulations of Green's functions. One where properties of GFs are defined and the GF determined using them. The other where GFs are determined as an eigenfunction expansion, in the case of self-adjoint problems (i.e. the BCs are pivotal). We will follow the former. The eigenfunction expansion form is given in non-examinable appendix H for completeness.

10.1 Properties of Green's Functions

If we have a second-order, linear, differential equation of the general form

$$\hat{L}[y(x)] = a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y(x) = f(x) \qquad (a \le x \le b) \tag{10.1}$$

then if we can solve for the complementary function (i.e. solve the equation with the RHS=0), the Green's function approach can give us the solution *for any* $f(x)$. The Green's function method changes a *differential* equation into an *integral* solution, which we assume we can evaluate, if necessary numerically.

How does this work? The *Green's function* $G(x,z)$ is defined to be the solution to the equation (10.1), but with the RHS set to a Dirac delta function, located at $x = z$:

$$a_2(x)\frac{d^2G}{dx^2} + a_1(x)\frac{dG}{dx} + a_0(x)G = \delta(x-z) \ . \tag{10.2}$$

The Green's function $G(x,z)$ is the *response at* x to an *impulse* at z. Once the Green's function is determined the solution to the the original inhomogeneous ODE, (10.1), can be found via

$$\boxed{y(x) = \int_a^b G(x,z)f(z)dz \ .} \tag{10.3}$$

This Green's function integral solution also works beyond 1D, by simply replacing $x \to \mathbf{x}$ and $z \to \mathbf{z}$ and integrating over all the independent variables. Why does it work? It works because we can write

$$f(x) = \int_a^b \delta(x-z)f(z)dz \tag{10.4}$$

in other words, a little loosely, the driving term is a sum of impulses, each with a weight $f(z)$, and since the equation is linear, we can simply add up the solutions for each δ function, each with a weight $f(z)$.

That the RHS of (10.3) really does yield the solution we seek, can be shown more rigorously as follows. Equation (10.2) has the form $\hat{L}[G(x,z)] = \delta(x-z)$. Multiplying through by $f(z)$ and taking it into the differential operator (which can be done since $\hat{L}$ involves derivatives w.r.t. x) we have

$$\hat{L}[f(z)G(x,z)] = f(z)\delta(x-z) \ .$$

Then integrating with respect to z, taking the integral inside the operator (which can be done since the operator and integral involve different variables, and the integral limits do not depend on x) and using the sifting property of the Dirac delta function, yields

$$\int \hat{L}[f(z)G(x,z)]\,dz = \int f(z)\delta(x-z)dz \qquad \Rightarrow \qquad \hat{L}[\int f(z)G(x,z)dz] = f(x) \ .$$

Comparing to (10.1), the inhomogeneous equation to be solved for $y(x)$, we identify the expression inside the differential operator with $y(x)$.

Note that the boundary conditions are important. For problems with homogeneous BCs, the GF solution – equation (10.3) – will satisfy them. For example, if $y = 0$ on the boundaries, then the Green's function will also be zero on the boundaries, by construction. Then any integral of G w.r.t. z will also be zero on the boundary and satisfy the conditions. Similar arguments hold for initial conditions of the type $y = 0$, $dy/dt = 0$ (at $t = 0$).

Defining properties of GFs

The properties below are general and apply to an nth order ODE. However, for the purposes of this course, the $n = 1$ and $n = 2$ cases will be of most interest.

1. The GF obeys the DE with the RHS a Dirac delta function, e.g.,

$$\hat{L}[G(x,z)] = \delta(x-z) \tag{10.5}$$

   where $\hat{L}$ here is an nth order linear differential operator, i.e. $\hat{L} = \sum_{j=0}^{n} a_j(x)\frac{d^j}{dx^j}$.

2. $G(x,z)$ satisfies the auxiliary conditions, e.g., BCs in x:

$$G(a,z) = y(a) = y_a \qquad \text{and} \qquad G(b,z) = y(b) = y_b \ . \tag{10.6}$$

3. Derivatives of the GF, $\frac{d^m G(x,z)}{dx^m}$, obey the following at $x = z$:

   - For $m = n$, the derivative at $x = z$ is infinite, as needed for the LHS to yield a Dirac delta function demanded by the RHS.
   - For $m = n-1$ the derivative is discontinuous. As $\epsilon \to 0$

$$\left.\frac{d^{n-1}G(x,z)}{dx^{n-1}}\right|_{x=z+\epsilon} - \left.\frac{d^{n-1}G(x,z)}{dx^{n-1}}\right|_{x=z-\epsilon} = \frac{1}{a_n(z)} \tag{10.7}$$

     (where $a_n(x)$ is the coefficient of the $d^n y/dx^n$ in the nth-order ODE).
   - For $m < n-1$ the derivative is ***continuous***. As $\epsilon \to 0$

$$\left.\frac{d^m G(x,z)}{dx^m}\right|_{x=z+\epsilon} - \left.\frac{d^m G(x,z)}{dx^m}\right|_{x=z-\epsilon} = 0 \ .$$

     This holds all the way down to $m = 0$.

4. GFs for self-adjoint problems possess symmetry:

$$G(x,z) = G^*(z,x) \ . \tag{10.8}$$

The above properties are expressed for Green's functions for 1D problems (i.e. problems with 1 independent variable). Properties 1 and 4 can be applied straightforwardly to multi-dimensional Green's functions with $x \to \mathbf{x}$ and $z \to \mathbf{z}$. However, properties 2 and 3 need some adaptation for 2D, 3D and even higher dimensions.

For $n = 2$, that we consider here, the first derivative dG/dx jumps by $1/a_2(z)$ whilst the zeroth derivative (i.e. G itself) is continuous at $x = z$. The finite jump of the first derivative yields an infinite value when differentiated once more; i.e. the second derivative is infinite.

Figure 10.1 shows an example of a 1D Green's function that will be derived shortly. You can see that it appears to be formed from two pieces that join up at $x = z$ but with different gradients as one approaches this point from either the left or the right.

To show the finite jump in the first derivative at $x = z$ for our second-order ODE, we integrate the equation between $z - \epsilon$ and $z + \epsilon$, and let $\epsilon \to 0$:

$$\int_{z-\epsilon}^{z+\epsilon} a_2(x)\frac{d^2G}{dx^2}dx + \int_{z-\epsilon}^{z+\epsilon} a_1(x)\frac{dG}{dx}dx + \int_{z-\epsilon}^{z+\epsilon} a_0(x)G(x)dx = \int_{z-\epsilon}^{z+\epsilon} \delta(x-z)dx \ . \tag{10.9}$$

If we assume that G and G' are finite over the integration range, then the second and third terms vanish as $\epsilon \to 0$. Noting that the RHS integrates to 1 and applying integration by parts to the first term on the LHS of (10.9) yields

$$\left.\frac{dG}{dx}\right|_{z+\epsilon} - \left.\frac{dG}{dx}\right|_{x=z-\epsilon} = \frac{1}{a_2(z)} \ . \tag{10.10}$$

Note that when integrating by parts, only the boundary term remains. The integrand of the integral term is finite, so this integral part vanishes when $\epsilon \to 0$. Importantly, note that the final result (10.10) is consistent with the assumptions made: It requires G' and G to be finite at and around $x = z$. A finite jump in G' at $x = z$ implies that G' must be finite either side of this point, right up to the point. If G' is finite in the vicinity of z, then G must be finite. The above procedure can also be applied to higher (or lower) order ODEs. Integrating the nth order ODE (10.5) over the range $z - \epsilon < x < z + \epsilon$ yields (10.7), the jump condition for $d^{n-1}G/dx^{n-1}$.

The proof of GF symmetry for self-adjoint problems is given in non-examinable appendix G, which also explores the interconnection between GFs and Wronskians.

10.2 Green's Functions for Boundary Value Problems

Let us consider a driven, 1D wave equation on a finite-length domain with homogenous boundary conditions, to illustrate the procedure for finding a GF and using it. It is driven at a single frequency ω.

The specification of the problem for $u(x,t)$ is

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} &= f(x,t) \\ u(0,t) &= u(\pi/2,t) = 0 \\ f(x,t) &= x\,\exp(i\omega t)\ . \end{aligned}$$

Let us remove the time dependence by assuming that $u(x,t) = \exp(i\omega t)X(x)$ so that $\partial^2 u/\partial t^2 = -\omega^2 u$. Then, using the phase speed relation, $c = \omega/k$, we get

$$\left[\frac{d^2}{dx^2} + k^2\right] X(x) = x\ . \tag{10.11}$$

With the RHS=0, this is known as ***Helmholtz's equation***. Hence, we have an inhomogeneous Helmholtz equation. There is a range of k that would satisfy this BVP, but for simplicity, let us set $k = 1$.

Finding the GF

The procedure is

1. Find the complementary function (CF) for the homogeneous equation (if you don't already know it).
2. The GF is formed from the CF in two pieces: one for $a \le x < z$ and one for $z < x \le b$.
3. Apply the boundary conditions. Apply the left BC(s) to the left 'piece' only and the right BC(s) to the right piece only.
4. Match the two pieces at $x = z$ using the continuity properties of GFs there. This will fix the z dependence of the GF.

From the properties of the Dirac delta function, except at $x = z$, the Green's function satisfies

$$\frac{d^2 G(x,z)}{dx^2} + G(x,z) = 0\ .$$

Hence away from $x = z$ the GF will have the form of the CF (with particular constants to fit the BCs). (Strictly, we might want to make this a partial derivative, at fixed z. I have written it this way so it looks like the equation for X).

Steps 1 & 2: Get CF and form GF from two pieces – Our example is a harmonic equation, with solution $X_{\rm CF}(x) = c_1 \sin x + c_2 \cos x$. The GF is formed from

$$G(x,z) = \begin{cases} A(z)\sin x + B(z)\cos x & (x < z) \\ C(z)\sin x + D(z)\cos x & (x > z)\ . \end{cases}$$

Notice that the constants depend on z (but of course not x).

Step 3: Apply BCs – The boundary condition $y = 0$ at $x = 0$ means that $B(z) = 0$, and $y = 0$ at $x = \pi/2$ implies that $C(z) = 0$. Hence

$$G(x,z) = \begin{cases} A(z)\sin x & (x < z) \\ D(z)\cos x & (x > z) \end{cases} .$$

Step 4: Match the pieces of GF at $x = z$ – Continuity of G and a discontinuity of 1 in its derivative at $x = z$ imply

$$\begin{aligned} &G \text{ continuity:} && A(z)\sin z - D(z)\cos z &&= 0 \\ &dG/dx \text{ discontinuity:} && -D(z)\sin z - A(z)\cos z &&= 1 . \end{aligned} \tag{10.12}$$

We have 2 equations in two unknowns, so we can eliminate A or D. Eliminating D

$$-A(z)\frac{\sin^2 z}{\cos z} - A(z)\cos z = 1 \qquad \Rightarrow \qquad A(z) = \frac{-\cos z}{\sin^2 z + \cos^2 z} = -\cos z$$

and consequently $D(z) = -\sin z$. Hence the Green's function is

$$\boxed{G(x,z) = \begin{cases} -\cos z \sin x & (x < z) \\ -\sin z \cos x & (x > z) \end{cases} .} \tag{10.13}$$

GF symmetry – The problem considered here, $\hat{L} = \frac{d^2}{dx^2} + k^2$ with homogeneous BCs, is a self-adjoint problem. Notice that the GF above is symmetrical in x and z. Interchanging x with z on the RHS, taking care to do this *both* in the expressions *and* the inequalities (i.e. $x < z \;\rightarrow\; z < x$), leaves the RHS unchanged. This is in accord with equation (10.8).

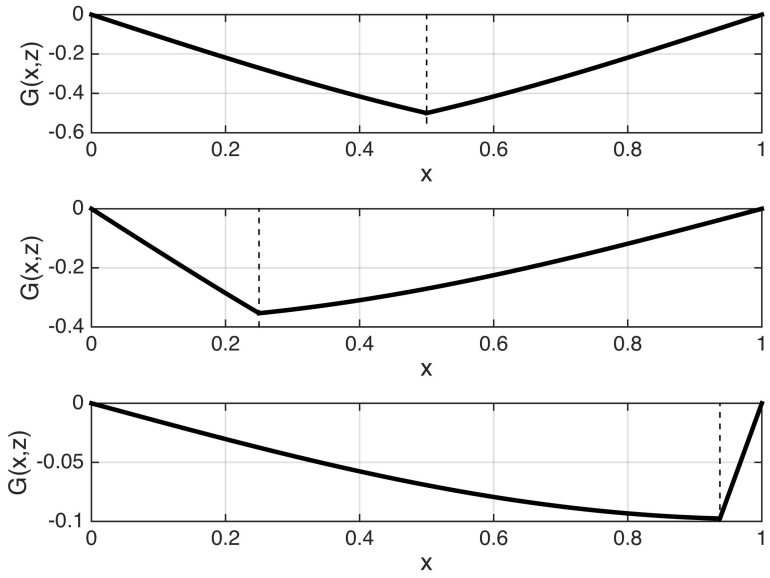


Figure 10.1: The Green's function $G(x,z)$ for the Helmholtz equation $X'' + X = 0$ with homogeneous BCs. (G solves $G'' + G = \delta(x - z)$, where $' = d/dx$.) Each of the three graphs has a different value of z, which is indicated by the vertical dashed line. The axis is scaled: $x/(\pi/2)$.

Figure 10.1 shows this Green's function plotted against x for three different values of z. You can see

that for these homogenous boundaries on a finite domain, the GF changes with z. But notice that G is always continuous at $x = z$. Also notice that it does look as if the gradient change across $x = z$ is the same in each case (as it should be).

Using the GF

We use equation (10.3) with $f(z) = z$, $y \to X$ and appropriate limits ($a = 0$, $b = \pi/2$). Since GFs have two pieces, the integral splits into two parts:

$$\begin{aligned} X(x) &= \int_0^{\pi/2} z\, G(x,z)\, dz = \int_0^x z\, G_{x>z}(x,z)\, dz + \int_x^{\pi/2} z\, G_{x<z}(x,z)\, dz \\ &= -\cos x \int_0^x z \sin z\, dz - \sin x \int_x^{\pi/2} z \cos z\, dz \ . \end{aligned} \tag{10.14}$$

Note: Two common mistakes to avoid: (1) Putting the wrong piece of G in each integral. Remember, we integrate w.r.t z and *not* x and thus in the $\int_0^x \ldots dz$ integral $x > z$, etc. (2) Inserting $f(x)$ (e.g. x in this case) by mistake instead of $f(z)$.

Finally, integrating by parts to evaluate the integrals

$$\begin{aligned} X(x) &= (x\cos x - \sin x)\cos x - \frac{1}{2}(\pi - 2\cos x - 2x \sin x)\sin x \\ &= x - \frac{\pi}{2}\sin x \ . \end{aligned}$$

Hence $u(x,t)$ goes as

$$u(x,t) = \exp(ikct)\left[x - \frac{\pi}{2}\sin x\right] \qquad \text{(with } k = 1) \ .$$

Infinite domain

We consider $-\infty < x < \infty$ and we assume that we only have outgoing waves – nothing is incoming from infinity. Helmholtz's equation, with arbitrary k and now using complex exponentials, has a CF that can be pieced together to form a GF, as follows

$$G(x,z) = \begin{cases} A(z)e^{ikx} + B(z)e^{-ikx} & (x < z) \\ C(z)e^{ikx} + D(z)e^{-ikx} & (x > z) \ . \end{cases}$$

The outgoing wave BCs mean that for $x < z$ we have no forward propagating wave $\exp[i(\omega t - kx)]$ hence $B = 0$. For $z < x$ we have no backwards propagating wave $\exp[i(\omega t + kx)]$ hence $C = 0$. It can readily be shown, in a similar fashion to above, that the Green's function is then

$$\boxed{G(x,z) = \frac{i}{2k}\exp(-ik|x-z|) \ .} \tag{10.15}$$

Notice that the modulus of the displacement between the source z and observation point x, i.e. $|x-z|$, is what splits up $G(x,z)$ into two pieces either size of $x = z$. Note that this has the form $G(x-z)$, i.e., x & z appear in the combination $x - z$. Thus

$$\begin{aligned} X(x) &= \int_{-\infty}^{+\infty} G(x,z) f(z) dz = \int_{-\infty}^{+\infty} G(x-z) f(z) dz \\ &= G(x) * f(x) \ , \end{aligned} \tag{10.16}$$

so that the solution is the ***convolution*** of the Green's function and driving term. This is an important feature of GFs on infinite domains, that is used frequently in Physics.

10.3 Green's Functions for Initial Value Problems

Green's function can also be used to solve initial value problems. The auxiliary conditions used to determine the GFs are now initial conditions (rather than BCs). We will not go through any examples here. (See the problem sheet for some.) But we just give the form of the GF for d/dt and d^2/dt^2 in isolation, as these operators are relevant to physics, i.e., appearing in diffusion, wave, equations, etc.

In both cases, we take the system to be at rest before the impulse $\delta(t-\tau)$ at $t=\tau$, so the initial conditions are:

$$y(t)=0 \quad \text{and} \quad \frac{dy}{dt}=0 \qquad (t<\tau)\ . \tag{10.17}$$

The impulsive 'drive' will affect $y(t)$ and dt/dy for $\tau < t$.

d/dt – The CF for $dy/dt=0$ is $y(t)=c$. Since this is a first order ODE, $n=1$ and $G(t,\tau)$ at $t=\tau$ is discontinuous, jumping by 1, according to (10.7). Satisfying the IC $y(t<\tau)=0$ and ensuring unity jump in G at $t=\tau$ results in

$$\boxed{\begin{aligned} G(t,\tau) &= \begin{cases} 0 & (t<\tau) \\ 1 & (t>\tau) \end{cases} \\ &= \Theta(t-\tau)\ , \end{aligned}} \tag{10.18}$$

which defines the ***Heaviside step function*** (or Heaviside unit function) $\Theta(t-\tau)$.

d^2/dt^2 – The CF for $d^2y/dt^2=0$ is $y(t)=at+b$. Satisfying the ICs $y=0$ and $dy/dt=0$ for $t<\tau$, and applying the two restrictions on G at $x=z$ results in

$$\boxed{\begin{aligned} G(t,\tau) &= \begin{cases} 0 & (t<\tau) \\ t-\tau & (t>\tau) \end{cases} \\ &= (t-\tau)\Theta(t-\tau)\ . \end{aligned}} \tag{10.19}$$

Both of the above Green's functions ensure ***causality***; the effect of the impulse applied at time τ manifests afterwards and not before!

Note that the Green's function of $\frac{d^2}{dt^2}+\frac{d}{dt}$ is *not* the sum of the Green's functions found above, since its complementary function is different to the sum of that for $\frac{d^2}{dt^2}$ alone and $\frac{d}{dt}$ alone.

10.4 Green's Functions in Multiple Dimensions

In this section, we look at ***fundamental Green's functions***, which are Green's functions for boundaries which are infinitely far away. Fundamental Green's functions typically tend to zero infinitely far from the source point $\mathbf{x}'$ (i.e. the location of the Dirac delta function on the RHS of the ODE) or outgoing wave boundary conditions are used. We saw an example of a fundamental Green's function, when considering the 1D wave equation on an infinite spatial domain (on page 99).

An important fundamental Green's function in three spatial dimensions will be derived. Other important fundamental GFs commonly occurring in physics problems are given in non-examinable appendix I for reference.

We then consider how to make Green's functions that can meet homogeneous spatial boundary conditions in more than one dimension, using the ***method of images***.

Whatever the problem is, once you have found the Green's function $G(\mathbf{x},\mathbf{z})$, the solution to the problem (i.e. $\hat{L}[u(\mathbf{x})] = f(\mathbf{x})$ with some BCs) is

$$\boxed{u(\mathbf{x}) = \int_V G(\mathbf{x},\mathbf{z}) f(\mathbf{z})\, d^3z \ .} \tag{10.20}$$

Before forging on, let us note that the Dirac delta function in 3D can be expressed as

$$\begin{aligned} \delta(\mathbf{x}-\mathbf{x}') &= \delta(x-x')\delta(y-y')\delta(z-z') \\ \text{or} \qquad &= \frac{\delta(r-r')}{r^2}\frac{\delta(\theta-\theta')}{\sin\theta}\delta(\phi-\phi') \end{aligned} \tag{10.21}$$

which emphasises that each and every one of the components of the observation point $\mathbf{x}$ must match that of the source point $\mathbf{x}'$. Also note that $\int_V \delta(\mathbf{x}-\mathbf{x}')d^3x = 1$ where the integral is over all space.

10.4.1 Laplace Operator in 3D

We will first derive the form of the fundamental Green's function for

$$\nabla^2 G(\mathbf{x},\mathbf{x}') = \delta(\mathbf{x}-\mathbf{x}') \tag{10.22}$$

in three spatial dimensions, subject to the boundary condition that $G \to 0$ infinitely far from the source point $\mathbf{x}'$. This yields the familiar $1/r$ "potential" from electrostatics. We will use the full machinery of Chapter 9.

With infinitely removed boundaries, the GF will be some function of displacement of the observation point from the source; $\mathbf{r} = \mathbf{x} - \mathbf{x}'$. (There is no other 'point of reference'!) Furthermore, for a point source in 3D, symmetry demands that the GF must be spherically symmetric about the source and only depend on $|\mathbf{r}| = r$. Thus we choose to put the source at the origin of the spherical polar coordinate system and seek $G(\mathbf{r},\mathbf{0}) = G(r,\theta,\phi)$. We already know the general solution to Laplace's equation in spherical polar coordinates; equation (9.14), which is a sum over normal modes

$$u_{\ell m}(r,\theta,\phi) = \left[A_{\ell m} r^{\ell} + B_{\ell m} r^{-(\ell+1)}\right] P_\ell^m(\cos\theta) e^{im\phi} \ .$$

Spherical symmetry implies $\ell = 0$ and $m = 0$ (which removes θ and ϕ dependence). Hence $G(r,\theta,\phi) = \sum_{\ell,m} u_{\ell m}(r,\theta,\phi)$ becomes $G(r,\theta,\phi) = u_{00}(r,\theta,\phi) = u_{00}(r)$ since all other modes are absent ($A_{\ell m} = 0$,

$B_{\ell m} = 0$, for $l > 0$). With no θ nor ϕ dependence, we therefore have

$$G(r) = A + \frac{B}{r} \ . \tag{10.23}$$

The next task is to determine the coefficients A and B. From the boundary conditions, we get $A = 0$ by demanding that $G \to 0$ as $r \to \infty$. We get B from the multi-dimensional analogue of the discontinuity/jump of the gradient of G at $x = z$ (seen previously for 1D, in section 10.1). To do this we integrate the PDE defining the GF for this problem, $\nabla^2 G(\mathbf{r}, \mathbf{0}) = \delta(\mathbf{r})$, over a finite volume V containing the origin (and therefore the delta function)

$$\int_V \nabla_r \cdot [\nabla_r G(\mathbf{r}, \mathbf{0})] \, d^3 r = \int_V \delta(\mathbf{r}) d^3 r \ , \tag{10.24}$$

where the r subscript on ∇_r emphasises that it is a derivative with respect to the first argument of the GF; $\mathbf{r}$. Also $\nabla^2 u = \nabla \cdot \nabla u$ has been used.

Then using Gauss's theorem, $\int_V \nabla \cdot \mathbf{A}(\mathbf{r}) \, d^3 r = \oint_S \mathbf{A} \cdot d\mathbf{s}$, results in

$$\oint_S \nabla_r G(\mathbf{r}, \mathbf{0}) \cdot d\mathbf{s} = 1 \ .$$

Due to the spherical symmetry, $\nabla_r G(\mathbf{r}, \mathbf{0}) = \hat{\mathbf{r}} \frac{\partial G(r)}{\partial r}$, and then choosing V to be a sphere of radius r yields

$$\frac{\partial G(r)}{\partial r} r^2 \oint_\Omega d\Omega = 1 \ .$$

Inserting (10.23) into this, remembering that $A = 0$ and noting that the integral over all solid angle $\oint_\Omega d\Omega = 4\pi$, yields

$$B = -\frac{1}{4\pi} \qquad \Rightarrow \qquad G(r) = -\frac{1}{4\pi r} \ . \tag{10.25}$$

Putting it back in vector form, allowing the location of the delta function to be arbitrary

$$\boxed{G(\mathbf{x}, \mathbf{x}') = -\frac{1}{4\pi} \frac{1}{|\mathbf{x} - \mathbf{x}'|} \ .} \tag{10.26}$$

This Green's function can be used to solve Poisson's equation

$$\nabla^2 u(\mathbf{x}) = f(\mathbf{x})$$

with BC $u \to 0$ at infinity, using

$$u(\mathbf{x}) \ = \ \int_V G(\mathbf{x}, \mathbf{x}') f(\mathbf{x}') \, d^3 x' \ = \ -\frac{1}{4\pi} \int_V \frac{f(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \, d^3 x' \ .$$

Fundamental Green's functions which commonly occur in physics are listed in non-examinable appendix I along with some points about them. These include fundamental GFs for the Laplace and Helmholtz equations in 2D and 3D, and the wave and diffusion equations in 1D & 3D.

10.4.2 Method of Images

The ***method of images*** is the method of choice for determining Green's functions for solving PDEs with homogeneous boundary conditions on finite (or semi-infinite) domains, especially when the boundary has a simple geometry. You have probably used it before in electrostatic problems, such as finding the electrical potential due to a point charge next to a grounded, conducting metal plate!

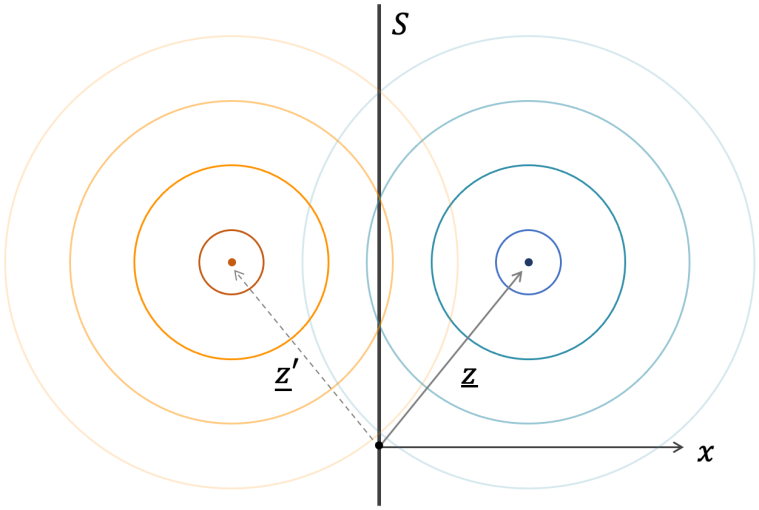


Figure 10.2: Method of images. The domain of the problem is $x \geq 0$, with homogeneous BC $u(0, y, z) = 0$. The Green's function for this problem is formed from a fundamental GF located at $\mathbf{z}$ inside the domain and an image GF located outside the domain at $\mathbf{z}'$.

If one wants to solve $\hat{L}[u(\mathbf{r})] = f(\mathbf{r})$ subject to the BC $u(\mathbf{r}) = 0$ on some finite boundary S, the procedure is to build a Green's function from the superposition of an appropriate fundamental Green's function [which will now be denoted as $F(\mathbf{x}, \mathbf{z})$] inside the domain of the problem, plus fundamental GFs located *outside* the domain. The latter can be throught of as images (or reflections) of the GF inside the domain. This is illustrated in figure 10.2.

Mathematically,

$$\boxed{G(\mathbf{x}, \mathbf{z}) = F(\mathbf{x}, \mathbf{z}) + H(\mathbf{x}, \mathbf{z})} \tag{10.27}$$

where $H(\mathbf{x}, \mathbf{z})$, a solution to the homogeneous PDE $\hat{L}[H] = 0$, represents the image GFs. Since the image GFs are located outside the domain, then $\hat{L}[H(\mathbf{x}, \mathbf{z})] = 0$ for all observation points $\mathbf{x}$ inside the domain. Thus, H serves to correct F so as to enforce the homogeneous BCs, but does not interfere with the singular behaviour of F. So

$$\begin{aligned} \hat{L}[G] &= \hat{L}[F] + \hat{L}[H] \\ &= \delta(\mathbf{x} - \mathbf{z}) + 0 \\ \therefore \quad \hat{L}[G] &= \delta(\mathbf{x} - \mathbf{z}) \end{aligned}$$

as required. H is

$$\boxed{H(\mathbf{x}, \mathbf{z}) = \sum_i \alpha_i F(\mathbf{x}, \mathbf{z}_i)} \tag{10.28}$$

where α_i and $\mathbf{z}_i$ are the amplitude and location of image GF i. These quantities depend on the location $\mathbf{z}$ of the GF inside the domain and on the nature of the BCs, and need to be determined.

More advanced aspects of GFs and BCs in multiple dimensions can be found in non-examinable appendix J for interested readers.

Example

For the 3D Poisson equation $\nabla^2 u(\mathbf{r}) = f(\mathbf{r})$ on $x \geq 0$ with $u(0, y, z) = 0$, the Green's function is

$$G(\mathbf{x}, \mathbf{z}) = -\frac{1}{4\pi|\mathbf{x} - \mathbf{z}|} + \frac{1}{4\pi|\mathbf{x} - \mathbf{z}_1|} \ , \tag{10.29}$$

where the first term on the RHS is the fundamental GF $F(\mathbf{x}, \mathbf{z})$ and the second term is the image GF $H(\mathbf{x}, \mathbf{z})$. If $\mathbf{z} = (x_0, y_0, z_0)$, then $\mathbf{z}_1 = (-x_0, y_0, z_0)$. Note that $\alpha_1 = -1$ here. The on $x = 0$

$$G = -\frac{1}{4\pi}\left(\left[x_o^2 + (y - y_0)^2 + (z - z_0^2)\right]^{-1/2} - \left[(-x_o)^2 + (y - y_0)^2 + (z - z_0^2)\right]^{-1/2}\right) = 0 \qquad \text{(as required)}$$

Appendix A

Exact first-order ODEs

Non-examinable

These encompass a wider range of equations than linear or separable equations. They have the form

$$\boxed{\frac{dy}{dx} = -\frac{f(x,y)}{g(x,y)} \qquad \text{where} \qquad \frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}} \tag{A.1}$$

which can be nonlinear. They can be re-expressed as a ***total differential***

$$f dx + g dy = du = 0 \tag{A.2}$$

where the problem then becomes finding $u(x,y)$ which is equal to a constant, since $du = 0$. This yields an implicit relationship between x, y and a single integration constant. This is the solution. The ***integrability condition***

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x} \tag{A.3}$$

guarantees that a function $u(x,y)$ satisfying $f dx + g dy = du$, does exist. You have already seen the machinery for finding total differentials in your year 1, Vector Calculus course. The only really new thing here is linking 1st-order ODEs to the differential expression $f dx + g dy = du = 0$.

For completeness, let us recap the key year 1 concepts and give an example. The total differential of a function $u(x,y)$ is

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \ . \tag{A.4}$$

If $f(x,y)dx + g(x,y)dy = du(x,y)$ we immediately can identify

$$f = \frac{\partial u}{\partial x} \quad , \qquad g = \frac{\partial u}{\partial y} \ . \tag{A.5}$$

Since $\partial^2 u/\partial x \partial y = \partial^2 u/\partial y \partial x$ must hold, the integrability condition (A.3) follows. Conversely, if we find that $\partial f/\partial y \neq \partial g/\partial x$ then there is no $u(x,y)$ function that can be found and our ODE (A.1) is not integrable[1].

[1] It might be possible to make it exact using an appropriate integrating factor. There is a prescription for attempting to find this. (It is not always possible.) It will not be covered in this course.

If our (rearranged) ODE is integrable, then we integrate up (A.5): [2]

$$f = \frac{\partial u}{\partial x} \quad \rightarrow \quad u(x,y) = \int^x f(X,y)dX + h(y) \tag{A.6}$$

$$\text{or} \quad g = \frac{\partial u}{\partial y} \quad \rightarrow \quad u(x,y) = \int^y g(x,Y)dY + j(x) \ . \tag{A.7}$$

Notice that because u depends on two variables and each integration of u is with respect to just one variable (x, say), we obtain an arbitrary function of the other variable (y, say) after the integration (e.g. $h(y)$). To determine $h(y)$ we differentiate equation (A.6) w.r.t. y, set the result equal to $g(x,y)$, rearrange for $\partial h/\partial y$ and then integrate back up w.r.t. y. If we can do the integrals, we get the desired closed form expression.

Example:

$$\frac{dy}{dx} = -\frac{x}{y} \quad \rightarrow \quad xdx + ydy = 0 \ .$$

We identify $f(x,y) = x$ and $g(x,y) = y$. They satisfy the integrability condition, i.e, $\partial f/\partial y = 0 = \partial g/\partial x$, so we carry on to determine $u(x,y)$. Using (A.6) we get

$$u = \int^x XdX + h(y) = \frac{x^2}{2} + h(y) \ .$$

Note that $h(y)$ may as well absorb the integration constant. Taking the partial y derivative;

$$\frac{\partial u}{\partial y} = 0 + \frac{\partial h}{\partial y} = g(x,y) = y \ .$$

Integrating w.r.t. y;

$$h = \int ydy = \frac{y^2}{2} + c_h \ .$$

But $u(x,y) = c_u$ (since $du = 0$). Putting it all together yields,

$$x^2 + y^2 = C \tag{A.8}$$

where $C = 2(c_u - c_h)$ is the overall single integration constant that we expect from a first-order ODE. The result is the equation of circle of radius $\sqrt{C}$, centred at the origin. Since the original ODE was separable, the solution could have been more easily obtained that way. But this technique is more general and applies to ODEs that are not separable.

Formally, putting all the machinery together yields

$$\boxed{\int^x f(X,y)dX + \int^y \left[g(Y) - \int^x \frac{\partial f(X,Y)}{\partial Y} dX \right] dY = c \ .} \tag{A.9}$$

A technical point is that no integration constant should be introduced after evaluating $\int^x \frac{\partial f(X,Y)}{\partial Y} dX$.
You are not expected to memorise this formula! It is better to know the procedure that it was obtained by.

[2] Take care: Capitalized variables X and Y denote dummy integration variables here. They should be distinguished from the independent variables x and y.

Appendix B

Other uses of the Wronskian

Non-examinable

For $y'' + p(x)y' + q(x)y = 0$ with continuous $p(x)$ and $q(x)$, the Wronskian is also given by

$$W(x) = c\exp\left(-\int^x p(z)dz\right) \tag{B.1}$$

where c is a constant. This was already revealed in section 3.6. It can also be obtained directly from the Wronskian determinant, formed from the independent solutions y_1 and y_2:

$$\begin{aligned} W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1' \\ \therefore \quad W' &= \cancel{y_1'y_2'} + y_1y_2'' - \left(\cancel{y_2'y_1'} + y_2y_1''\right) \ . \end{aligned}$$

Since y_1 and y_2 are solutions of the homogeneous equation, $y_1'' = -[py_1' + qy_1]$ and similarly for y_2. Substituting these into W' yields

$$\begin{aligned} \frac{dW}{dx} &= -\left[py_1y_2' + \cancel{qy_1y_2}\right] + \left[py_2y_1' + \cancel{qy_2y_1}\right] \\ \therefore \quad \frac{dW}{dx} &= -p(x)W \end{aligned}$$

which is a separable 1st-order ODE for $W(x)$, that can easily be solved to yield equation (B.1).

[QED]

The importance of equation (B.1) is that we do not even need to know what y_1 and y_2 are in order to evaluate the Wronskian (to within an arbitrary factor c), in constrast to the determinant form of the Wronskian! From (B.1) we can deduce that

$$W(x) = W(x_0)\exp\left(-\int_{x_0}^x p(z)dz\right)$$

by partitioning the integral into $\int^x = \int^{x_0} + \int_{x_0}^x$. Then because $\exp(\ldots) > 0$, for any real-valued argument (i.e. -ve, 0, +ve), it follows that determining whether W is zero or not at one point is sufficient to tell you whether it is zero or not across the whole interval where $p(x)$ and $q(x)$ are continuous. These insights are collectively known as ***Abel's theorem***.

Variation of Parameters can be used in the solution of linear 2nd-order ODEs with variable coefficients, where methods for constant coefficient ODEs do not work. It turns out that the results of the

variation of parameter method can succinctly be expressed in terms of Wronskians. As seen in section 3.6, for homogeneous equations, variation of parameters can be used to find the second solution from the first via

$$y_2(x) = u(x)y_1(x) \ .$$

For inhomogeneous equations, variation of parameters can also be used to find the PI when both basis functions y_1 and y_2 (forming the CF) are known, by assuming

$$y_{\text{PI}}(x) = u(x)y_1(x) + v(x)y_2(x) \ ,$$

substituting into the IE and then grinding through the maths to determine $u(x)$ and $v(x)$. The results of these procedures can be expressed in terms of the Wronskian:

$$y_2(x) = y_1(x) \int^x \frac{W(z)}{[y_1(z)]^2}\,dz \tag{B.2}$$

and (not proven here)

$$y_{\text{PI}}(x) = -y_1(x) \int^x \frac{y_2(z)g(z)}{W(z)}\,dz + y_2(x) \int^x \frac{y_1(s)g(s)}{W(s)}\,ds \ . \tag{B.3}$$

In fact, if the integration constant is included when evaluating each indefinite integral in equation (B.3), then the result includes $-c_1y_1 + c_2y_2$ and therefore includes the complementary function as well as the particular integral. Thus (B.3) gives the whole general solution.

Note that the determinant form of $W(x)$ should be used in (B.3), since the Wronskian obtained by (B.1) is only known to within a multiplicative constant. Unfortunately, multiplying a PI by a constant (other than 1) stops it solving the IE.

Appendix C

Gauss' test

Non-examinable

A more powerful and sensitive convergence test than the ratio test is ***Gauss' test***. It is usually suitable for determining if an infinite series converges or diverges at the limiting value $|x - x_0| = R$, i.e. right on the radius of convergence. It will work for all the series solutions that will be considered in this course!

Gauss' test – Let t_n be a sequence of ***positive*** numbers. If

$$\frac{t_n}{t_{n+1}} = 1 + \frac{\alpha}{n} + \frac{B_n}{n^2} \tag{C.1}$$

where α is a constant, and the sequence B_n is *bounded* (as $n \to \infty$), then

$$\sum_n t_n \;\to\; \begin{cases} \text{converges if} & \alpha > 1 \ , \\ \text{diverges if} & \alpha \leq 1 \ . \end{cases} \tag{C.2}$$

We will not consider the proof in this course. For the purposes of testing series found in this course, a convenient and practical specialisation of Gauss' test is to consider

$$\frac{t_n}{t_{n+1}} = \frac{n^2 + a_1 n + a_0}{n^2 + b_1 n + b_0} \ . \tag{C.3}$$

Then

$$\sum_n t_n \;\to\; \begin{cases} \text{converges if} & a_1 - b_1 > 1 \ , \\ \text{diverges if} & a_1 - b_1 \leq 1 \ . \end{cases} \tag{C.4}$$

Take care: Gauss' test (as presented) deals with the ratio of terms the other way round compared to the ratio test! Also note that Gauss' test applies when all terms are positive. However, it does not matter if some initial terms in a series are negative. As long as there is a certain point beyond which *all remaining* terms in an infinite series are positive, then Gauss' test indicates the convergence property of that remaining infinite series. So is still applicable.

Example – Legendre series
Let's consider the infinite series solutions to Legendre's differential equation. The recurrence relation generating the power series coefficients is

$$c_{n+2} = c_n \frac{n(n+1) - k}{(n+2)(n+1)} \qquad (n = 0, 1, 2, \ldots)$$

hence consecutive terms (in both odd & even series) are $c_n x^n$ and $c_{n+2} x^{n+2}$. Therefore the Legendre

series splits into even and odd parts. Let $n = 2m$. Then for the series with even powers, $t_m = c_{2m}x^{2m}$ and $t_{m+1} = c_{2(m+1)}x^{2(m+1)}$ such that

$$\frac{t_m}{t_{m+1}} = \frac{c_{2m}x^{2m}}{c_{2(m+1)}x^{2(m+1)}} = \frac{(2m+2)(2m+1)}{2m(2m+1)-k}\frac{1}{x^2} \ .$$

At $|x| = 1$:

$$\frac{t_m}{t_{m+1}} = \frac{4m^2+6m+2}{4m^2+2m-k} = \frac{m^2+\frac{6}{4}m+\frac{2}{4}}{m^2+\frac{2}{4}m-\frac{k}{4}} \ .$$

Comparing to the form of Gauss' test (C.3) we can read off the important coefficients; $a_1 = 3/2$, $b_1 = 1/2$. Thus

$$a_1 - b_1 = 1$$

since $a_1 - b_1 \leq 1$, the infinite series is **divergent** at $|x| = 1$.

What about the series involving only odd powers? Then $n = 2p+1$ $(p \in \mathbb{Z})$ and

$$\begin{aligned}\frac{t_p}{t_{p+1}} &= \frac{c_{2p+1}x^{2p+1}}{c_{2p+3}x^{2p+3}} = \frac{(2p+3)(2p+2)}{(2p+1)(2p+2)-k}\frac{1}{x^2} \\ &= \frac{p^2+\frac{10}{4}p+\frac{6}{4}}{p^2+\frac{6}{4}+\frac{(2-k)}{4}} \ .\end{aligned}$$

Both a_1 and b_1 have increased by 1, but $a_1 - b_1 = 1$ still, hence the odd series has exactly the same convergence properties as the even; it **diverges** at $|x| = 1$.

(*Given that the odd and even infinite power series arise from the same recurrence relation, it would be strange if they did not have the same convergence properties!*)

Appendix D

Proof of Legendre Polynomial orthogonality

The particular proof below is ***non-examinable***. However, you are expected to be to apply the concepts of section 5.4 to show that Legendre's equation together with truncated (i.e. polynomial) solutions form a Sturm-Liouville problem, which guarantees that these Legendre polynomials are orthogonal.

Proof of orthogonality – Use the Legendre equation itself for $P_\ell(x)$:

$$(1-x^2)\frac{d^2P_\ell}{dx^2} - 2x\frac{dP_\ell}{dx} + \ell(\ell+1)P_\ell = 0 \ . \tag{D.1}$$

Note that the first two terms can be written as a perfect derivative:

$$\frac{d}{dx}\left[(1-x^2)\frac{dP_\ell}{dx}\right] + \ell(\ell+1)P_\ell = 0 \ . \tag{D.2}$$

Now multiply by $P_n(x)$:

$$P_n\frac{d}{dx}\left[(1-x^2)\frac{dP_\ell}{dx}\right] + \ell(\ell+1)P_\ell P_n = 0 \ , \tag{D.3}$$

and similarly (by exchanging ℓ and n):

$$P_\ell\frac{d}{dx}\left[(1-x^2)\frac{dP_n}{dx}\right] + n(n+1)P_\ell P_n = 0. \tag{D.4}$$

Subtract these. First note that (from $d(AB)/dx = BdA/dx + AdB/dx$)

$$\frac{d}{dx}\left[(1-x^2)\frac{dP_\ell}{dx}P_n\right] = P_n\frac{d}{dx}\left[(1-x^2)\frac{dP_\ell}{dx}\right] + (1-x^2)\frac{dP_\ell}{dx}\frac{dP_n}{dx} \tag{D.5}$$

so if we interchange the indices ($\ell \leftrightarrow n$) in equation (D.5) and then subtract that from equation (D.5), the last terms on the RHS cancel and we get

$$P_n\frac{d}{dx}\left[(1-x^2)\frac{dP_\ell}{dx}\right] - P_\ell\frac{d}{dx}\left[(1-x^2)\frac{dP_n}{dx}\right] = \frac{d}{dx}\left[(1-x^2)\left(\frac{dP_\ell}{dx}P_n - \frac{dP_n}{dx}P_\ell\right)\right] \ . \tag{D.6}$$

If we insert this into the equation (D.3) minus equation (D.4) and integrate, we get

$$\int_{-1}^{1} dx\,\frac{d}{dx}\left[(1-x^2)\left(\frac{dP_\ell}{dx}P_n - \frac{dP_n}{dx}P_\ell\right)\right] + \int_{-1}^{1} dx\,P_\ell P_n\left[\ell(\ell+1) - n(n+1)\right] = 0. \tag{D.7}$$

Now the first integral is zero, since it is the difference between the square bracket value at $x = -1$ and $x = 1$, and these are both zero because of the $1 - x^2$ factor. Hence

$$[\ell(\ell+1) - n(n+1)] \int_{-1}^{1} dx\, P_\ell P_n = 0 \tag{D.8}$$

and so, if $\ell \neq n$,

$$\int_{-1}^{1} dx\, P_\ell(x) P_n(x) = 0 \qquad \ell \neq n \ . \tag{D.9}$$

Hence the Legendre polynomials are orthogonal. **[QED]**

Appendix E

Bessel functions of the 2nd kind

Non-examinable

The Bessel function of the second Kind can be written as

$$Y_p(x) = \lim_{\mu \to p} \left[\frac{J_\mu(x)\cos\mu\pi - J_{-\mu}(x)}{\sin\mu\pi} \right] .$$

It is linearly independent to $J_p(x)$. For non-integer p the limit is not really needed. You can then see that $Y_p(x)$ is linear combination of J_p and J_{-p}, which makes it linearly independent to J_p. The limit is needed for integer p since the numerator and denominator both evaluate to zero there. For reference, it can be shown that

$$Y_0(x) = \frac{2}{\pi}\left[\ln x + \gamma - \ln 2\right] J_0(x) + \frac{2}{\pi}\left\{ \frac{x^2}{2^2} - \frac{x^4}{2^2 4^2}\left(1 + \frac{1}{2}\right) + \frac{x^6}{2^2 4^2 6^2}\left(1 + \frac{1}{2} + \frac{1}{3}\right) - \ldots \right\} ,$$

where $\gamma = 0.5772156\ldots$ is known as *Euler's constant*. This $p = 0$ case is interesting as it corresponds to the indicial equation having a repeated root $r = \pm p = 0$. Notice the term proportional to $J_0(x)\ln x$ in the above expression for $Y_0(x)$. For series solutions about regular singular points, the appearance of a term proportional to $y_1(x)\ln x$ in the expression for $y_2(x)$ is typical when the roots of the indicial equation are repeated or are two distinct integers.[1]

[1] See Boyce, DiPrima & Meade, theorem 5.6.1, page 227.

Appendix F

Semi-infinte rod problem: Extension to non-axisymmetric case

Non-examinable

Here we again consider a semi-infinte rod ($0 \leq z < \infty$, radius a) with homogeneous BCs at $r = a$ (and $z \to \infty$) as in section 9.2.1. However, we now allow the inhomogeneous BC (i.e. applied temperature at the end $z = 0$) to have ϕ dependence, i.e., $u(r, \phi, 0) = f(r, \phi)$. Thus u inside the rod will also depend on ϕ.

Now we have to lift the restriction made in section 9.2.1 that $m = 0$. Normal modes with J_1, J_2, etc., are now permitted, not just J_0. However, the zeros of J_0, J_1, J_2, etc., do not coincide, so we need to carefully distinguish them; $x_{m,n}$ is the nth zero of the mth order Bessel function, i.e., $J_m(x_{m,n}) = 0$. Thus setting the argument of J_m to $ka = x_{m,n}$ to ensure it meets the homogeneous BC sets the separation constant (or eigenvalue) to $k_{m,n} \equiv x_{m,n}/a$. Thus the normal modes are

$$u_{m,n} = J_m(k_{m,n} r) e^{im\phi} e^{-k_{m,n} z} \tag{F.1}$$

and the general solution (9.35) extends to

$$u(r, \phi, z) = \sum_{m=-\infty}^{\infty} \sum_{n=1}^{\infty} c_{m,n} J_m(k_{m,n} r) e^{im\phi} e^{-k_{m,n} z} \ . \tag{F.2}$$

(Remember that $J_{-m}(x) = (-1)^m J_{-m}(x)$, thus the sum $\sum_{m=-\infty}^{\infty}$ involving $e^{im\phi}$ could be replaced with $\sum_{m=0}^{\infty}$ with $\{\cos m\phi, \sin m\phi\}$.) Also note that eigenfunction orthogonality (9.36) extends to

$$\int_0^{2\pi} \int_0^a \left\{ J_m(k_{m,n} r) e^{im\phi} \right\} \left\{ J_{m',n'}(k_{m',n'} r) e^{-im'\phi} \right\} r \, dr \, d\theta = A_{m,n} \, \delta_{m,m'} \delta_{n,n'} \ , \tag{F.3}$$

where $A_{m,n}$ is a normalisation factor (that can be found from mathematical tables) that generalises that on the RHS of equation (9.36). The normal mode coefficients are found via

$$c_{m,n} = \frac{1}{A_{m,n}} \int_0^{2\pi} \int_0^a J_m(k_{m,n} r) e^{im\phi} f(r, \phi) \, r \, dr \, d\theta \ . \tag{F.4}$$

This expression is found in the familiar way, by setting $z = 0$ in the general solution, eqn (F.2), equating this to the boundary function $f(r, \phi)$, multiplying through by the normal mode $u^*_{m',n'}$, integrating over r and ϕ and invoking the orthogonality condition (F.3).

Appendix G

Green's Functions in terms of Wronksian and ODE basis functions

Non-examinable

For a ***self-adjoint BVP***, the Green's function is related to the Wronskian determinant W and the basis functions y_1, y_2 in the following, elegant way

$$G(x,z) = \frac{1}{W(z)a_2(z)} \begin{cases} y_1(x)y_2(z) & (x < z) \\ y_2(x)y_1(z) & (x > z) \end{cases} . \tag{G.1}$$

For example, the problem in section 10.2, $\hat{L} = \frac{d^2}{dx^2} + k^2$ (with $k = 1$) with homogeneous BCs, is indeed in self-adjoint form, with $y_1(x) = \sin x$, $y_2(x) = \cos x$ and $a_2(x) = 1$. Thus $W(x) = -1$. Inserting these into equation (G.1) recovers the GF found 'manually' as given in equation (10.13):

$$G(x,z) = \begin{cases} -\cos z \sin x & (x < z) \\ -\sin z \cos x & (x > z) \end{cases} .$$

It can be shown that the Wronksian form of the GF is symmetrical; it satisfies

$$G(x,z) = G(z,x) .$$

This involves using Abel's theorem to show that Wa_2 is a constant. (See below.) With this, we can see that interchanging $x \leftrightarrow z$ in equation G.1 leaves it unchanged.

Proof – To prove equation (G.1), consider the self-adjoint operator (i.e. one that yields an ODE in Sturm-Liouville form)

$$\hat{L}[y] = \frac{d}{dx}\left(a_2(x)\frac{dy}{dx}\right) + a_0(x)y(x)$$

on $a \leq x \leq b$, together with homogeneous boundary conditions $y(a) = y(b) = 0$. This operator expands to

$$\hat{L}[y] = a_2(x)\frac{d^2y}{dx^2} + a_2'(x)\frac{dy}{dx} + a_0(x)y(x) . \tag{G.2}$$

Suppose that the general solution to $\hat{L}[y] = 0$ is known, and furthermore, suppose that the two basis functions are chosen such that $y_1(a) = 0$ and $y_2(b) = 0$. The CF for the ODE $\hat{L}[y] = 0$ is

$$y_{CF} = c_1 y_1(x) + c_2 y_2(x)$$

where c_1 and c_2 are arbitrary constants. The GF therefore has the form

$$G(x,z) = \begin{cases} A(z)y_1(x) + B(z)y_2(x) & (x < z) \\ C(z)y_1(x) + D(z)y_2(x) & (x > z) \end{cases} .$$

To meet the left BC, $y(a) = 0$ means $A\cdot 0 + By_2(a) = 0$ hence $B = 0$. To meet the right BC, $y(b) = 0$ means $Cy_1(b) + D\cdot 0 = 0$ hence $C = 0$. Thus

$$G(x,z) = \begin{cases} A(z)y_1(x) & (x < z) \\ D(z)y_2(x) & (x > z) \end{cases} .$$

The derivative of the GF is

$$\frac{dG(x,z)}{dx} = \begin{cases} A\,y_1'(x) & (x < z) \\ D\,y_2'(x) & (x > z) \end{cases} .$$

The discontinuity of G' at $x = z$ is

$$\lim_{\epsilon\to 0} \; G'|_{z+\epsilon} - G'|_{z-\epsilon} = \frac{1}{a_2(z)} \quad .$$

Continuity of G and the jump in G' at $x = z$ imply

$$\begin{aligned} &G \text{ continuity:} && Dy_2(z) - Ay_1(z) \;=\; 0 \\ &G' \text{ discontinuity:} && Dy_2'(z) - Ay_1'(z) \;=\; [a_2(z)]^{-1} \quad , \end{aligned}$$

which in matrix form is

$$\mathbf{M}\mathbf{v} \;=\; \begin{bmatrix} y_1(z) & y_2(z) \\ y_1'(z) & y_2'(z) \end{bmatrix} \begin{pmatrix} -A \\ D \end{pmatrix} = \begin{pmatrix} 0 \\ a_2^{-1} \end{pmatrix} .$$

We need the inverse of $\mathbf{M}$ to find A and D. This is

$$\mathbf{M}^{-1} \;=\; \frac{1}{\det(\mathbf{M})} \begin{bmatrix} y_2'(z) & -y_1'(z) \\ -y_2(z) & y_1(z) \end{bmatrix}^T \;=\; \frac{1}{W_{y_1,y_2}(z)} \begin{bmatrix} y_2'(z) & -y_2(z) \\ -y_1'(z) & y_1(z) \end{bmatrix} .$$

You can see how the Wronskian determinant W has emerged from finding the inverse of the matrix! (i.e. in the process of solving the simultaneous equations) Thus

$$\begin{pmatrix} -A \\ D \end{pmatrix} \;=\; \frac{1}{W_{y_1,y_2}(z)} \begin{bmatrix} y_2'(z) & -y_2(z) \\ -y_1'(z) & y_1(z) \end{bmatrix} \begin{pmatrix} 0 \\ a_2^{-1} \end{pmatrix} \;=\; \frac{1}{W} \begin{pmatrix} 0 - y_2(z)/a_2 \\ 0 + y_1(z)/a_2 \end{pmatrix} .$$

Hence

$$A = \frac{y_2(z)}{W(z)a_2(z)} \quad , \qquad D = \frac{y_1(z)}{W(z)a_2(z)} \quad .$$

This determines the GF to be

$$G(x,z) = \frac{1}{W(z)a_2(z)} \begin{cases} y_1(x)y_2(z) & (x < z) \\ y_2(x)y_1(z) & (x > z) \end{cases}$$

[QED]

Proof – $W(x)a_2(x) = c$ for an ODE in Sturm-Liouville form – To prove this we use Abel's theorem which allows us to determine W (to within a multiplicative constant) from the coefficients of the ODE:

$$W(x) = ce^{-\int^x p(z)dz}$$

when the ODE is written in standard form $y'' + p(x)y' + q(x)y = g(x)$. Coverting the ODE with self-adjoint $\hat{L}[y]$ given by (G.2) to standard from yields

$$y'' + \frac{a_2'(x)}{a_2(x)}y' + \frac{a_0(x)}{a_2(x)}y(x) = \text{RHS} \ .$$

Thus

$$W \ = \ c\exp\left(-\int^x \frac{1}{a_2(z)}\frac{da_2}{dz}dz\right) \ = \ ce^{-\ln|a_2(x)|} \ = \ \frac{c}{a_2(x)} \ .$$

Thus $W(x)a_2(x) = c$.

[QED]

Appendix H

Eigenfunction expansion of Green's Functions

Non-examinable

Here we consider how Green's functions can be built from a superposition of eigenfunctions. The benefit of this is that it will highlight the phenomenon of ***resonance***.

We would like to solve the following inhomogeneous DE

$$\boxed{\hat{L}[y(x)] - \mu\rho(x)y(x) = f(x)} \tag{H.1}$$

on $[a,b]$ subject to homogeneous boundary conditions. Here μ is a parameter, set by the physics of the problem under consideration. We take the case where the linear differential operator $\hat{L}$ is such that together with the BCs, it forms a Sturm-Liouville problem. The eigenfunctions and eigenvalues satisfy

$$\hat{L}[y_n(x)] = \lambda_n\rho(x)y_n(x) \tag{H.2}$$

and the homogeneous BCs. Since the problem is a Sturm-Liouville one, we know that the eigenfunctions $y_n(x)$ are orthogonal and complete, whilst the eigenvalues λ_n are real. We will denote normalised versions of the eigenfunctions as $\hat{y}_n(x)$.

The solution to the problem formed by the inhomogeneous DE (H.1) and homogeneous BCs can be formed from a superposition of eigenfunctions;

$$y(x) = \sum_{n=0}^{\infty} c_n\hat{y}_n(x) \ . \tag{H.3}$$

The task is to find the necessary c_n. To do this, we insert (H.3) into (H.1), and use the linearity of $\hat{L}$ to bring it inside the summations;

$$\begin{aligned} f(x) &= \sum_n c_n\left[\hat{L}[\hat{y}_n(x)] - \mu\rho\hat{y}_n\right] \\ &= \sum_n c_n\left[(\lambda_n - \mu)\rho(x)\hat{y}_n(x)\right] \ . \end{aligned} \tag{H.4}$$

We now multiply through by $\hat{y}_m^*$, integrate and use the orthogonality of the eigenfunctions to isolate

c_m:

$$\begin{aligned}\int_a^b \hat{y}_m^*(z)f(z)dz &= \sum_n c_n(\lambda_n-\mu)\int_a^b \hat{y}_m^*(z)\hat{y}_n(z)\rho(z)dz \\ &= \sum_n c_n(\lambda_n-\mu)\delta_{nm} \\ &= c_m(\lambda_m-\mu)\ .\end{aligned} \tag{H.5}$$

Inserting c_m into (H.3) yields

$$y(x) = \int_a^b \left\{\sum_n \frac{\hat{y}_n^*(z)\hat{y}_n(x)}{\lambda_n-\mu}\right\} f(z)dz$$

and we identify the quantity in $\{\ldots\}$ as the Green's function:

$$\boxed{G(x,z) = \sum_n \frac{\hat{y}_n^*(z)\hat{y}_n(x)}{\lambda_n-\mu}\ .} \tag{H.6}$$

Note that this expression G possesses *adjoint symmetry* in x and z; $G(z,x) = G^*(x,z)$, since λ_n are real and assuming μ is real too. If the eigenfunctions are real, then it is symmetrical: $G(z,x) = G(x,z)$.

For the case of continuous eigenvalues (e.g. for infinite domains)

$$\boxed{G(x,z) = \int_\lambda \frac{y_\lambda^*(z)y_\lambda(x)}{\lambda-\mu}\,d\lambda\ .} \tag{H.7}$$

Resonance

For the Helmholtz equation example in section 10.2

$$\left[\frac{d^2}{dx^2}+k^2\right]X(x) = f(x)$$

we identify $\hat{L} = \frac{d^2}{dx^2}$, $\mu = -k^2$ and $\rho = 1$. We see that the parameter $\mu = -(\omega/c)^2$ is set by the chosen, fixed, driving frequency ω and the phase speed c (determined by the properties of the system). The eigenfunctions and eigenvalues of this $\hat{L}$ together with the homogenous BCs that we had ($X(0) = X(L) = 0$) are the now very familiar sine profiles $X_n(x) = \sin(n\pi x/L)$ with $\lambda_n = -(n\pi/L)^2$. Hence in this case

$$G(x,z) = \frac{4}{L}\sum_{n=1}^{\infty} \frac{\sin(n\pi z/L)\sin(n\pi x/L)}{\frac{\omega^2}{c^2}-\left(\frac{n\pi}{L}\right)^2}\ . \tag{H.8}$$

It is clear here, that $G(x,z)$ does not change if x and z are interchanged, so is symmetrical. Notice that if one of the eigenvalues, say λ_k, happens to equal μ (i.e. $k\pi/L = \omega/c$) then the denominator for that term in the sum goes to zero and the GF becomes unbounded. The means that the response $X(x)$ to the driver will become infinitely large too. This is ***resonance*** which occurs when the driving frequency precisely matches the frequency of a normal mode. This example is undamped. With the addition of damping, the denominator will vanish and large, but finite vibration will occur instead. Also, if the source profile $f(x)$ contains no component of the $\hat{y}_k$ mode, then $\int \hat{y}_k^*(z)f(z)dz = 0$ and resonance will be avoided.

Appendix I

Common Green's Functions in Physics

Non-examinable

Table I.1 shows the fundamental Green's function for the Laplace, Helmholtz and modified Helmholtz equations, in 1D, 2D and 3D.

	$\hat{L}$	1D	2D	3D
Laplace:	∇^2	$\frac{1}{2}\lvert x-z\rvert$	$\frac{1}{2\pi}\ln\lvert\mathbf{x}-\mathbf{z}\rvert$	$-\frac{1}{4\pi\lvert\mathbf{x}-\mathbf{z}\rvert}$
Helmholtz:	∇^2+k^2	$\frac{i}{2k}\exp(-ik\lvert x-z\rvert)$	$-\frac{i}{4}H_0^{(1)}(k\lvert\mathbf{x}-\mathbf{z}\rvert)$	$-\frac{\exp(ik\lvert\mathbf{x}-\mathbf{z}\rvert)}{4\pi\lvert\mathbf{x}-\mathbf{z}\rvert}$
Modified Helmholtz:	∇^2-k^2	$\frac{1}{2k}\exp(-k\lvert x-z\rvert)$	$-\frac{1}{2\pi}K_0(k\lvert\mathbf{x}-\mathbf{z}\rvert)$	$-\frac{\exp(-k\lvert\mathbf{x}-\mathbf{z}\rvert)}{4\pi\lvert\mathbf{x}-\mathbf{z}\rvert}$

Table I.1: Fundamental Green's functions $G(\mathbf{x},\mathbf{z})$ for common, time independent operators in Physics, in 1D, 2D and 3D. $\mathbf{x}$ denotes the observation position, and $\mathbf{z}$ the source.

Points

- The modified Helmholtz GFs are the same as the Helmholtz ones with $k \rightarrow ik$.
- The 1D Laplace GF can easily be derived using the techniques of section 10.2.
  The 2D Laplace GF can be derived using the same procedure used in section 10.4.1. [*Try it!*]
- The 3D Helmholtz GF will be derived in your (Year 2) Electromagnetism – Part 1. You can see it is an oscillating function that falls off as $1/r$.
- The 2D Helmholtz GFs involve Bessel functions (as could be expected).
  - $H_0^{(1)}(x)$ is the Hankel function of the first kind of order zero. It is essentially the Bessel function equivalent of $e^{ix} = \cos(x) + i\sin(x)$:

$$H_0^1(x) = J_0(x) + iY_0(x) \approx \begin{cases} a\sqrt{x}e^{ix} & (x >> 1) \\ i\frac{2}{\pi}\ln\left(\frac{x}{2}\right) & (x \rightarrow 0) \end{cases}$$

    where a is a constant. For large x this GF oscillates. For small x it diverges.

– $K_0(x)$ is the modified Bessel function of the second kind (of order 0). It is related to the Hankel function; $K_0(x) \propto H_0^{(1)}(ix)$. It is the solution of the modified Bessel equation

$$x^2y'' + xy' - (x^2 + m^2)y = 0$$

with $m = 0$. (The other independent solution is the modified Bessel function of the first kind, $I_m(x)$.)

I.1 Green's Function for PDEs with Time Dependence

For equations involving space and time, the GF satisfies

$$\hat{L}[G(x,t;z,\tau)] = \delta(x-z)\delta(t-\tau) \ . \tag{I.1}$$

The solution to $\hat{L}[u(x,t)] = f(x,t)$ would be given by

$$u(x,t) = \int_{\tau=0}^{\infty}\int_{z=-\infty}^{\infty} G(x,t;z,\tau)f(z,\tau)\,dz\,d\tau \ . \tag{I.2}$$

Table I.2 shows some important cases.

	$\hat{L}$	1D	3D
Wave:	$\frac{\partial^2}{\partial t^2} - c^2\frac{\partial^2}{\partial x^2}$	$\frac{1}{2c}\Theta\left[(t-\tau) - \frac{\lvert x-z\rvert}{c}\right]$	$\frac{1}{4\pi c\lvert\mathbf{x}-\mathbf{z}\rvert}\delta\left[(t-\tau) - \frac{\lvert\mathbf{x}-\mathbf{z}\rvert}{c}\right]$
Diffusion:	$\frac{\partial}{\partial t} - D\frac{\partial^2}{\partial x^2}$	$\Theta(t-\tau)\left(\frac{1}{4\pi D(t-\tau)}\right)^{1/2}\exp\left(-\frac{(x-z)^2}{4D(t-\tau)}\right)$	$\Theta(t-\tau)\left(\frac{1}{4\pi D(t-\tau)}\right)^{3/2}\exp\left(-\frac{\lvert\mathbf{x}-\mathbf{z}\rvert^2}{4D(t-\tau)}\right)$

Table I.2: Fundamental Green's functions for common time-dependent operators in Physics, in 1D and 3D. (The 2D versions are different.) $G(x,t;z,\tau)$ where x,t denotes the observation position & time, and z,τ the source of an impulse.

Points

Diffusion equation

- The 1D case is the solution seen in section 8.1, but with the 'initial' time and source position shifted to τ and z, respectively.
- The 3D case merely has a different normalising amplitude $(\ldots)^{3/2}$ rather than $(\ldots)^{1/2}$ in the 1D case.
- The Heaviside step function ensures the GF is zero before the impulse is applied at $t = \tau$.

Wave equation

- The Green's functions for the wave equation are essentially restricting the influence at location x,t to points in the past within (or on) the 'light-cone'.

- The GF for the wave equation in 3D will be derived in (Year 2) Electromagnetism – Part 1. It will be important for solving the inhomogeneous wave equations for electric potential ϕ and vector potential $\mathbf{A}$, where charge and electrical currents are sources of waves in ϕ and $\mathbf{A}$, respectively. In this context, this GF is known as the retarded potential.
- In general, the derivation of these GFs requires Laplace transforms in time and FT in space.

Appendix J

Advanced aspects of GFs and BCs in multiple dimensions

Non-examinable

Here we generalise things to two or three dimensions more rigorously.

J.1 Eigenfunction expansion

A linear, partial differential operator $\hat{L}$ is self-adjoint iff

$$\int_V v^*(\mathbf{x})\hat{L}[u(\mathbf{x})]d^3x = \int_V \left\{\hat{L}[v(\mathbf{x}]\right\}^* u(\mathbf{x})d^3x \tag{J.1}$$

where $v(\mathbf{x})$ and $u(\mathbf{x})$ are solutions to the PDE $\hat{L}[y(\mathbf{x})] = 0$ and homogeneous boundary conditions. Eigenfunctions of the PDE satisfy

$$\hat{L}[y_n(\mathbf{x})] = \lambda_n \rho(\mathbf{x}) y_n(\mathbf{x}) \ .$$

With this, the same eigenfunction expansion seen in 1D carries over. If we wish to solve

$$\hat{L}[y(\mathbf{x})] - \mu\rho(\mathbf{x})y(\mathbf{x}) = f(\mathbf{x})$$

together with homogeneous BCs, then the GF is

$$G(\mathbf{x},\mathbf{z}) = \sum_n \frac{\hat{y}_n^*(\mathbf{z})\hat{y}_n(\mathbf{x})}{\lambda_n - \mu} \ . \tag{J.2}$$

J.2 BCs in Multiple Dimensions

We take Poisson's equation as an example to illustrate this. The Green's function for Poisson's equation $\nabla^2 u(\mathbf{x}) = f(\mathbf{x})$ must satisfy $\nabla^2 G(\mathbf{x},\mathbf{z}) = \delta(\mathbf{x}-\mathbf{z})$. To understand how BCs behave in multiple dimensions we start with Green's second theorem

$$\int_V \left(\phi\nabla^2\psi - \psi\nabla^2\phi\right) d^3x = \oint_S (\phi\nabla\psi - \psi\nabla\phi)\cdot d\mathbf{s} \tag{J.3}$$

for a volume V bounded by a surface S. Setting $\phi = u(\mathbf{x})$ and $\psi = G(\mathbf{x},\mathbf{z})$ and inserting into Green's second theorem, yields

$$\int_V \left[u(\mathbf{x})\nabla^2 G(\mathbf{x},\mathbf{z}) - G(\mathbf{x},\mathbf{z})\nabla^2 u(\mathbf{x})\right] d^3x = \oint_S \left[u(\mathbf{x})\nabla G(\mathbf{x},\mathbf{z}) - G(\mathbf{x},\mathbf{z})\nabla u(\mathbf{x})\right]\cdot d\mathbf{s} \ .$$

Using $\nabla^2 G(\mathbf{x},\mathbf{z}) = \delta(\mathbf{x}-\mathbf{z})$ and $\nabla^2 u(\mathbf{x}) = f(\mathbf{x})$ in the first and second terms on the lefthand side:

$$\int_V \left[u(\mathbf{x})\delta(\mathbf{x}-\mathbf{z}) - G(\mathbf{x},\mathbf{z})f(\mathbf{x})\right] d^3x = \oint_S \left[u(\mathbf{x})\nabla G(\mathbf{x},\mathbf{z}) - G(\mathbf{x},\mathbf{z})\nabla u(\mathbf{x})\right]\cdot d\mathbf{s} \ .$$

If $\mathbf{z}$ lies within V then $\int_V u(\mathbf{x})\delta(\mathbf{x}-\mathbf{z})d^3x = u(\mathbf{z})$, thus

$$u(\mathbf{z}) = \int_V G(\mathbf{x},\mathbf{z})f(\mathbf{x})d^3x + \oint_S \left[u(\mathbf{x})\nabla G(\mathbf{x},\mathbf{z}) - G(\mathbf{x},\mathbf{z})\nabla u(\mathbf{x})\right]\cdot d\mathbf{s} \ .$$

Finally, interchanging labels z and x and using the symmetry of the Green's function yields the solution to Poisson's equation,

$$u(\mathbf{x}) = \int_V G(\mathbf{x},\mathbf{z})f(\mathbf{z})d^3z + \oint_S \left[u(\mathbf{z})\nabla G(\mathbf{x},\mathbf{z}) - G(\mathbf{x},\mathbf{z})\nabla u(\mathbf{z})\right]\cdot d\mathbf{s}_z \tag{J.4}$$

where the surface integral is with respect to $\mathbf{z}$. The surface integral vanishes for the case of an infinite domain (for physically sensible cases), but is crucial in meeting inhomogeneous boundary conditions in multiple dimensions, for finite (or semi-infinite domains).

Dirichlet problems

For Dirichlet BCs – $u(\mathbf{x}) = w(\mathbf{x})$ for $\mathbf{x}$ on S – one chooses $G(\mathbf{x},\mathbf{z}) = 0$ for $\mathbf{x}$ on S and the solution to Poisson's equation $\nabla^2 u(\mathbf{x}) = f(\mathbf{x})$ is

$$u(\mathbf{x}) = \int_V G(\mathbf{x},\mathbf{z})f(\mathbf{z})d^3z + \oint_S w(\mathbf{z})\nabla G(\mathbf{x},\mathbf{z})\cdot d\mathbf{s}_z \ . \tag{J.5}$$

You can see that the boundary term communicates the imposed value of u on the boundary $u = w(\mathbf{x})$ into the domain via the Green's function $\nabla G(\mathbf{x},\mathbf{z})$.

Neumann problems

For Neumann BCs – $\nabla u(\mathbf{x}) = w(\mathbf{x})$ for $\mathbf{x}$ on S – see §21.5.4 of Riley, Hobson and Bence, "Mathematical Methods for Physics and Engineering".

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